TAMIL NADU AGRICULTURAL UNIVERSITY

Agricultural College and Research Institute · Coimbatore

B.Sc. (Hons.) Agriculture — Third Semester

SWE 211

Fundamentals of Soil and Water Conservation Engineering

Lecture-wise Theory Study Material (2025 Syllabus)

Course title: Fundamentals of Soil and Water Conservation Engineering
Course code: SWE 211
Credit load: 2 + 1 (Theory + Practical)
Semester: Third (2025 syllabus)
Coverage: 17 lectures, fully worked examples, tables, diagrams, glossary, formula sheet and references

Prepared for self-study and examination revision. All numericals follow standard textbook constants and published ranges. This is study material, not a substitute for the official syllabus.

How to Use This Material

This handbook follows the official SWE 211 lecture sequence one-for-one. Every lecture is a self-contained chapter with the same internal structure so that revision is predictable and fast.

Structure of every chapter

- **Headed sections** expand each syllabus sub-topic in the order given in the syllabus.
- **Definitions** are placed in green boxes — these are the sentences to reproduce in a definition question.
- **Formulas** are placed in yellow boxes with every symbol explained.
- **Tables** summarise comparisons and standard values for one-glance revision.
- **Text diagrams** (ASCII/box art) show the layout of instruments, structures and sections.
- **Worked numericals** are set out step by step with the formula, the substitution and the answer with units.
- **Exam focus** closes each chapter with key points, short-answer questions with model answers, and long-answer questions with answer outlines.

Exam strategy in brief

- Most theory papers weight *definitions, comparisons and causes/effects*; the numericals in this course are usually direct one-formula problems — memorise the formula sheet at the end.
- Units carry marks. Always write the unit (m, cm, ha, cumec, t/ha/yr, hp) with the final answer.
- In "list and explain" questions, write the heading, then one crisp line per point — not paragraphs.
- Draw the relevant text diagram wherever a figure is asked; a labelled sketch earns marks even if small.

Note on values: Where a soil, crop or structure parameter varies across textbooks, a standard published range is given instead of a single false-precise number. Use the range and state the assumption.

Contents

1 Introduction — Land Surveying, Uses in Agriculture, Chain and Cross Staff

2 Dumpy Level — Setting, Types of Levelling, Computation of Land Slope and Difference in Elevation

3 Computation of Area and Volume — Simpson's Rule and Trapezoidal Rule

4 Soil Erosion — Causes, Evil Effects, Geologic and Accelerated Erosion

5 Water Erosion — Causes, Erosivity and Erodibility, Stages, Ravines, Landslides, USLE

6 Wind Erosion — Factors, Mechanics, Effects of Water and Wind Erosion

7 Erosion Control for Agricultural Lands — Biological Measures

8 Mechanical Measures — Bunds, Terraces, Trenches, Stone Walls

9 Mid Semester Examination — Revision and Model Paper

10 Rainwater Harvesting — Runoff Water Harvesting and Runoff Computation

11 Ground Water Recharge Techniques

12 Farm Ponds and Percolation Ponds

13 Irrigation — Measurement of Flow in Open Channels

14 Water Duty and Irrigation Efficiencies

15 Agricultural Drainage — Need, Surface and Subsurface Systems, Drainage Coefficient

16 Types of Wells and Sizes — Suitability

17 Types of Pumps and Working Principles

A Glossary of Key Terms

B Formula Sheet (One-Page Revision)

C Reference List and Further Reading

1 Introduction — Land Surveying, Uses in Agriculture, Chain and Cross Staff

1.1 What is Surveying?

> **Surveying** is the art and science of determining the relative positions of points on, above or below the surface of the earth, and of representing these positions on a map or plan to a suitable scale.

Surveying answers two practical questions for a farm engineer: *where is a point* (its position) and *how high is it* (its elevation). The branch that fixes only positions on a horizontal plane is called **plane surveying** (used for small areas, where the earth's curvature is neglected). When the area is very large and curvature must be considered, it is **geodetic surveying**.

1.2 Classification of Surveying

- **By purpose:** land survey, engineering survey, topographic survey, cadastral (revenue) survey, city survey, mine survey, hydrographic survey.
- **By instrument:** chain survey, compass survey, plane-table survey, theodolite survey, level survey, tacheometric survey, total-station / GNSS survey.
- **By nature of ground:** plain survey (flat ground), geodetic survey (curved earth).

1.3 Uses of Surveying in Agriculture

Agricultural engineering depends on accurate ground measurement at every stage of land and water development. The main uses are:

- **Farm layout and land levelling:** designing field boundaries, irrigation and drainage layouts on a proper gradient.
- **Land grading and levelling for irrigation:** fixing cuts and fills so water spreads evenly (especially for basin and furrow irrigation).
- **Contour mapping:** laying out contour bunds, contour trenches and terraces for soil conservation.
- **Irrigation and drainage design:** deciding canal alignments, channel slopes and drain gradients.
- **Soil conservation structures:** siting check dams, farm ponds, gully-control structures and waterways.
- **Watershed development:** computing catchment area, stream length and slope for runoff estimation.
- **Computing area of fields:** needed for input calculation, land records and crop planning.
- **Computing volume of earthwork:** for bunds, terraces, ponds, tanks and channels.
- **Locating wells, tube wells and pump houses** and laying pipe lines for sprinkler/drip systems.
- **Farm roads, culverts and bridges:** setting alignments and grades.

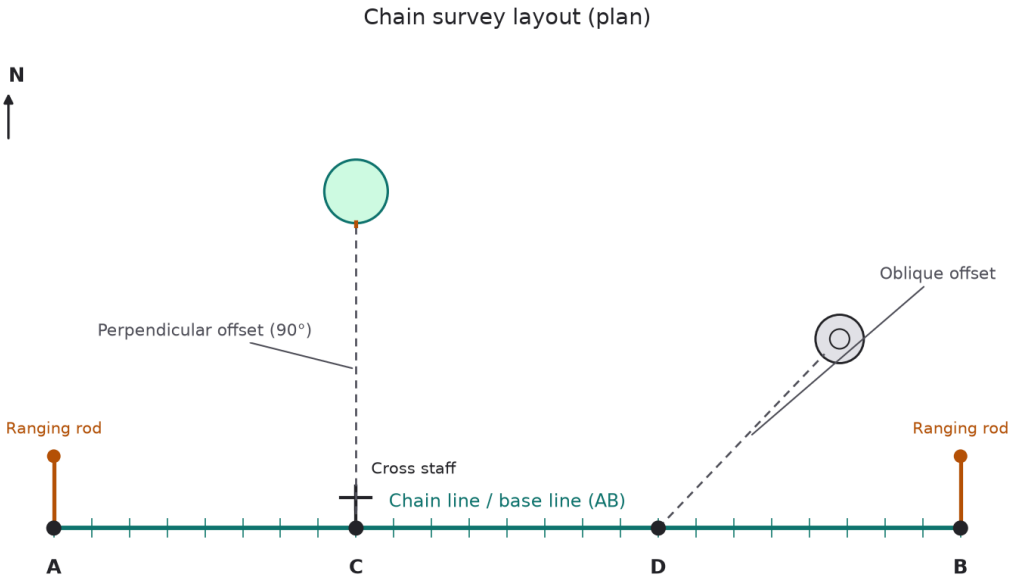


Figure 1.1 Chain survey layout (plan)

Chain line A–B with stations, cross staff and perpendicular / oblique offsets.

1.4 Scales and Ranging

> **Scale** is the fixed ratio between the distance on the plan and the corresponding distance on the ground. Example: 1 cm = 10 m means a scale of 1:1000.

- **Plain scale:** measures two dimensions (e.g., metres and decimetres).
- **Diagonal scale:** measures three successive units with greater accuracy (e.g., m, dm, cm).
- **Representative Fraction (RF):** RF = distance on plan / distance on ground. A scale of 1 cm = 5 m has RF = 1/500.

Ranging is the process of fixing intermediate points on a straight survey line. **Direct ranging** is used when both ends are visible; **reciprocal ranging** is used when the ends are not inter-visible (e.g., a hill in between).

1.5 Chain Surveying

Chain surveying is the simplest and oldest method of land measurement. Only linear measurements are taken; no angular measurement is made. It suits small, fairly level and open areas.

> **Chain surveying** is the method of surveying in which the area is divided into a network of triangles and only the sides (linear measurements) of these triangles are measured with a chain or tape.

Principle

A triangle is the only figure that can be plotted from the lengths of its sides alone. Hence the whole area is split into a chain of triangles (a **triangulation network** or framework).

Instruments used

- **Chain / Tape:** for measuring distances.
- **Arrows:** to mark the ends of a chain length on the ground.
- **Ranging rods:** to range and mark stations and intermediate points.
- **Cross staff:** to set out right angles (offsets).
- **Pegs, plumb bob, mallet:** to mark and hold stations.
- **Field book:** to record measurements in the field.

Types of chain

Table 1.1 — Common chains and tapes used in land surveying

Type	Length	No. of links	Link length	Typical use
Metric chain	20 m	100	0.2 m	Standard land survey (most common)
Metric chain	30 m	150	0.2 m	Large open fields
Gunter's chain	66 ft	100	0.66 ft	Old revenue/cadastral work
Engineer's chain	100 ft	100	1 ft	Engineering surveys (imperial)
Revenue chain	33 ft	16	2.0625 ft	Cadastral (revenue) survey
Steel band / metallic tape	20/30 m	—	Graduated	Accurate measurement
Invar tape	30/50 m	—	Graduated	Highest precision (low expansion)

Technical terms

- **Main survey line (chain line):** the primary line joining two main stations.
- **Base line:** the longest and most important main line, measured most carefully; the framework is built from it.
- **Tie line / subsidiary line:** a line joining two fixed points to check or take offsets.
- **Check line:** a line measured to verify the accuracy of the framework.
- **Offset:** the lateral distance of a detail (tree, corner, well) from the chain line. **Perpendicular offset** is at 90°; **oblique offset** is at any other angle.
- **Station:** a point on the chain line where measurement begins or ends (main station or subsidiary station).

Conducting a chain survey — steps

1. Reconnaissance (walk over the area to plan the framework).
2. Draw a rough sketch (index sketch) showing stations and lines.
3. Fix the base line and other main lines; measure them.
4. Take offsets to all details; enter in the field book.
5. Measure check lines to verify the framework.
6. Plot the survey to scale and compute the area.

1.6 Cross Staff and Offsetting

> **Cross staff** is a simple instrument used to set out a right angle (90°) at a point on the chain line, so that perpendicular offsets can be taken to nearby objects.

Types of cross staff

- **Open cross staff:** four vertical slits in two pairs at right angles; simplest and cheapest.
- **French cross staff:** an octagonal brass block with slits at 45° and 90°; can set out 45° also.
- **Adjustable cross staff:** can set out any angle, not only 90°.

Setting a perpendicular offset with a cross staff

1. Hold the staff vertically on the chain line near the object.
2. Look through one pair of slits and align it with the chain line.
3. Look through the other pair; if the object is not seen, move along the chain line until the object appears exactly in the second slit.
4. The point on the chain line below the staff is the foot of the perpendicular. Measure the offset.

Accuracy check: To test whether an offset is truly perpendicular, the sum of the two chainages at the foot of the offset should not change when the foot is re-measured after reversing the staff. Also the offsets to a station taken from two different chain lines must agree.

1.7 Text Diagram — Chain Survey Layout

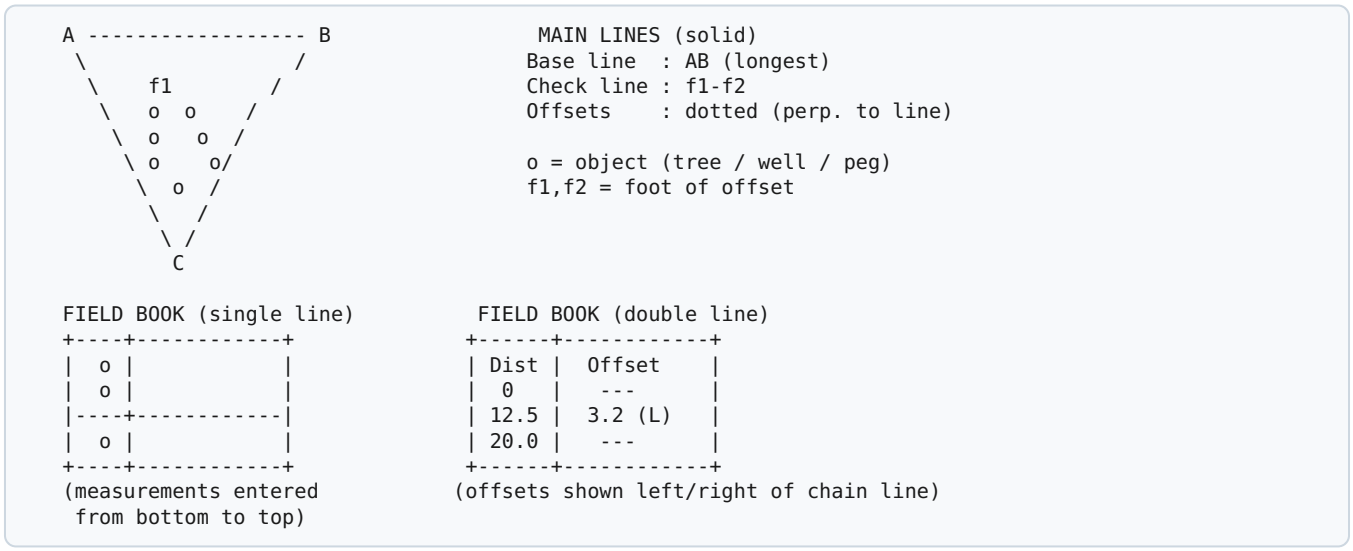


Fig. 1.1 — Chain survey framework with offsets and field-book entry

1.8 Chain Survey — Precautions and Errors

- **Personal errors:** wrong reading, mis-counting arrows, holding chain out of line.
- **Instrumental errors:** chain longer or shorter than standard (compare with a standard tape).
- **Natural errors:** sag, temperature change, slope, wind.
- **Corrections:** apply correction for incorrect chain length: *True length = (L' / L) × Measured length*, where L' is the actual chain length and L is the nominal length.

Correction for chain length:

Correct distance = Measured distance × (L' / L)

where L' = actual (true) length of chain, L = nominal (designated) length.

Worked Example 1.1 — Chain correction

A field is measured with a 20 m chain and the recorded length of a line is 350 m. On checking, the chain is found to be 0.10 m too long. Find the true length of the line.

Given: Measured length = 350 m; L = 20 m; L' = 20 + 0.10 = 20.10 m.

Step 1: True length = Measured × (L'/L) = 350 × (20.10 / 20.0)

Step 2: = 350 × 1.005 = **351.75 m**

Since the chain is too long, each measured length is a little short of the truth, so the true length is greater. Answer: 351.75 m.

Worked Example 1.2 — Chain too short

A line measured with a 30 m chain gives 240 m. The chain is found to be 0.15 m too short. Find the true length.

Given: L = 30 m; L' = 30 − 0.15 = 29.85 m; Measured = 240 m.

True length = 240 × (29.85 / 30.0) = 240 × 0.995 = **238.80 m**

1.9 Uses of Chain and Cross Staff in Agriculture

- Measuring field boundaries for area calculation and land records.
- Laying out contour bunds, trenches and field channels at measured intervals.
- Setting out planting geometry (row and plant spacing) in orchards and plantations.
- Marking experimental plots for field trials with exact dimensions.
- Fixing the alignment of farm roads, drains and pipelines.
- Taking offsets to trees, wells, buildings and other farm details for farm maps.

1.10 Survey Instruments and Their Uses

Table 1.2 — Common survey instruments and their uses

Instrument	Main use
Chain / steel tape	Measuring horizontal distances on the ground
Arrows (10 per chain)	Marking each chain length during measurement
Ranging rods	Ranging lines and marking stations (3–4 m long, banded)
Cross staff	Setting out right angles for perpendicular offsets
Optical square	Setting out right angles more accurately than the cross staff
Pegs and mallet	Marking stations and fixing them in the ground
Plumb bob	Transferring a point vertically to the ground
Field book	Recording all field measurements
Dumpy / auto level	Finding differences of elevation
Levelling staff	Reading heights on the level
Prismatic compass	Measuring bearings (angles from north)
Plane table	Preparing a map in the field directly
Theodolite	Measuring horizontal and vertical angles
Total station	Electronic measurement of angles and distances together
GNSS / GPS receiver	Fixing position by satellite (latitude, longitude, elevation)

1.11 Obstacles in Chain Surveying

An obstacle is any feature that prevents direct measurement or ranging. The method of working depends on the type of obstacle:

- **Obstacle to ranging but not chaining** (e.g., a hill between two stations): use **reciprocal ranging** with ranging rods.
- **Obstacle to chaining but not ranging** (e.g., a pond or river): measure a perpendicular offset and use geometry, or take a tie line, and apply Pythagoras' theorem.
- **Obstacle to both chaining and ranging** (e.g., a building): take offsets and use a tie line or the method of parallels.

Distance across an obstacle (Pythagoras):
If a perpendicular offset d is taken from each end so that the points A and B are inter-visible:
$AB = \sqrt{(\text{base}^2 + (\text{difference of offsets})^2)}$

1.12 Errors in Chaining and Their Corrections

- **Positive errors (chain too short):** measured length is more than the true length.
- **Negative errors (chain too long):** measured length is less than the true length.

Correction for slope (horizontal distance):

Horizontal distance = $\sqrt{(\text{measured length}^2 - \text{height difference}^2)}$

Correction = measured − horizontal (always subtractive)

Correction for temperature:

$C_t = \alpha \cdot L \cdot (T_m - T_s)$

α = coefficient of thermal expansion, L = length, T_m = field temp., T_s = standard temp.

Correction for pull/tension:

$C_p = (P - P_s)\, L / (A\, E)$

1.13 Modern Surveying Methods

- **Total station:** combines an electronic theodolite and distance meter; records angles and distances electronically and computes coordinates; widely used for farm layout, levelling and earthwork.
- **GNSS (GPS):** gives position (latitude, longitude, elevation) by satellite; useful for large watersheds, tank boundaries and cadastral mapping.
- **Remote sensing and GIS:** satellite imagery and maps for land use, watershed delineation and planning.
- **Drones / UAV:** rapid aerial survey for topography and crop monitoring.

1.14 Plotting and Computing Area from the Survey

After the survey, the framework is plotted to scale, offsets are laid off, and the area is computed by dividing the plot into triangles and trapeziums, or by using the trapezoidal/ Simpson's rules (Lecture 3). Plotting to a large scale (e.g., 1:500 or 1:1000) gives better accuracy for small fields.

1.15 Conventional Signs and Units in Surveying

Table 1.3 — Some conventional signs used on survey plans

Feature	Conventional sign (idea)
Boundary (field)	Dashed or chain-dotted line
Contour	Continuous thin line with its value marked
Road	Two parallel lines with cross ticks
Stream / canal	Wavy / straight double line with flow arrow
Well	Small circle with a cross
Tree	Small circle with a dot
Building	Solid outline, hatched
Bench mark	Triangle with a dot, RL written alongside

Units of linear and area measurement

Table 1.4 — Common units in land measurement

Unit	Equivalent

1 metre (m)	100 cm = 1000 mm = 3.281 ft
1 kilometre (km)	1000 m
1 hectare (ha)	10,000 m² = 2.471 acres
1 acre	4046.86 m² = 0.4047 ha
1 cent (TN land unit)	40.47 m²; 100 cents = 1 acre
1 ground (TN)	222.97 m² (about 2400 sq ft)

Tamil Nadu revenue records often use **cents, grounds and acres**; engineering designs use hectares and m². Convert carefully between them.

1.16 Measurement of a Line — Ranging Procedure

1. Fix the two end stations with ranging rods.
2. Stand a few metres behind one station and sight the other rod.
3. Direct the assistant to move the intermediate rod left or right until it is exactly in line.
4. Mark the point with a peg and repeat at the required interval.
5. Measure the line with the chain along the ranged points, keeping the chain straight and horizontal.

Exam Focus — Lecture 1

Key points

- Surveying = fixing relative positions of points on/above/below the earth's surface.
- Chain survey uses only linear measurement; principle is the triangle (three sides fix a figure).
- Metric chain: 20 m, 100 links, link = 0.2 m.
- Base line = longest, most carefully measured line; check line verifies accuracy.
- Cross staff sets out 90° for perpendicular offsets.
- Offset = lateral distance of a detail from the chain line.
- RF = plan distance / ground distance.
- Chain correction: True length = Measured × (L'/L).

Short-answer questions

Q1. Define surveying and state its main uses in agriculture.
Model answer: Surveying is the art and science of determining the relative positions of points on, above or below the earth's surface and representing them on a plan to scale. Uses: farm layout, land levelling for irrigation, contour mapping, canal/drain alignment, siting conservation structures, area computation, volume of earthwork, and setting out roads and pipelines.

Q2. What is a cross staff? Name its types.
Model answer: A cross staff is a simple instrument for setting out right angles on a chain line to take perpendicular offsets. Types: open cross staff, French cross staff (45° and 90°), and adjustable cross staff.

Q3. Distinguish between a base line and a check line.
Model answer: A base line is the longest and most important main survey line, measured

most accurately, from which the whole framework is built. A check line joins two fixed points (often not measured directly) purely to verify that the plotted framework is correct.

Long-answer questions

L1. Describe chain surveying with its principle, instruments and field procedure. Draw the field book.
Outline: Definition → principle (triangle is rigid) → instruments (chain, arrows, ranging rods, cross staff, pegs, field book) → steps (reconnaissance, index sketch, base line, offsets, check lines, plotting) → field-book sketch → precautions.

L2. Explain the uses of surveying in agriculture with examples from Tamil Nadu.
Outline: Introduction to agricultural surveying → uses (levelling for the Cauvery delta canal irrigation, contour bunding in the TN uplands, tank irrigation layout, farm ponds under TN watershed schemes, drip layout in Coimbatore banana/vegetable plots) → conclusion.

2 Dumpy Level — Setting, Types of Levelling, Computation of Land Slope and Difference in Elevation

2.1 Levelling — Basic Idea

> **Levelling** is the operation of determining the difference in elevation between two or more points on the earth's surface, or of measuring heights of points relative to a common datum.

Datum is a permanently fixed reference surface from which elevations are measured (mean sea level is the common datum). **Reduced Level (RL)** of a point is its height above or below the datum.

Levelling is the single most important survey operation in soil and water conservation because bunds, channels, terraces, drains and ponds are all laid out on designed gradients.

2.2 The Dumpy Level and its Parts

> **Dumpy level** is a surveying instrument with a telescope rigidly fixed to the level tube (the telescope cannot be rotated in its bearings), used with a levelling staff to find differences of elevation.

Main parts

- **Telescope:** with eyepiece, object glass and a diaphragm carrying the cross hairs (stadia hairs + central horizontal and vertical hairs).
- **Level tube (bubble tube):** fixed parallel to the line of collimation; the bubble must be central.
- **Foot screws:** three (or four) screws used to level the instrument.
- **Levelling head:** the base with tribrach, foot screws and a clamp for the spindle.
- **Vertical spindle:** allows the telescope to rotate in a horizontal plane.
- **Tripod stand:** three legs supporting the instrument.
- **Levelling staff:** graduated rod (self-reading) held on the point whose level is required.

2.3 Setting up the Dumpy Level

1. **Fix the tripod:** spread the legs on firm ground, roughly levelled, tripod head about chest height.
2. **Mount the level** on the tripod and tighten the clamp.
3. **Rough levelling:** bring the foot screws to a central position; adjust the tripod legs to bring the bubble near the centre.
4. **Fine levelling (foot-screw levelling):** turn the foot screws to bring the bubble exactly to the centre of the tube. The rule is: *bubble moves in the direction of the left thumb*. With three screws, turn two together and the third separately.
5. **Focusing:** first focus the eyepiece to get sharp cross hairs, then focus the object glass to get a sharp image of the staff (eliminate parallax).
6. **Reading the staff:** read the central hair to 0.001 m by estimation.

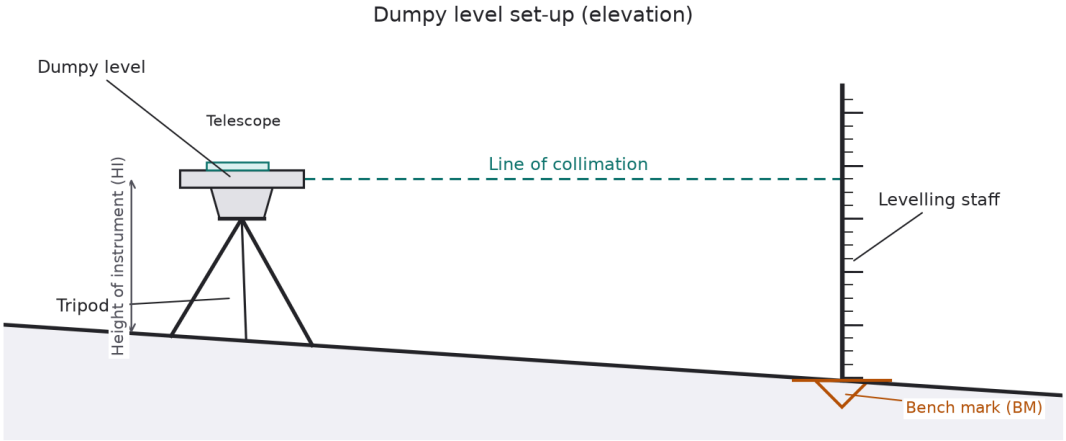


Figure 2.1 Dumpy level set-up (elevation)

Tripod, telescope, levelling staff, line of collimation, height of instrument and bench mark.

2.4 Text Diagram — Dumpy Level Set-up

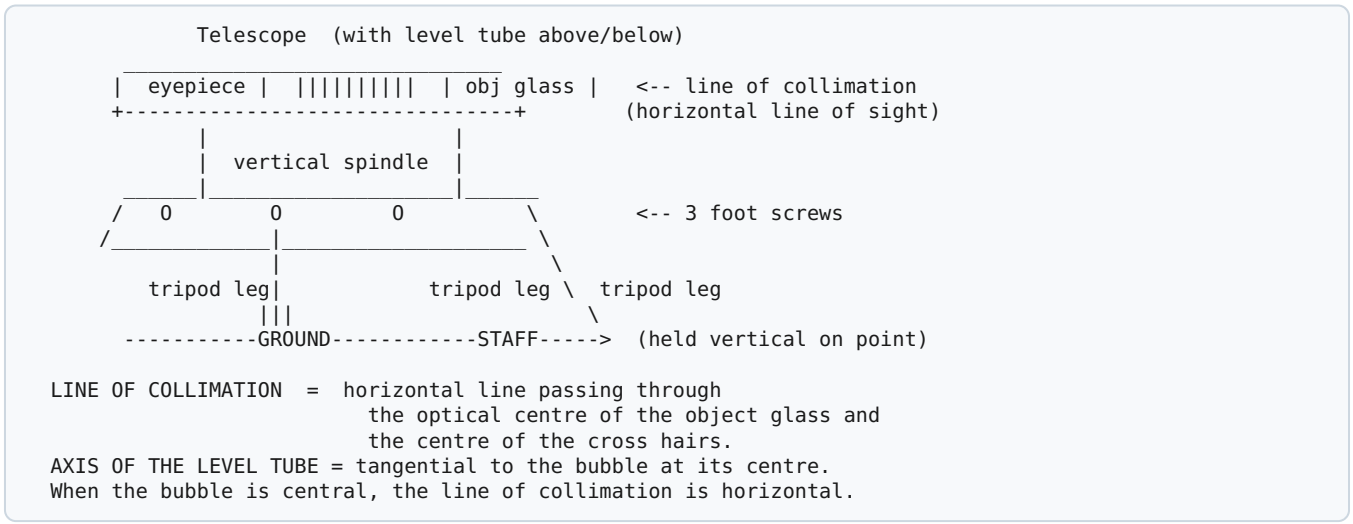


Fig. 2.1 — Dumpy level set-up and line of collimation

2.5 Levelling Staff and Readings

A self-reading staff is graduated in metres, decimetres and centimetres, with the decimetre figures marked. The reading is taken to three decimal places (last digit estimated).

- **Back sight (BS):** the first reading after setting the instrument; taken on a known-RL point (bench mark).
- **Fore sight (FS):** the last reading before shifting the instrument; taken on the point whose RL is required.
- **Intermediate sight (IS):** any reading taken between BS and FS on other points.
- **Change point (CP):** a point common to two instrument settings, on which both FS and the next BS are taken.

2.6 Types of Levelling

Table 2.1 — Types of levelling and their use

Type	Method	Typical use
Simple levelling	One instrument setting; only BS and FS.	Difference of level between two nearby points.
Differential levelling	Several settings with change points; series of BS/IS/FS.	Long profiles, longitudinal sections of canals/ roads.
Fly levelling	Rapid differential levelling without side details; only BS and FS.	Carrying levels to a distant bench mark; reconnaissance.
Profile / longitudinal levelling	Levels taken along a line at fixed intervals.	Alignment of channels, drains, roads, pipelines.
Cross-sectioning	Levels taken at right angles across the line.	Earthwork volume, canal/road cross-sections.
Reciprocal levelling	Two settings with instrument near each end in turn; mean taken.	Across a river/valley; eliminates curvature, refraction, collimation errors.
Precise levelling	Very accurate instruments and staves, short sights.	Bench-mark establishment, geodetic work.
Check levelling	Repeating levels at intervals.	Verifying previous levels.

Barometric levelling	From change in atmospheric pressure with height.	Rough reconnaissance, hilly areas.
Trigonometric levelling	From vertical angle and horizontal distance (theodolite).	Inaccessible and hilly points.

2.7 Reduction of Levels

Reducing a level book means converting staff readings into RLs. Two standard methods are used.

(a) Height of Collimation (Height of Instrument) Method

Height of Collimation:

```
Height of Collimation (HC) = RL of Bench Mark + Back Sight
RL of a point = Height of Collimation − Staff reading on that point
(for a change point: new HC = RL of CP + BS on CP)
```

Best when many intermediate sights are taken from one setting.

(b) Rise and Fall Method

Rise and Fall:

```
If a reading is LESS than the previous reading → RISE = previous − present.
If a reading is MORE than the previous reading → FALL = present − previous.
RL of a point = RL of previous point ± Rise/Fall
Arithmetic check: ΣBS − ΣFS = ΣRise − ΣFall = Last RL − First RL
```

Best because every reading is checked, giving greater confidence.

2.8 Worked Numerical — Differential Levelling (Both Methods)

Worked Example 2.1 — RL by Height of Collimation and Rise & Fall

The following readings were taken with a dumpy level. The first reading was on a bench mark of RL 100.000 m. Reduce the levels by both methods and apply the arithmetic checks.

Station	BS (m)	IS (m)	FS (m)
A (BM)	1.585	—	—
B	—	1.250	—
C	—	1.640	—
D (CP)	1.300	—	2.105
E	—	1.475	—
F	—	—	1.955

(a) Height of Collimation Method

1st setting HC = RL of BM + BS = 100.000 + 1.585 = **101.585 m**

Station	BS	IS	FS	HC (m)	RL (m)
A (BM)	1.585	—	—	101.585	100.000
B	—	1.250	—	101.585	100.335
C	—	1.640	—	101.585	99.945
D (CP)	1.300	—	2.105	100.780	99.480
E	—	1.475	—	100.780	99.305
F	—	—	1.955	—	98.825

RL of D = 101.585 − 2.105 = 99.480 m. New HC = RL of D + BS = 99.480 + 1.300 = 100.780 m. RL of F = 100.780 − 1.955 = 98.825 m.

Check: ΣBS − ΣFS = (1.585 + 1.300) − (2.105 + 1.955) = 2.885 − 4.060 = −1.175 m. Last RL − First RL = 98.825 − 100.000 = −1.175 m. ✓ The check is satisfied.

(b) Rise and Fall Method

Station	BS	IS	FS	Rise	Fall	RL (m)
A (BM)	1.585	—	—	—	—	100.000
B	—	1.250	—	0.335	—	100.335
C	—	1.640	—	—	0.390	99.945
D (CP)	1.300	—	2.105	—	0.465	99.480
E	—	1.475	—	0.825	—	99.305
F	—	—	1.955	—	0.480	98.825

A→B: 1.585 − 1.250 = +0.335 rise. B→C: 1.250 − 1.640 = −0.390 fall. C→D: 1.640 − 2.105 = −0.465 fall. D→E: 2.105 − 1.475 = +0.825 rise. E→F: 1.475 − 1.955 = −0.480 fall.

Check: ΣBS − ΣFS = −1.175 m. ΣRise − ΣFall = (0.335 + 0.825) − (0.390 + 0.465 + 0.480) = 1.160 − 1.335 = −1.175 m. ✓

Answer: RL of B = 100.335, C = 99.945, D = 99.480, E = 99.305, F = 98.825 m. The ground falls from A to F by 1.175 m overall.

2.9 Computation of Difference in Elevation

Simple levelling:

```
Difference in elevation between two points
= (Back Sight on the higher/first point) − (Fore Sight on the second point)
Δh = BS − FS   (positive = second point lower than first)
```

Worked Example 2.2 — Simple difference of level

With the instrument set between two points P and Q, the BS on P is 1.420 m and the FS on Q is 0.980 m. Find the difference in elevation and state which point is higher.

Step 1: Δh = BS − FS = 1.420 − 0.980 = +0.440 m.

Step 2: A positive value means the FS point is lower. Hence Q is lower than P.

Answer: Difference in elevation = 0.440 m; P is higher than Q by 0.440 m.

2.10 Computation of Land Slope

Land slope is the inclination of the ground surface, expressed as a percentage, a ratio (e.g., 1 in 100), or an angle in degrees.

Slope formulas:

```
Slope (%) = (Difference in elevation / Horizontal distance) × 100
Slope (ratio) = Horizontal distance : Difference in elevation = 1 : (HD / Δh)
Slope (degrees) = tan⁻¹ (Δh / HD)
Also: Slope (%) = tan(θ) × 100
```

For soil conservation design, slope classes used in India are broadly: nearly level (0–1%), very gently sloping (1–3%), gently sloping (3–5%), moderately sloping (5–10%), strongly sloping (10–15%), moderately steep (15–30%), steep (30–60%) and very steep (>60%).

Worked Example 2.3 — Land slope in per cent, ratio and degrees

Two points on a field are 40 m apart horizontally. The difference in elevation between them is 2.4 m. Find the land slope in per cent, as a ratio and in degrees.

Step 1 (per cent): Slope = (2.4 / 40) × 100 = **6 %**.

Step 2 (ratio): 1 : (40 / 2.4) = 1 : 16.7, i.e. **1 in 16.7**.

Step 3 (degrees): $\theta = \tan^{-1}(2.4/40) = \tan^{-1}(0.06) = \mathbf{3.43°}$.

A 6% slope falls in the "moderately sloping" class; contour bunding is usually recommended on such land.

Worked Example 2.4 — Horizontal distance from slope

A field has a uniform slope of 4%. The difference in elevation between the top and bottom is 1.8 m. Find the horizontal distance and the sloping distance.

Step 1: HD = (Δh / slope%) × 100 = (1.8 / 4) × 100 = **45 m**.

Step 2: Sloping distance = $\sqrt{(HD^2 + \Delta h^2)} = \sqrt{(45^2 + 1.8^2)} = \sqrt{(2025 + 3.24)} = \mathbf{45.04\ m}$.

For gentle slopes the sloping distance is almost equal to the horizontal distance.

2.11 Levelling in Soil Conservation Practice

- **Contour bunding:** the level contour line is traced with the level and the bund is built along it.
- **Land levelling/grading:** existing and designed levels are compared to compute cut and fill.
- **Canal and drain alignment:** longitudinal levels fix the bed slope (normally 1 in 1000 to 1 in 5000 for field channels).
- **Farm ponds and tanks:** depth and storage computed from levels.
- **Terraces:** vertical interval between terraces derived from the slope.

2.12 Bench Marks

Bench mark (BM) is a permanently fixed point of known reduced level, used as a reference for levelling.

- **GTS bench mark:** established by the Survey of India; the most reliable reference.
- **Permanent bench mark:** fixed on a durable structure (plinth, culvert) with a cut mark.
- **Temporary bench mark:** a point of known level used for one job and then discarded.
- **Arbitrary bench mark:** a convenient point whose RL is assumed (e.g., 100.000 m) for local work.

2.13 Curvature and Refraction

- **Curvature correction:** the earth's surface curves away from the horizontal line of sight, so a staff appears higher than it is: $C_c = 0.0673\ d^2$ (d in km, C_c in m). Always subtractive.
- **Refraction correction:** the line of sight bends downward due to atmospheric refraction: $C_c = -0.0112\ d^2$. The combined correction is about $0.0673\ d^2 - 0.0112\ d^2 = 0.0561\ d^2$.
- These corrections are negligible for short sights but matter for long sights (hundreds of metres) and in **reciprocal levelling**.

2.14 Reciprocal Levelling

Reciprocal levelling is the method of determining the true difference of level between two distant points by taking levels with the instrument set up close to each end in turn; it eliminates errors of curvature, refraction and collimation.

Reciprocal levelling (true difference of level):

With instrument near A: $\Delta h_1 = BS_1 - FS_1$ (FS_1 on B)

With instrument near B: $\Delta h_2 = BS_2 - FS_2$ (FS_2 on A)

True difference of level $h = (\Delta h_1 + \Delta h_2) / 2$

Worked Example 2.5 — Reciprocal levelling across a river

In levelling across a river: instrument near A — BS on A = 1.200 m, FS on B = 1.700 m. Instrument near B — BS on B = 1.500 m, FS on A = 1.000 m. Find the true difference in elevation and the collimation error.

Step 1: $\Delta h_1 = 1.200 - 1.700 = -0.500$ m.

Step 2: $\Delta h_2 = 1.500 - 1.000 = +0.500$ m.

Step 3 (true difference): h = (−0.500 + 0.500)/2 = **0.000 m** (the points are at the same level).

The discrepancy between −0.5 and +0.5 indicates the combined error; the true difference is their mean.

2.15 Text Diagram — Levelling Staff Readings

```
LEVELLING STAFF (self-reading, front face)
+--------+   E-pattern / block pattern graduations
|  1.0   |   figures in decimetres
|  .1    |
|  .2    |   <-- black on white and red on white blocks
|  .3    |
|  1.5   |   the metre and decimetre figures marked
|  .6    |
|  .7    |
+--------+

READING THE STAFF (through the level):
  -------- upper stadia hair  ------  1.585  (top hair)
  ~~~~~~~~ central hair  ~~~~~~~~~~  1.500  (actual reading taken)
  -------- lower stadia hair  ------  1.415  (bottom hair)
The CENTRAL hair reading is the staff reading; to 3 decimal places.

SIGN CONVENTION:
  BS on BM (known RL)  -->  HC = RL + BS
  IS / FS on any point -->  RL = HC - reading
```

Fig. 2.2 — Levelling staff graduation and how a reading is taken

2.16 Difficulties in Levelling and Sources of Error

- **Instrument out of adjustment:** the line of collimation is not truly horizontal (collimation error) — check by the two-peg test.
- **Parallax:** relative movement of the image and cross hairs — focus properly.
- **Staff not held vertical:** gives a reading that is too large (foresight/backsight both affected); use a staff bubble.
- **Bubble not central:** line of collimation not horizontal.
- **Long sights:** curvature and refraction effects.
- **Settlement of the tripod** during reading.
- **Reading and booking mistakes:** guard against by the arithmetic check.

Two-peg test (collimation check):
Set the level midway between two pegs A and B (say 30 m apart) and find the true difference. Then set the level close to A and read both staves. If the apparent difference differs from the true difference, the instrument has a collimation error and must be adjusted.

2.17 Precautions in Levelling

- Keep the backsight and foresight distances roughly equal (this cancels curvature, refraction and collimation errors).
- Always level the instrument accurately before reading.
- Hold the staff vertical and on firm ground.
- Take the reading to three decimal places and book it immediately.
- Apply the arithmetic check before leaving the field.

Exam Focus — Lecture 2

Key points

- Levelling finds differences of elevation; RL is height above datum (usually MSL).
- Dumpy level: telescope rigidly fixed to the level tube.
- Bubble rule: bubble moves in the direction of the left thumb.
- Line of collimation = horizontal line through object glass centre and cross hairs.
- BS = first reading (known RL); FS = last reading; CP = common point of two settings.
- Two reduction methods: Height of Collimation and Rise & Fall.
- Check: ΣBS − ΣFS = ΣRise − ΣFall = Last RL − First RL.
- Slope (%) = (Δh / HD) × 100.

Short-answer questions

Q1. Define levelling, datum and reduced level.
Model answer: Levelling is the operation of finding the difference of elevation between points. Datum is the fixed reference surface from which heights are measured. Reduced level is the height of a point above or below the datum.

Q2. What is a change point and why is it needed?
Model answer: A change point is a point on which both a fore sight and the next back sight are taken, common to two successive instrument settings. It is needed when the instrument is shifted, to carry the level forward without losing continuity.

Q3. State the two methods of reducing levels and their arithmetic check.
Model answer: (i) Height of Collimation method — HC = RL + BS; RL = HC − reading. (ii) Rise and Fall method. The common check is ΣBS − ΣFS = ΣRise − ΣFall = Last RL − First RL.

Long-answer questions

L1. Describe the dumpy level, its parts and the procedure of setting it up. Explain the two methods of reducing levels with a numerical example.
Outline: Parts diagram → setting up steps (tripod, rough levelling, foot-screw levelling, focusing) → HC method with table → Rise and Fall method with table → arithmetic checks → comparison.

L2. Explain how levelling is used to compute land slope and to lay out contour bunds in a Tamil Nadu upland field.
Outline: Slope formula → measuring Δh by differential levelling → classifying slope → tracing contour with the level → spacing of bunds → worked slope example.

3 Computation of Area and Volume — Simpson's Rule and Trapezoidal Rule

3.1 Why Area and Volume Computation?

In soil and water conservation the engineer repeatedly needs the **area** of a field or catchment and the **volume** of earth or water. Examples: area of a field for input calculation, area of a catchment for runoff, volume of a bund or terrace (earthwork), volume of a farm pond (storage), and volume of soil lost to erosion.

Areas are found by (a) dividing the land into regular geometrical figures, (b) measuring offsets from a base line and using the trapezoidal or Simpson's rule, or (c) planimeter / digital methods. Volumes are found from cross-sectional areas (trapezoidal or prismoidal rule) or from contour maps.

3.2 Area from Offsets — Mid-ordinate, Trapezoidal and Simpson's Rules

Offset (ordinate) is the perpendicular distance measured from the chain (base) line to the boundary. When offsets are taken at equal intervals, the area between the base line and the boundary can be computed by approximate rules.

(a) Mid-ordinate Rule

Mid-ordinate rule:

Area = d × (Sum of mid-ordinates)

where d = common interval, mid-ordinates are measured midway between the ordinates.

(b) Average Ordinate Rule

Average ordinate rule:

Area = [$(O_0 + O_1 + O_2 + ... + O_n)$ / (n+1)] × L

where L = total length of base line = n × d, and $O_0 ... O_n$ are (n+1) ordinates.

(c) Trapezoidal Rule

Joins successive ordinates by straight lines (trapeziums). Works for any number of ordinates.

Trapezoidal rule:

$A = (d/2) [O_0 + O_n + 2(O_1 + O_2 + ... + O_{n11})]$

where d = common interval, O_0 and O_n are the first and last ordinates.

(d) Simpson's Rule (one-third rule)

Joins successive ordinates by arcs of a parabola; more accurate than the trapezoidal rule but requires an **odd number of ordinates** (i.e., an even number of intervals).

Simpson's rule:

$A = (d/3) [(O_0 + O_n) + 4(O_1 + O_3 + ...) + 2(O_2 + O_4 + ...)]$

i.e., A = (d/3)[(first + last) + 4(sum of even-positioned ordinates) + 2(sum of odd-positioned interior ordinates)]

Conditions for Simpson's rule: (1) the ordinates must be equally spaced; (2) the number of ordinates must be odd (intervals even). If the number of ordinates is even, apply Simpson's rule to all but the last strip and add the last strip by the trapezoidal rule.

Table 3.1 — Comparison of area rules

Rule	Assumed shape between ordinates	Condition	Accuracy
Trapezoidal	Straight line (trapezium)	Any number of ordinates	Fair
Simpson's 1/3	Parabolic arc	Odd no. of ordinates	High (best for curves)
Mid-ordinate	Rectangle	Ordinates at mid-points	Rough
Average ordinate	Rectangle of mean height	Any number	Rough

3.3 Text Diagram — Offsets from a Base Line

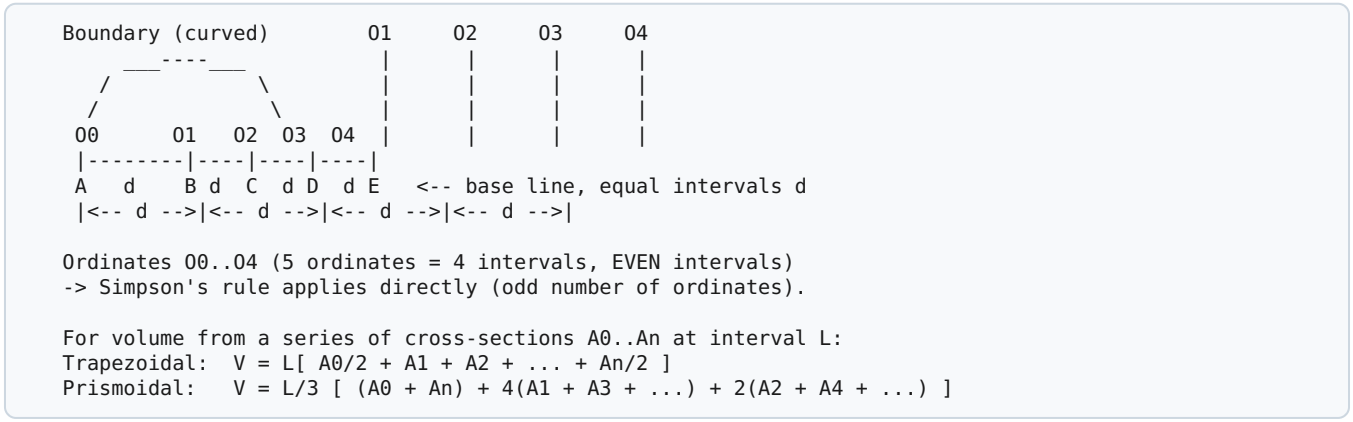


Fig. 3.1 — Offsets from a base line and volume from cross-sections

3.4 Worked Numerical — Area by Trapezoidal and Simpson's Rules

Worked Example 3.1 — Area of a strip of land

The following offsets were taken at intervals of 15 m from a chain line to a boundary: 0, 3.2, 5.6, 6.8, 5.4, 4.0, 0 m. Compute the area by (a) the Trapezoidal rule and (b) Simpson's rule.

Given: d = 15 m; O_0=0, O_1=3.2, O_2=5.6, O_3=6.8, O_4=5.4, O_5=4.0, O_6=0 m (7 ordinates = 6 intervals, even → Simpson's rule applies).

(a) Trapezoidal rule

$A = (d/2)[\ O_0 + O_6 + 2(O_1+O_2+O_3+O_4+O_5)\ ]$

Sum of interior = 3.2+5.6+6.8+5.4+4.0 = 25.0

$A = (15/2)[\ 0 + 0 + 2\times25.0\ ] = 7.5 \times 50.0 =$ **375.0 m²**

(b) Simpson's rule

$A = (d/3)[\ (O_0+O_6) + 4(O_1+O_3+O_5) + 2(O_2+O_4)\ ]$

Even-positioned (O_1,O_3,O_5) = 3.2+6.8+4.0 = 14.0; ×4 = 56.0

Odd-positioned interior (O_2,O_4) = 5.6+5.4 = 11.0; ×2 = 22.0

Ends = 0 + 0 = 0

$A = (15/3)[\ 0 + 56.0 + 22.0\ ] = 5 \times 78.0 =$ **390.0 m²**

Simpson's rule gives 390.0 m², slightly larger because it follows the curved boundary better. Answer: Trapezoidal 375 m², Simpson's 390 m².

Worked Example 3.2 — Simpson's rule with unequal last strip

Offsets at 10 m intervals are: 2.0, 4.0, 6.0, 5.0, 3.0, 1.0 m (6 ordinates = 5 intervals, odd intervals). Compute the area.

Step 1: 6 ordinates is an even number, so apply Simpson's rule to the first 5 ordinates (2.0, 4.0, 6.0, 5.0, 3.0) and add the last strip by the trapezoidal rule.

Simpson's part: A_1 = (10/3)[(2.0+3.0) + 4(4.0+5.0) + 2(6.0)] = (10/3)[5 + 36 + 12] = (10/3)(53) = 176.67 m²

Trapezoidal last strip (O_4=3.0 to O_5=1.0): A_2 = (10/2)(3.0+1.0) = 5 × 4.0 = 20.0 m²

Total area = 176.67 + 20.0 = **196.67 m²**

3.5 Volume Computation

Volume of earthwork is the quantity of soil to be cut or filled, obtained as the product of cross-sectional area and length (with suitable correction for varying sections).

Rules for volume from cross-sections

(a) Trapezoidal (average end area) rule:

$V = L\ [\ (A_0 + A_n)/2 + A_1 + A_2 + \ldots + A_{n11}\ ]$

(b) Prismoidal rule:

$V = (L/3)[(A_0 + A_n) + 4(A_1 + A_3 + ...) + 2(A_2 + A_4 + ...)]$
where $A_0 ... A_n$ are cross-sectional areas at interval L (requires odd number of sections).

Volume by contour method (for ponds and reservoirs)

Contour area method:
$V = (h/3)[A_0 + A_n + 4(A_1 + A_3 + ...) + 2(A_2 + A_4 + ...)]$ (Simpson's / prismoidal form)
where h = contour interval, A = water-spread area at each contour.

Worked Example 3.3 — Volume of earthwork (bund/channel)

Cross-sectional areas of a channel at 20 m intervals are 2.0, 3.5, 4.8, 3.0 and 1.5 m². Compute the volume of earthwork by the trapezoidal and prismoidal rules.

(a) Trapezoidal rule (5 sections, interval L = 20 m):

V = 20 [(2.0 + 1.5)/2 + 3.5 + 4.8 + 3.0] = 20 [1.75 + 11.3] = 20 × 13.05 = **261.0 m³**

(b) Prismoidal rule (odd number of sections = 5):

V = (20/3)[(2.0 + 1.5) + 4(3.5 + 3.0) + 2(4.8)] = (20/3)[3.5 + 26.0 + 9.6] = (20/3)(39.1) = **260.67 m³**

The two rules agree closely here. The prismoidal rule is more accurate; the trapezoidal rule slightly over-estimates.

Worked Example 3.4 — Volume of a farm pond from contour areas

A farm pond has water-spread areas of 0, 400, 900, 1200 and 1300 m² at depths of 0, 0.5, 1.0, 1.5 and 2.0 m from the bottom. Find the storage volume.

Given: h = 0.5 m; areas A_0=0, A_1=400, A_2=900, A_3=1200, A_4=1300 m² (5 areas, even intervals).

Prismoidal (Simpson's) rule:

V = (0.5/3)[(0 + 1300) + 4(400 + 1200) + 2(900)] = 0.1667 × [1300 + 6400 + 1800] = 0.1667 × 9500 = **1583.3 m³**

For comparison, the trapezoidal rule gives V = 0.5[0/2 + 400 + 900 + 1200 + 1300/2] = 0.5[0 + 400 + 900 + 1200 + 650] = 1575 m³. The pond stores about 1580 m³ (1.58 million litres).

3.6 Area by Geometrical Figures

- **Triangle:** A = ½ × base × height, or A = √[s(s−a)(s−b)(s−c)] where s = (a+b+c)/2.
- **Rectangle:** A = length × breadth.
- **Trapezium:** A = ½(sum of parallel sides) × height.
- **Circle:** A = πr² = πd²/4.
- **Land units:** 1 ha = 10,000 m² = 2.471 acres; 1 acre = 4046.86 m²; 1 hectare = 100 ares.

3.7 Planimeter and Area of Irregular Fields

Planimeter is an instrument for measuring the area of a figure drawn on a plan by tracing its boundary with a pointer.

- **Working:** the area is proportional to the number of revolutions of the tracing wheel: Area = M × (final reading − initial reading), where M is the multiplier.
- Used to measure the area of irregular fields, tank water-spread and catchment boundaries from a map.
- For accuracy, the boundary is traced twice (both clockwise and anticlockwise) and the mean taken.

Area from coordinates (shoelace / coordinate method)

Coordinate (shoelace) method:

$2A = |\Sigma\ (x_i\ y_{i+1} - x_{i+1}\ y_i)|$

where (x_i, y_i) are the coordinates of the boundary points, taken in order.

3.8 Worked Numerical — Area from coordinates

Worked Example 3.5 — Area of a quadrilateral by coordinates

The corners of a field are A(0,0), B(40,0), C(50,30), D(10,40) (metres). Find the area.

Step 1: Apply $2A = \Sigma(x_i y_{i+1} - x_{i+1} y_i)$, closing back to A.

Step 2: A→B: 0×0 − 40×0 = 0; B→C: 40×30 − 50×0 = 1200; C→D: 50×40 − 10×30 = 2000−300 = 1700; D→A: 10×0 − 0×40 = 0.

Step 3: 2A = 0 + 1200 + 1700 + 0 = 2900; A = 1450 m².

Answer: **1450 m²** = 0.145 ha.

3.9 Worked Numerical — Volume of a tank/reservoir by contour areas

Worked Example 3.6 — Capacity of a tank

Water-spread areas of a tank at successive 1 m contour intervals are 0, 1500, 3500, 6000, 8000 and 9000 m² (6 areas, at depths 0 to 5 m). Find the storage volume.

Given: h = 1 m; 6 areas (even number), so apply Simpson's rule to the first 5 and the trapezoidal rule to the last strip.

Simpson's (A_0..A_4 = 0,1500,3500,6000,8000): V_1 = (1/3)[(0+8000) + 4(1500+6000) + 2(3500)] = (1/3)[8000 + 30000 + 7000] = (1/3)(45000) = 15000 m³.

Trapezoidal last strip (8000 to 9000): V_2 = (1/2)(8000 + 9000) = 8500 m³.

Total: V = 15000 + 8500 = **23,500 m³**.

This is the live storage of the tank, used to plan the ayacut (command area) under tank irrigation.

3.10 Earthwork for Land Levelling

Land levelling requires cutting high spots and filling low spots. The cut-and-fill volume is computed from a grid of levels:

Grid method (mean area / borrow pit):
For a grid of squares of side a with corner levels, the volume = a^2 × (sum of corner depths weighted by the number of squares meeting at each corner, /4).
Cut volume = Fill volume (for a balanced design, the earth is moved, not imported)

3.11 Worked Numerical — Nine-ordinate Simpson's Rule

Worked Example 3.7 — Area with nine ordinates

Offsets at 10 m intervals are 0, 2.0, 4.5, 7.0, 8.5, 7.5, 5.0, 2.5, 0 m. Find the area by Simpson's rule.

Given: d = 10 m; 9 ordinates (odd number) → Simpson's rule applies directly.

Even-positioned (O_1,O_3,O_5,O_7): 2.0 + 7.0 + 7.5 + 2.5 = 19.0; ×4 = 76.0.

Odd-positioned interior (O_2,O_4,O_6): 4.5 + 8.5 + 5.0 = 18.0; ×2 = 36.0.

Ends: 0 + 0 = 0.

Area: A = (10/3)[0 + 76.0 + 36.0] = (10/3)(112) = **373.3 m²**.

Trapezoidal rule would give A = (10/2)[0+0+2(2.0+4.5+7.0+8.5+7.5+5.0+2.5)] = 5 × 2 × 37.0 = 370 m². The two differ by only about 1%.

3.12 Prismoidal Correction

Prismoidal correction (for volume):
$C_p = (L/12)\,(D - d)(h_1 - h_2)$ (for level sections)
Volume (corrected) = Trapezoidal volume − Prismoidal correction
where D, d are the top and bottom widths and h_1, h_2 the heights at the two sections.

The prismoidal rule gives a volume smaller than the trapezoidal rule; the difference is the prismoidal correction, which is subtracted from the trapezoidal volume.

Exam Focus — Lecture 3

Key points

- Trapezoidal rule works for any number of ordinates; Simpson's needs an odd number.
- Trapezoidal: A = (d/2)[O_0+O_n+2(Σinterior)].
- Simpson's: A = (d/3)[(O_0+O_n)+4(Σeven-pos)+2(Σodd-pos-interior)].
- Volume, trapezoidal: V = L[(A_0+A_n)/2 + Σinterior].
- Volume, prismoidal: V = (L/3)[(A_0+A_n)+4Σodd+2Σeven].
- 1 ha = 10,000 m²; 1 acre = 4046.86 m².

Short-answer questions

Q1. State the Simpson's rule for area and its conditions.
Model answer: A = (d/3)[(O_0+O_n) + 4(sum of even-positioned ordinates) + 2(sum of odd-positioned interior ordinates)]. Conditions: ordinates equally spaced and an odd number of ordinates (even number of intervals).

Q2. Why is the prismoidal rule more accurate than the trapezoidal rule?
Model answer: The trapezoidal rule assumes straight-line (linear) variation between

sections, while the prismoidal rule assumes a parabolic (curved) variation, which matches natural ground surfaces better; hence it is more accurate.

Q3. Convert 2.5 ha to acres and m².
Model answer: 2.5 ha = 2.5 × 10,000 = 25,000 m² = 2.5 × 2.471 = 6.18 acres.

Long-answer questions

L1. Describe the methods of computing the area of an irregular field from offsets, and work one numerical by both the trapezoidal and Simpson's rules.
Outline: Offsets and base line → rules (mid-ordinate, average ordinate, trapezoidal, Simpson's) → conditions → fully worked numerical → comparison of accuracy.

L2. Explain how the volume of earthwork for a bund/channel and the storage volume of a farm pond are computed.
Outline: Cross-section method → trapezoidal and prismoidal volume rules → contour-area method for ponds → worked examples → practical relevance to farm ponds in TN watershed schemes.

4 Soil Erosion — Causes, Evil Effects, Geologic and Accelerated Erosion

4.1 What is Soil Erosion?

> **Soil erosion** is the detachment and transportation of soil particles from their place of origin by an agent of erosion (water, wind, ice or gravity) and their deposition elsewhere.

Erosion has three stages: **detachment** of soil particles, **transportation** of the detached particles, and **deposition** at a new location. Soil erosion is a natural geological process, but it becomes a serious problem when human activity accelerates it far beyond the rate at which soil forms.

Soil formation is extremely slow — roughly 1 cm of topsoil may take hundreds of years to form — whereas erosion can remove the same depth in a single season. This imbalance is the root of the problem.

4.2 Causes of Soil Erosion

The causes are conveniently grouped into two broad classes: **natural (geological) causes** and **human-induced causes**. Both are powered by the two main erosive agents — **water** and **wind** — with gravity (mass movement) as a third.

(a) Natural causes

- **Rainfall:** raindrop impact detaches soil; runoff transports it. High-intensity storms are the most destructive.
- **Steep slope:** greater slope and longer slope length increase the velocity and erosive power of runoff.
- **Erodible soil:** fine sandy and silty soils with weak structure and low organic matter erode easily.
- **Sparse vegetation:** lack of plant cover exposes soil to raindrop impact and wind.
- **Wind:** strong dry winds remove loose, fine, bare soil.
- **Intense storms and high temperature:** cause heavy runoff and dry the soil surface.
- **Mass movement:** landslides and slips on steep, saturated slopes.

(b) Human-induced (anthropogenic) causes

- **Deforestation** and removal of natural vegetation.
- **Overgrazing** by livestock, which destroys the protective grass cover.
- **Faulty cultivation practices:** ploughing up and down the slope, clean cultivation, excessive tillage.
- **Shifting cultivation (jhum / podu)** on hill slopes.
- **Improper land use** and cultivation of steep, marginal lands.
- **Faulty construction of roads, canals and buildings** without drainage control.
- **Forest fires** and mining/quarrying activities.

4.3 Evil Effects of Soil Erosion

On-site (on the eroding field)

- **Loss of fertile topsoil** rich in organic matter, nitrogen and available nutrients.
- **Reduced soil depth** and reduced water-holding capacity.

- **Loss of soil structure** and decline in infiltration.
- **Lower crop yields** and reduced productivity.
- **Formation of rills, gullies and badlands** making land unfit for cultivation.
- **Removal of seeds and young seedlings** and exposure of plant roots.
- **Reduced effectiveness of fertilizers** as applied nutrients are washed away.

Off-site (downstream and elsewhere)

- **Siltation of reservoirs, tanks and ponds**, reducing their storage capacity and life.
- **Siltation and blocking of canals, drains and river channels**, causing floods.
- **Water pollution** from sediment, nutrients and agro-chemicals carried with eroded soil.
- **Deposition of infertile sand** on fertile downstream lands (sand casting).
- **Damage to roads, bridges and culverts** and higher maintenance cost.
- **Eutrophication** of lakes due to nutrient-laden sediment.
- **Reduced ground water recharge** as infiltration falls and runoff rises.

Economic effect: India loses a large quantity of topsoil (commonly quoted as roughly 5 billion tonnes of soil annually in national estimates) and a substantial amount of plant nutrients each year, translating into huge crop-production losses and costly reservoir siltation.

4.4 Geologic (Natural) Erosion

Geologic erosion (also called normal or natural erosion) is the slow, continuous wearing away of the land surface by natural agents under a protective vegetative cover, balancing roughly against soil formation.

It is a constructive as well as destructive process: it carves valleys and plains and forms fertile deltas (for example, the Cauvery delta soils of Tamil Nadu were built by geologic erosion and deposition over ages). Geologic erosion proceeds at a rate equal to or less than the rate of soil formation, so the soil resource is maintained.

4.5 Accelerated Erosion

Accelerated erosion is the rapid removal of soil by erosion agents at a rate much faster than the natural rate of soil formation, mainly due to human interference and loss of vegetation.

Accelerated erosion is the form that concerns the conservation engineer. It far exceeds soil formation and results in net loss of the resource. It includes sheet, rill, gully, stream-bank and wind erosion as well as mass movement.

4.6 Geologic vs Accelerated Erosion

Table 4.1 — Comparison of geologic and accelerated erosion

Point	Geologic erosion	Accelerated erosion
Also called	Normal / natural erosion	Man-induced erosion
Rate	Very slow; ≈ rate of soil formation	Very fast; far exceeds soil formation

Cause	Natural agents under vegetation cover	Human activity: deforestation, overgrazing, faulty cultivation
Vegetative cover	Present / protective	Removed or reduced
Effect on soil resource	Balanced — soil resource maintained	Net loss of topsoil and productivity
Nature	Constructive and destructive	Mainly destructive
Control needed	Not normally needed	Essential — the subject of conservation

4.7 Forms / Types of Erosion (Overview)

Table 4.2 — Types of erosion with their agents

Agent	Type	Brief description
Water	Splash erosion	Detachment by raindrop impact
	Sheet erosion	Uniform removal of a thin layer of soil by sheet flow
	Rill erosion	Small, shallow channels (rills) formed by concentrated flow
	Gully erosion	Deep, wide channels too large to be crossed by ploughing
Wind	Surface creep	Coarse particles rolled along the surface
	Saltation	Particles bouncing along in short hops
	Suspension	Fine particles carried high in the air
Gravity	Landslide / slip / mass movement	Down-slope movement of saturated soil mass

4.8 Text Diagram — Erosion Process and Slope Positions

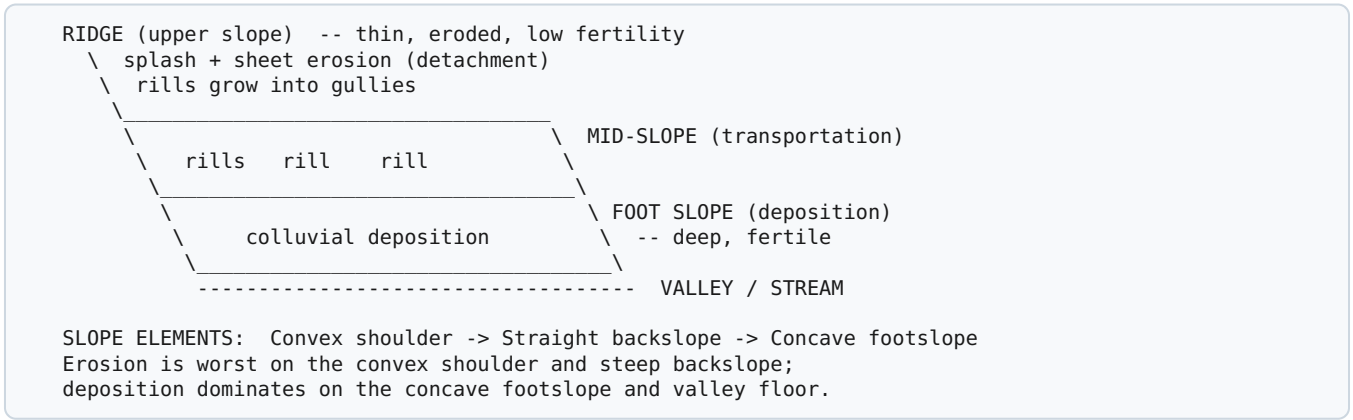


Fig. 4.1 — Slope-position pattern of erosion and deposition

4.9 Erosion and Soil Formation — The Balance

Soil erosion balance:

Net change in soil depth = Rate of soil formation − Rate of soil erosion

If erosion ≤ formation → stable soil (geologic erosion).

If erosion > formation → net loss (accelerated erosion).

Tolerable soil loss (T-value) is the maximum rate of erosion that can be permitted without causing long-term decline in productivity; it is commonly taken as 4 to 11 t/ha/yr depending on soil depth, and about 5 t/ha/yr is a widely used working figure for medium-deep Indian soils.

Worked Example 4.1 — Loss of soil depth over a field

A 5 ha field loses soil at the rate of 20 tonnes/ha/yr. The soil bulk density is 1.35 g/cm³. Find the depth of soil lost per year and the total soil loss.

Step 1 (total soil loss): 20 × 5 = 100 t/yr = 100,000 kg/yr.

Step 2 (depth): Depth = mass / (area × bulk density). Area = 5 ha = 50,000 m²; bulk density = 1350 kg/m³.

Depth = 100,000 / (50,000 × 1350) = 100,000 / 6.75×10^7 = 0.00148 m = **1.48 mm/yr**.

This is about 15 times the tolerable limit — hence the field needs immediate conservation treatment.

4.10 Factors Affecting Soil Erosion

The rate of soil erosion depends on the interaction of four groups of factors:

Table 4.3 — Factors affecting soil erosion

Group	Factors	Effect
Climate	Rainfall amount & intensity, wind, temperature	High-intensity rain and dry winds increase erosion
Soil	Texture, structure, organic matter, permeability, depth	Fine sandy/silty, weakly structured soils erode most
Topography	Slope steepness, slope length, shape, drainage	Steeper and longer slopes erode more
Vegetation / land use	Cover, crop type, tillage, grazing	Dense cover protects; bare land erodes

4.11 Soil Loss Tolerance and Erosion Classes

Erosion is often classified by the annual soil loss and the stage of degradation:

Table 4.4 — Broad erosion classes

Class	Annual soil loss (t/ha/yr)	State of land
Slight	< 5	Within tolerable limit; little damage
Moderate	5–10	Topsoil loss begins; yield decline
Severe	10–50	Marked topsoil loss, rills and gullies
Very severe	> 50	Land largely degraded, unproductive

Values are indicative; the tolerable limit depends on soil depth and formation rate (commonly 4–11 t/ha/yr, with about 5 t/ha/yr used for medium-deep Indian soils).

4.12 Erosion and Land Degradation in India

- Water erosion affects the largest area of degraded land in India, followed by wind erosion.
- Gully and ravine erosion, water-logging and salinity are major problems in specific regions.
- In Tamil Nadu, erosion is significant in the uplands of Coimbatore, Erode, Dharmapuri and Krishnagiri, in the hill areas of the Nilgiris, and in the coastal sandy belts.
- Conservation programmes (watershed development, contour bunding, farm ponds) aim to bring erosion within the tolerable limit.

4.13 The Erosion Process in Detail

Erosion by water proceeds through three linked sub-processes:

- **Detachment:** soil particles are separated from the mass by raindrop impact or by the shear of flowing water.
- **Transportation:** the detached particles are carried by splash, sheet flow or concentrated flow; the ability to transport depends on the velocity and turbulence of the flow.
- **Deposition:** when the velocity falls, the coarser particles settle first; fine particles are carried farther. Deposition builds up fertile foot-slope soils but silts up reservoirs.

4.14 Principles of Soil Erosion Control

- **Reduce raindrop impact:** maintain a cover (vegetation, mulch) over the soil.
- **Increase infiltration:** keep the soil open and permeable so that rain enters the soil instead of running off.
- **Reduce runoff velocity:** shorten the slope length and lay out structures across the slope.
- **Dispose of surplus runoff safely:** lead excess water to a protected outlet (grassed waterway, drain).
- **Increase soil resistance:** add organic matter and improve structure.

Every conservation measure — biological or mechanical — works through one or more of these five principles.

Exam Focus — Lecture 4

Key points

- Soil erosion = detachment + transportation + deposition of soil by water, wind, ice or gravity.
- Main agents: water and wind; gravity causes mass movement.
- Causes: natural (rainfall, slope, erodible soil, sparse vegetation) and human (deforestation, overgrazing, faulty cultivation, shifting cultivation).
- Evil effects: on-site (topsoil and nutrient loss, lower yield, gullies) and off-site (siltation, floods, pollution).
- Geologic erosion is slow and balanced by soil formation; accelerated erosion is fast and man-induced.
- Accelerated erosion is the concern of the conservation engineer.

Short-answer questions

Q1. Define soil erosion and name its three stages.
Model answer: Soil erosion is the detachment and transportation of soil particles from their place of origin by an agent of erosion, with deposition elsewhere. Stages: detachment, transportation and deposition.

Q2. Differentiate geologic and accelerated erosion.
Model answer: Geologic (normal) erosion is slow, natural and balanced by soil formation under vegetation cover; accelerated erosion is rapid, mainly man-induced, exceeds soil formation, and causes net loss of topsoil and productivity.

Q3. List any five evil effects of soil erosion.
Model answer: Loss of fertile topsoil; reduced soil depth and water-holding capacity;

formation of rills and gullies; lower crop yields; siltation of reservoirs and canals causing floods and pollution.

Long-answer questions

L1. Explain the causes of soil erosion and its evil effects, giving examples from Tamil Nadu.
Outline: Definition → natural and human causes → on-site and off-site effects → TN examples (upland erosion in Coimbatore/Dharmapuri, siltation of tanks in the Cauvery basin, sand casting on delta lands) → need for conservation.

L2. Distinguish between geologic and accelerated erosion and explain why accelerated erosion is the focus of soil conservation.
Outline: Definitions → comparison table → soil formation vs erosion balance → tolerable soil loss → role of vegetation → conclusion.

5 Water Erosion — Causes, Erosivity and Erodibility, Stages, Ravines, Landslides, USLE

5.1 Water Erosion — Definition and Causes

Water erosion is the detachment and transportation of soil particles by the action of water, chiefly rainfall and the resulting surface runoff.

The two forces at work are the **impact of raindrops** (which detaches soil) and the **shear of flowing water** (which transports it). The main causes are:

- **Rainfall amount and intensity:** high-intensity storms produce more splash and runoff energy.
- **Runoff:** volume and velocity of surface flow determine transporting power.
- **Slope steepness and length:** steeper, longer slopes give faster, more erosive flow.
- **Soil properties:** texture, structure, organic matter and permeability govern erodibility.
- **Vegetative cover:** absence of cover leaves soil exposed to raindrop impact.
- **Land use and management:** faulty tillage, deforestation and overgrazing.

5.2 Erosivity and Erodibility

Erosivity (E or R) is the potential ability of rain and runoff to cause erosion; it is a property of the *erosive agent* (rainfall/runoff) and depends on rainfall amount, intensity and energy.

Erodibility (K) is the susceptibility of a soil to erosion; it is a property of the *soil* and depends on texture, structure, organic matter and permeability.

Table 5.1 — Erosivity vs Erodibility

Point	Erosivity	Erodibility
Property of	The erosive agent (rain/runoff)	The soil
Depends on	Rainfall amount, intensity, energy, runoff	Texture, structure, organic matter, permeability
Symbol (USLE)	R (rainfall-runoff erosivity factor)	K (soil erodibility factor)
Can be changed by	Not directly (climate)	Yes — by management (organic matter, cover)
Typical behaviour	High in monsoon and coastal TN	High for fine sandy and silty soils

5.3 Stages of Water Erosion

Water erosion proceeds in four successive stages of increasing severity:

(a) Splash Erosion

- Detachment of soil particles by the impact of falling raindrops.
- The raindrop breaks soil aggregates and throws particles up and outward (up to about 0.6 m high and 1.5 m away).
- It is the first and basic stage; a dense canopy or mulch reduces it sharply.
- Fine particles (silt and clay) are preferentially removed, leaving the surface coarse and sealed.

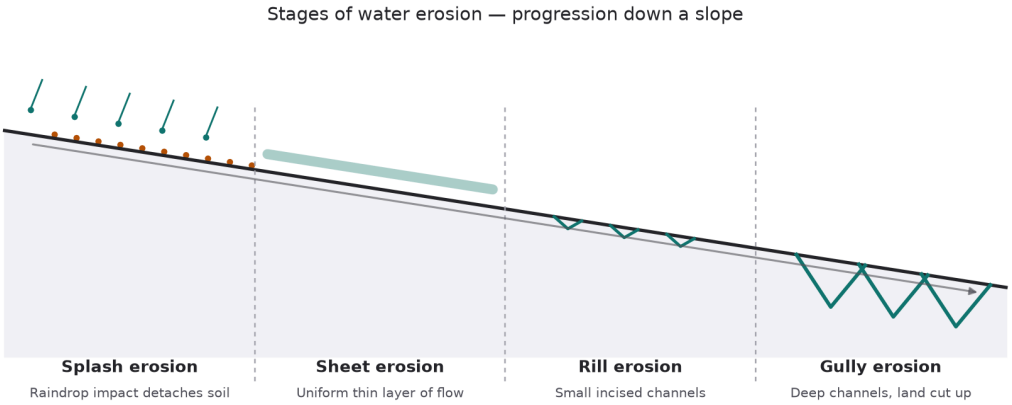


Figure 5.1 Stages of water erosion

Splash, sheet, rill and gully erosion as a progression down a slope.

(b) Sheet Erosion

- Uniform removal of a thin layer of soil from the entire slope by unchanneled sheet flow.
- Least obvious — often unnoticed until productivity falls or lighter-coloured subsoil appears.
- Most damaging to soil fertility because it removes the fertile topsoil evenly.
- Detected by exposed roots, stones on the surface and a lighter soil colour.

(c) Rill Erosion

- Concentrated flow forms small, well-defined channels called **rills** (a few centimetres deep).
- Rills are shallow enough to be removed by ordinary tillage; they are the warning sign of advancing erosion.
- Rills form a finger-like or dendritic pattern down the slope.

(d) Gully Erosion

- When rills deepen and widen beyond the reach of tillage, they become **gullies** (deeper than about 30 cm, uncrossable by a plough).
- Gully erosion is the most advanced and spectacular stage; it cuts into subsoil and even parent material.
- A gully advances headward (head-cut) and by side slumping; the eroded material is deposited as a fan at the outlet.

Classification of gullies: (i) by shape — V-shaped, U-shaped, trapezoidal; (ii) by state of development — active and stabilized; (iii) by size — small (<1 m deep), medium (1–3 m) and large (>3 m).

5.4 Text Diagram — Stages of Water Erosion

```
(a) SPLASH      (b) SHEET        (c) RILL         (d) GULLY
  |  |  |        ~~~flow~~~      \  |  /  |       \         /
  \/ \/ \/       ~~~~~~~~~~       \ | /  |         \_______/
 ~~~ ~~~ ~~~     ~~~~~~~~~~        \|/   |          \       /
 particles up    thin layer       small channels    deep, wide
 and outward     removed evenly   (tillage can fix) channel (needs
                                                      structures)

 PROGRESSION:  Splash -> Sheet -> Rill -> Gully  (increasing severity)
 Once gullies form, reclamation needs engineering structures (check dams).

 GULLY HEAD-CUT MIGRATION:
      head cut
         v
    _____|_____
   /            \____ advancing headward (upslope)
  /                  \________
 /_____________________________\___ outlet fan
```

Fig. 5.1 — Four stages of water erosion and gully head-cut migration

5.5 Ravines

Ravines are a network of deep, wide gullies with steep sides, developed on flat alluvial land along river banks, forming an intricate system of ridges and channels that make the land unfit for cultivation.

Ravines develop where flat, deep alluvial soil with an easily eroded substratum is subjected to heavy runoff and where vegetation has been removed. They are common in India along the Yamuna, Chambal and Betwa rivers (the Chambal ravines of Madhya Pradesh, Uttar Pradesh

and Rajasthan are the best-known example). In Tamil Nadu, ravine-like land occurs locally along some river banks and in the black-soil tracts. Ravine lands are usually classified as **deep ravine** (>3 m deep), **medium ravine** (1–3 m) and **shallow ravine** (<1 m).

Control of ravines: peripheral bunding to keep out runoff, gully plugging with brushwood and check dams, spurs and bank protection, and revegetation with grasses and trees.

5.6 Landslides

> **Landslide** is the rapid down-slope movement of a mass of soil and rock under the influence of gravity, usually triggered when the slope becomes saturated with water and loses shear strength.

Causes: steep slopes; heavy or prolonged rainfall and saturation; removal of vegetation; under-cutting of the toe by streams or road-cutting; earthquakes; and weak, sloping rock or soil strata.

Types (by movement): falls, topples, slides (rotational/translational), spreads and flows (debris flow, mudflow).

Control measures: avoiding cultivation and construction on steep unstable slopes; contour trenching and terracing; retaining walls and breast walls; sub-surface drainage to reduce pore pressure; afforestation with deep-rooted species; and bio-engineering (vetiver, bamboo) for slope stabilisation. Landslides are a serious hazard in the Nilgiris and other hill districts of Tamil Nadu.

5.7 Universal Soil Loss Equation (USLE)

> **Universal Soil Loss Equation (USLE)** is an empirical model that estimates the average annual soil loss from sheet and rill erosion on a given field as the product of six factors.

USLE:

```
A = R × K × LS × C × P
A = average annual soil loss (t/ha/yr)
R = rainfall-runoff erosivity factor
K = soil erodibility factor
L = slope-length factor (dimensionless)
S = slope-steepness factor (dimensionless)
C = cover-management factor (dimensionless)
P = support-practice (conservation) factor (dimensionless)
(L and S are usually combined as LS.)
```

Table 5.2 — USLE factors and typical values

Factor	Meaning	Typical / range of values
R	Rainfall-runoff erosivity	~20–100 (varies widely); high in monsoon India; TN coastal/upland values are moderate to high
K	Soil erodibility	~0.01–0.06 t·h/(MJ·mm); sandy/silty soils higher, clay lower
L	Slope length	Ratio to standard 22.1 m plot; 1.0 at 22.1 m, increases with length
S	Slope steepness	Ratio to standard 9% slope; 1.0 at 9%, rises steeply with slope
C	Cover management	Bare soil ~1.0; good pasture ~0.01–0.1; dense forest ~0.001–0.01

P	Support practice	Up-down slope = 1.0; contouring ~0.5; strip cropping ~0.25–0.5; terracing ~0.1–0.3

Note: The USLE was revised as RUSLE (Revised USLE). The MUSLE (Modified USLE) estimates event-based sediment yield rather than average annual soil loss.

5.8 Worked Numerical — Soil Loss by USLE

Worked Example 5.1 — Average annual soil loss

A 2 ha field on a 6% slope with a slope length of 45 m has an R value of 55, K of 0.035, an LS factor of 1.8, a C factor of 0.30 (poor cover) and a P factor of 1.0 (up-and-down cultivation). Estimate the annual soil loss.

Given: R = 55; K = 0.035; LS = 1.8; C = 0.30; P = 1.0.

Step 1: A = R × K × LS × C × P

Step 2: A = 55 × 0.035 × 1.8 × 0.30 × 1.0

Step 3: 55 × 0.035 = 1.925; × 1.8 = 3.465; × 0.30 = 1.0395; × 1.0 = 1.04

Answer: A = **1.04 t/ha/yr** for the given plot factors.

Here A is expressed per hectare. Total soil loss from the 2 ha field = 1.04 × 2 = 2.08 t/yr.

Worked Example 5.2 — Effect of conservation practices

In Example 5.1 the farmer adopts contour cultivation (P = 0.5) and improves the cover to C = 0.12. Find the new soil loss and the per cent reduction.

Step 1: New A = 55 × 0.035 × 1.8 × 0.12 × 0.5 = 1.925 × 1.8 × 0.06 = **0.208 t/ha/yr**.

Step 2: Reduction = (1.04 − 0.208)/1.04 × 100 = (0.832/1.04) × 100 = **80 %**.

Conclusion: good cover and contouring together cut soil loss by about 80%. This shows why the C and P factors are the easiest levers for the farmer.

5.9 Stream Bank Erosion and Other Forms

- **Stream bank erosion:** cutting away of the banks of streams and rivers by flowing water; controlled by bank protection, vegetation and spurs.
- **Channel erosion:** deepening and widening of stream beds by scouring.
- **Coastal erosion:** loss of shore land by wave action; controlled by sea walls and vegetation (e.g., Casuarina and mangroves).
- **Mass movement:** landslides and slips (discussed above).

5.10 Gully Control Measures

Gully control has two aims — stopping the advance of the gully and stabilising the treated land.

- **Temporary structures:** brushwood dams, loose-rock check dams, woven-wire dams, earth plugs — cheap and quick, used at the gully head and along the bed.
- **Permanent structures:** masonry check dams, gabion structures, and drop spillways for larger gullies.
- **Vegetative measures:** planting of grasses (vetiver, Cenchrus, khus) and trees on the gully sides and bed.

- **Head treatment:** diverting runoff away from the gully head by a diversion channel (diversion drain), so that no water falls into the head.

5.11 Text Diagram — Gully Control

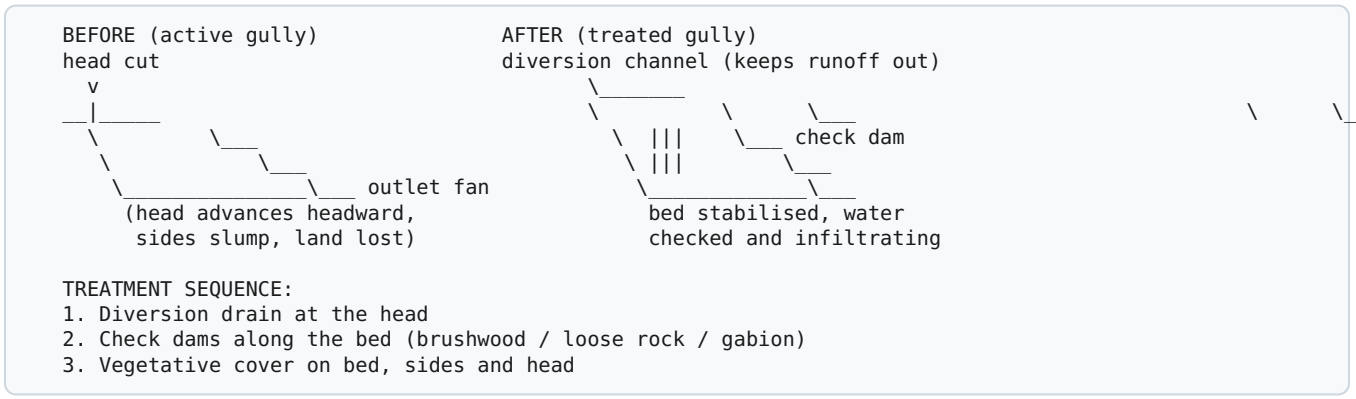


Fig. 5.2 — Gully control by head diversion, check dams and vegetation

5.12 Limitations of the USLE

- Estimates only **sheet and rill erosion**, not gully, stream-bank or wind erosion.
- Gives an **average annual** soil loss, not the loss from a single storm.
- Developed for a specific region and plot size; factor values must be calibrated locally.
- Does not estimate **sediment delivery** to a reservoir (use MUSLE for event sediment yield).

5.13 Total Soil Loss from a Field

Worked Example 5.3 — Total annual soil loss and nutrient loss

A 3 ha field has an estimated soil loss of 12 t/ha/yr. The topsoil contains 0.10% nitrogen and 0.06% phosphorus. Find the total soil loss and the nutrients lost per year.

Step 1 (soil): Total = 12 × 3 = **36 t/yr**.

Step 2 (N): 36 × 0.0010 = 0.036 t = **36 kg N/yr**.

Step 3 (P): 36 × 0.0006 = 0.0216 t = **21.6 kg P/yr**.

36 kg N is roughly equivalent to 78 kg of urea. Soil erosion is thus a hidden fertilizer drain on the farm.

5.14 Conservation Planning by USLE

The USLE is used in planning to bring soil loss within the tolerable limit. If the computed A exceeds T (tolerable loss), the C and P factors are changed (better cover, contouring, terracing) until A falls to or below T. The rainfall (R) and soil (K) cannot be changed, so management targets C and P.

Exam Focus — Lecture 5

Key points

- Water erosion is caused by raindrop impact and runoff; slope and cover are key controls.
- Erosivity (R) is a property of rain/runoff; erodibility (K) is a property of soil.
- Stages: splash → sheet → rill → gully (increasing severity).
- Rills can be removed by tillage; gullies cannot (need structures).

- Ravines = network of deep gullies on flat alluvium (e.g., Chambal).
- Landslides = gravity-driven mass movement, triggered by saturation.
- USLE: A = R K LS C P (average annual sheet & rill soil loss).

Short-answer questions

Q1. Distinguish between erosivity and erodibility.
Model answer: Erosivity is the potential of rain and runoff to cause erosion (a property of the agent, expressed as R); erodibility is the susceptibility of soil to erosion (a property of the soil, expressed as K).

Q2. Name the four stages of water erosion with one line each.
Model answer: Splash — detachment by raindrop impact; sheet — uniform removal of a thin soil layer by sheet flow; rill — small channels removable by tillage; gully — deep, wide channels beyond the reach of tillage.

Q3. Write the USLE and name its factors.
Model answer: A = R × K × LS × C × P, where A = annual soil loss, R = rainfall erosivity, K = soil erodibility, LS = slope length-steepness, C = cover management, P = support practice.

Long-answer questions

L1. Describe the stages of water erosion and explain how each is controlled.
Outline: Splash → sheet → rill → gully, with a diagram → detection signs → control of each (mulch/canopy for splash; contour farming and cover for sheet; tillage and small checks for rill; check dams, gully plugs and revegetation for gully) → conclusion.

L2. Explain the USLE with its factors and work an example showing the effect of conservation practices.
Outline: Concept of empirical model → six factors with units/typical values → worked example → effect of C and P → limitations (sheet & rill only, annual average) → mention RUSLE/MUSLE.

6 Wind Erosion — Factors, Mechanics, Effects of Water and Wind Erosion

6.1 What is Wind Erosion?

> **Wind erosion** is the detachment and transportation of loose, dry, finely divided soil particles by the action of wind.

Wind erosion occurs mainly in arid and semi-arid regions, and on bare, dry, smooth, loose and finely textured soils. It is a serious problem in the dryland tracts of Rajasthan, Gujarat, Haryana, Punjab and parts of Karnataka, Andhra Pradesh and Tamil Nadu (notably the black-soil and coastal sandy belts).

6.2 Factors Influencing Wind Erosion

The factors are conveniently grouped as climatic, soil and vegetation/management factors, plus field-surface and wind characteristics.

(a) Climatic factors

- **Wind velocity:** erosion begins only above a threshold velocity (about 5–6 m/s at 30 cm height); above it, the rate rises roughly with the cube of velocity.
- **Wind direction and duration:** prevailing dry winds blowing over long fetches do most damage.
- **Rainfall and humidity:** dry, droughty periods and low humidity make soil loose and erodible.
- **Temperature:** high temperature dries the surface and destroys aggregation.

(b) Soil factors

- **Texture:** fine sand (0.1–0.5 mm) is most erodible; clay and coarse gravel are less so.
- **Structure and aggregation:** well-aggregated soils resist wind; single-grain soils do not.
- **Moisture content:** moist soil has higher cohesion and resists erosion.
- **Organic matter:** higher organic matter improves aggregation and resistance.
- **Surface roughness:** cloddy, rough surfaces trap moving particles and reduce erosion.

(c) Vegetative / management factors

- **Vegetative cover:** the single most important control — cover reduces wind speed at the surface.
- **Windbreaks and shelterbelts:** rows of trees/shrubs that slow the wind.
- **Field width along the wind direction** (unsheltered distance).
- **Tillage practice:** over-tillage and clean fallow expose loose, fine soil.

6.3 Mechanics of Wind Erosion

Wind moves soil particles in three ways, depending on particle size and wind speed:

(a) Suspension

- Very fine particles (silt and clay, roughly <0.1 mm) are lifted and carried high into the air for long distances.
- They settle only when the wind drops or with rain (giving "dust storms").

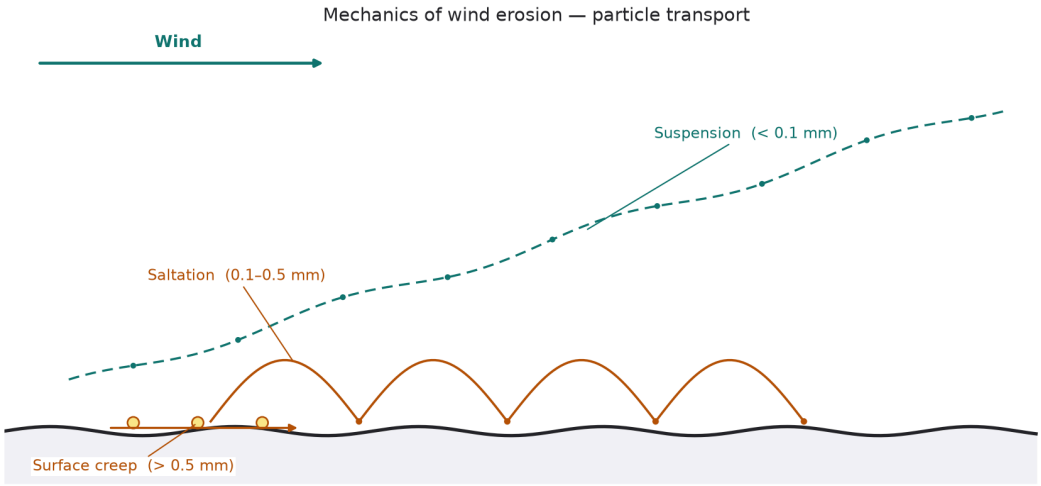


Figure 6.1 Mechanics of wind erosion

Suspension, saltation and surface creep particle paths over a soil surface.

- This is why fine, nutrient-rich fractions are lost far from the field.

(b) Saltation

- Particles of about 0.1–0.5 mm (fine sand) bounce or hop along the surface in short jumps (typically 10–30 cm above the ground).
- Saltation accounts for the largest share (commonly quoted as roughly three-quarters) of wind-borne soil movement.
- Impact of hopping grains dislodges more particles, so saltation is self-propagating.

(c) Surface Creep

- Coarser particles (about 0.5–2 mm, coarse sand) too heavy to be lifted are rolled and pushed along the surface.
- It accounts for a smaller fraction (roughly one-quarter) of movement.
- Creep is driven by the impact of saltating grains rather than by wind directly.

6.4 Text Diagram — Mechanics of Wind Erosion

```
   WIND DIRECTION ------>>>>>>>>>>>>>>>>>>>>>>>>

   ~~~ fine dust carried high .......... SUSPENSION (silt & clay, <0.1 mm)
        o     o         o          o        (long distance, high altitude)
       o  o      o   o        o

          ^v^    ^v^     ^v^     ^v^       SALTATION (fine sand 0.1-0.5 mm)
        bouncing / hopping along in short jumps (~10-30 cm high)
        (largest share of movement)

      o---o----o---o-----o---o---o---o     SURFACE CREEP (coarse sand 0.5-2 mm)
        rolled / pushed along the ground  (smaller share)


   ____________________________________________________________________
        SOIL SURFACE

   Approx. shares: Saltation ~3/4, Surface creep ~1/4, Suspension small but
   removes the finest, most fertile particles far away.
```

Fig. 6.1 — Three modes of wind transport: suspension, saltation, surface creep

6.5 Factors — Summary Table

Table 6.1 — Factors influencing wind erosion

Group	Factor	Effect on erosion
Climatic	Wind velocity	Erosion rises sharply above threshold (~5–6 m/s); ≈ cube of velocity
	Wind direction/duration	Long, dry prevailing winds increase loss
	Rainfall & humidity	Low rainfall/humidity → dry, loose soil → more erosion
	Temperature	High temperature dries and disintegrates surface
Soil	Texture	Fine sand most erodible; clay/gravel less
	Structure/aggregation	Aggregated soil resists erosion
	Moisture	Moist soil resists (higher cohesion)
	Organic matter	Improves aggregation → less erosion
Surface	Roughness	Cloddy surface traps moving grains
	Unsheltered field width	Greater width → more soil carried
Vegetative	Cover	Most important control; reduces surface wind speed

	Windbreaks/shelterbelts	Slow the wind and trap particles

6.6 Wind Erosion Equation (WEQ)

Wind Erosion Equation:

```
E = f (I, K, C, L, V)
E = potential average annual soil loss
I = soil erodibility index (climatic factor)
K = ridge roughness factor
C = climatic factor (wind speed, precipitation, temperature)
L = unsheltered field width (length along wind)
V = equivalent quantity of vegetative cover
(the function f is usually read from charts/nomographs, not multiplied directly)
```

6.7 Effects of Water and Wind Erosion

Table 6.2 — Effects of water erosion and wind erosion

Effect	Water erosion	Wind erosion
Detachment agent	Raindrop impact & runoff	Wind (moving grains)
Main landform damage	Rills, gullies, ravines, landslides	Drifting, dunes, blow-outs, denuded fields
On-site loss	Topsoil and nutrient loss, lower yield	Topsoil, seed and seedling loss, buried crops
Off-site effects	Reservoir/canal siltation, floods, water pollution	Dust storms, air pollution, burial of fertile land, damage to roads
Nutrients lost	Both dissolved and sediment-bound	Mainly the finest, nutrient-rich fractions
Typical region	High-rainfall and sloping areas	Arid/semi-arid, dryland, coastal sandy areas
Control emphasis	Contour farming, bunds, terraces, waterways	Vegetative cover, windbreaks, shelterbelts, mulching, stabilising dunes

Worked Example 6.1 — Soil lost in a dust storm

A 4 ha field loses topsoil at the rate of 12 tonnes/ha in one season of wind erosion. The soil contains 0.08% nitrogen and 0.05% phosphorus. Find the mass of soil and the nutrients lost.

Step 1 (soil loss): Total = 12 × 4 = **48 t**.

Step 2 (nitrogen): N lost = 48 × (0.08/100) = 0.0384 t = **38.4 kg N**.

Step 3 (phosphorus): P lost = 48 × (0.05/100) = 0.024 t = **24 kg P**.

Expressed as fertilizer: 38.4 kg N is about 83 kg of urea (46% N). This shows how wind erosion drains both soil and money.

Worked Example 6.2 — Threshold velocity idea

Wind erosion of a fine sandy soil begins at a threshold velocity of 5.5 m/s. If the wind speed doubles, by what factor does the erosive power rise (using the cube law)?

Step 1: Erosional power ∝ (velocity)³.

Step 2: Ratio = (2)³ = **8 times**.

Doubling wind speed raises the erosive power about eightfold — which is why strong summer winds in the dryland belts are so destructive.

6.8 Control of Wind Erosion

- **Vegetative cover:** keeping the soil covered with crops, stubble or mulch — the most effective and cheapest control.
- **Windbreaks and shelterbelts:** rows of trees/shrubs across the wind (Lecture 7).
- **Strip cropping:** strips across the wind direction (wind strip cropping).
- **Mechanical measures:** rough cloddy surface, ridge and furrow, and "basin listing" to trap particles.
- **Stabilising sand dunes:** planting of grasses (e.g., Cenchrus, Panicum), shrubs and trees, and mulching with crop residues to fix the surface.
- **Irrigation / moisture conservation:** keeping the surface moist reduces erodibility.
- **Reducing field width:** breaking large open fields into smaller strips to reduce the unsheltered distance.

6.9 Text Diagram — Wind Erosion Control by a Shelterbelt

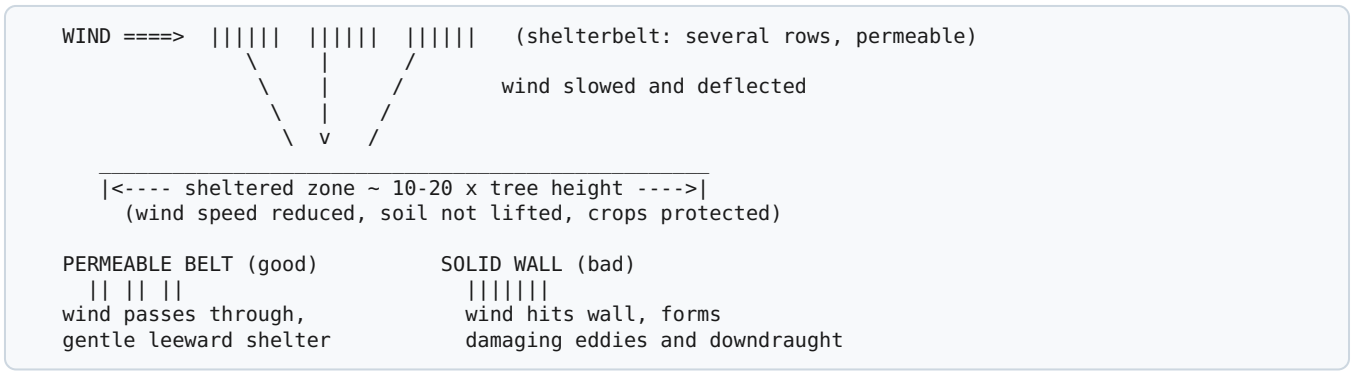


Fig. 6.2 — Shelterbelt and the effect of a permeable vs solid barrier

6.10 Wind Erosion vs Water Erosion — Contrast in Control

Table 6.3 — Contrasting features of wind and water erosion control

Aspect	Water erosion	Wind erosion
Driving force	Gravity + runoff	Wind velocity
Surface cover	Mulch, close crops	Stubble, mulch, roughness
Field arrangement	Along the contour	Across the wind
Permanent barriers	Bunds, terraces, check dams	Windbreaks, shelterbelts
Soil condition	Increase infiltration	Keep moist, aggregated, rough

6.11 Worked Numerical — Soil loss over a dryland field

Worked Example 6.3 — Wind erosion and nutrient drain

A 6 ha dryland field loses 8 t/ha of soil in a windy season. The soil contains 0.07% nitrogen and 0.04% phosphorus. Find the soil and nutrient loss.

Step 1 (soil): Total = 8 × 6 = **48 t**.

Step 2 (N): 48 × 0.0007 = 0.0336 t = **33.6 kg N**.

Step 3 (P): 48 × 0.0004 = 0.0192 t = **19.2 kg P**.

Because suspension carries away the finest, nutrient-rich particles, the nutrient loss per tonne is often higher than the soil loss suggests.

6.12 Sand Dune Stabilisation

- **Mechanical:** mulching with straw, brush, clay or bitumen to fix the surface; erection of brushwood or palm-leaf fences across the wind.
- **Vegetative:** planting of sand-binding grasses (Cenchrus, Panicum, Saccharum), shrubs (Prosopis, Cassia) and trees (Casuarina, Acacia).
- **Shelterbelts** around the stabilised dune to reduce wind speed further.

Exam Focus — Lecture 6

Key points

- Wind erosion = detachment and transport of loose, dry, fine soil by wind.
- Factors: climatic (velocity, dryness), soil (texture, moisture, organic matter), surface roughness, and vegetation cover.
- Three mechanics: suspension (fine), saltation (fine sand, largest share), surface creep (coarse).
- Erosion begins above a threshold wind velocity (~5–6 m/s); power rises with the cube of velocity.
- Vegetative cover and windbreaks are the best controls.
- Wind erosion mainly affects arid/semi-arid and coastal sandy regions.

Short-answer questions

Q1. What is wind erosion? Name the three ways wind moves soil.
Model answer: Wind erosion is the detachment and transportation of loose, dry, fine soil particles by wind. The three modes are suspension (fine silt/clay), saltation (fine sand hopping) and surface creep (coarse sand rolling).

Q2. State any five factors influencing wind erosion.
Model answer: Wind velocity; rainfall and humidity (dryness); soil texture and aggregation; soil moisture; vegetative cover; and surface roughness / unsheltered field width.

Q3. Why is saltation the most important mode of wind transport?
Model answer: Saltation moves the largest share of wind-borne soil (roughly three-quarters); the hopping grains strike the surface and dislodge more particles, so the process is self-propagating.

Long-answer questions

L1. Explain the mechanics of wind erosion with a diagram, and the factors that influence it.

Outline: Definition → suspension, saltation, surface creep (with diagram and approximate shares) → factors (climatic, soil, surface, vegetative) → threshold velocity and cube law → conclusion.

L2. Compare the effects of water erosion and wind erosion and state the control approach for each.

Outline: Table of effects (detachment, landform damage, on-site and off-site effects, regions) → control of water erosion (contour farming, bunds, terraces) → control of wind erosion (cover, windbreaks, mulching, dune stabilisation) → conclusion.

7 Erosion Control for Agricultural Lands — Biological Measures

7.1 Classification of Erosion Control Measures

Erosion control measures are broadly of two kinds:

- **Biological (vegetative) measures:** use plants and cropping practices to protect the soil — contour cultivation, strip cropping, cropping systems, vegetative barriers, windbreaks/ shelterbelts, mulching, afforestation and pasture development.
- **Mechanical (engineering) measures:** use structures to control runoff — contour bunds, graded bunds, terraces, trenches, check dams, stone walls (Lecture 8).

In practice the two are combined in a **watershed development programme**: biological measures treat the land surface, while mechanical measures handle concentrated runoff.

7.2 Contour Cultivation

> **Contour cultivation (contour farming)** is the practice of carrying out all tillage and crop operations, and planting crops, along the contour (across the slope) rather than up and down the slope.

How it works: Each contour row of crop forms a small barrier that slows runoff and traps soil, giving the water more time to infiltrate.

- Reduces runoff velocity and volume, hence erosion.
- Increases infiltration and soil moisture storage.
- Suits gentle to moderate slopes (roughly up to 8–10%); not enough alone on steep land.
- Contour furrows and ridges give better results than plain contour planting.

> **Limitation:** If the field is very irregular, contour rows may break and concentrate water into low spots, so contouring must be combined with a suitable outlet or grassed waterway.

7.3 Strip Cropping

> **Strip cropping** is the practice of growing crops in alternate strips (usually along the contour) of close-growing and row (erosion-permitting) crops, so that the close-growing strips filter runoff and trap soil from the row-crop strips.

Types

- **Contour strip cropping:** strips laid across the slope on the contour.
- **Field strip cropping:** strips of uniform width laid across the general slope but not exactly on the contour.
- **Buffer strip cropping:** strips of grass/legume (not rotated) alternating with regular crop strips.
- **Wind strip cropping:** strips laid perpendicular to the prevailing wind to check wind erosion.

Strip width depends on slope, soil erodibility and rainfall; steeper or more erodible land needs narrower strips. Typical strip widths range from about 10–20 m on moderate slopes to 20–50 m on gentle slopes.

7.4 Cropping Systems

Cropping systems and rotations are chosen to keep the soil covered and to maintain organic matter, thereby reducing erosion.

- **Cover crops:** quick-growing crops (e.g., cowpea, sunhemp, horsegram) that cover the soil between main crops.
- **Green manuring:** ploughing in a leguminous crop to add organic matter (e.g., sunhemp, daincha).
- **Mulching:** spreading crop residues or straw on the surface to absorb raindrop energy and reduce runoff.
- **Mixed/intercropping:** growing two or more crops together so the ground is covered for more of the year.
- **Strip/rotation with grasses and legumes:** alternating with sod crops.
- **Agroforestry:** trees with crops (e.g., in TN, coconut-based and mango-based systems).

7.5 Vegetative Barriers

> **Vegetative barriers** are dense strips of perennial grasses or shrubs planted along the contour (or across the wind) to act as living barriers that slow runoff and trap sediment.

- **Grasses used:** vetiver (Vetiveria zizanioides), khus, lemon grass, Cenchrus, Panicum, and in TN, vetiver and Cenchrus are common.
- **Advantages:** cheap, no land to be taken out permanently, roots bind the soil, biomass can be used for fodder/mulch.
- **Spacing:** barrier-to-barrier spacing depends on slope; on moderate slopes, roughly 15–30 m; closer on steeper land.
- **Vetiver system:** vetiver's deep, fibrous root system makes it a very effective bio-engineering barrier for erosion control and slope stabilisation.

7.6 Windbreaks and Shelterbelts

> **Windbreak** is a strip of trees and/or shrubs planted to protect an area from wind. When several rows are planted (usually at right angles to the prevailing wind) it is called a **shelterbelt**.

- **Function:** reduce wind velocity over the protected area, reducing wind erosion and protecting crops from mechanical damage and desiccation.
- **Effective shelter distance:** a good windbreak reduces wind speed on the leeward side for a distance of roughly 10–20 times its height.
- **Design:** a permeable (not solid) barrier is better than a solid wall, which creates damaging eddies.
- **Species (TN/dryland examples):** Casuarina, Eucalyptus, Acacia (A. nilotica, A. planifrons), Prosopis, Azadirachta indica (neem), Glyricidia, Tamarind, and in coastal areas Casuarina is widely used.
- **Shelterbelt vs windbreak:** a windbreak is generally a single strip protecting a field; a shelterbelt is a belt of several rows protecting a larger area, often along roads, canals and field boundaries.

7.7 Shifting Cultivation

> **Shifting cultivation** is the practice of clearing a patch of forest or hill land, cultivating it for a few years, and then abandoning it when fertility declines, moving on to a fresh patch. It is known as *jhum* in the north-east, *podu* in Andhra Pradesh/Telangana, and *kumari* in the Western Ghats.

Why it causes erosion: the land is left bare after the crop, the protective forest cover is removed by burning, and cultivation on steep slopes exposes soil to heavy monsoon erosion. It is one of the most destructive practices on hill slopes.

Control / replacement measures:

- **Terracing and contour bunding** of the cultivated patch.
- **Permanent settled cultivation** with a fixed, terraced field.
- **Agroforestry and horticulture** (fruit and plantation crops) to keep the slope under tree cover.
- **Contour trenching and check dams** to control runoff.
- **Improved cropping** with cover crops and green manures to shorten the bare period.

7.8 Text Diagram — Biological Measures Layout

```
CONTOUR STRIP CROPPING (plan view, slope downward)
==================================================
\\\\\\\\\\  close-growing strip (grass/legume)  \\\\\\\\\\
~~~~~~~~~~  row crop strip (maize/groundnut)  ~~~~~~~~~~
\\\\\\\\\\  close-growing strip  \\\\\\\\\\
~~~~~~~~~~  row crop strip  ~~~~~~~~~~
\\\\\\\\\\  close-growing strip  \\\\\\\\\\
     contour lines run across the slope (----->)

VEGETATIVE BARRIER (section on slope)
     grass barrier          grass barrier
          |||                    |||
slope -->|||--water slows, traps silt-->|||-->
________|||____________________________|||______  ground

WINDBREAK / SHELTERBELT (plan view)
  WIND ====>     |||||  |||||   (rows of trees)
             sheltered zone extends ~10-20 x tree height leeward
  WIND ====>     |||||  |||||
              ^ belt at right angles to prevailing wind
```

Fig. 7.1 — Contour strip cropping, vegetative barriers and shelterbelt layout

7.9 Biological Measures — Comparison Table

Table 7.1 — Biological erosion-control measures at a glance

Measure	Principle	Best suited to	Effect
Contour cultivation	Rows across slope trap water & soil	Gentle-moderate slopes (≤8-10%)	Reduces runoff & erosion, raises infiltration
Strip cropping	Close strips filter runoff from row strips	Sloping arable land	Cuts soil loss, spreads water
Cover crops / mulching	Protect surface from raindrop impact	All lands, orchards, plantations	Reduces splash & sheet erosion
Vegetative barriers	Living grass strips slow flow	Sloping fields, terraces	Traps sediment, forms natural terrace

Windbreaks/ shelterbelts	Reduce wind velocity	Dryland, coastal sandy areas	Cuts wind erosion, protects crops
Agroforestry	Tree cover over the slope	Hills, marginal lands	Stabilises soil, adds income

Worked Example 7.1 — Number of vegetative barrier rows

A field 120 m long up the slope is to be protected by vetiver barriers at 20 m vertical spacing measured along the slope. How many barrier rows are needed, and what is the total length if the field is 80 m wide along the contour?

Step 1 (number of intervals): 120 / 20 = 6 intervals.

Step 2 (rows): A barrier is planted at the top of each interval; allowing one at the bottom, the number of rows = 6 + 1 = **7 rows** (or 6 interior rows if the top is at the boundary).

Step 3 (total length): 7 × 80 = **560 m** of barrier.

Using the convention of 6 interior rows, the length is 480 m. State the convention assumed.

Worked Example 7.2 — Effective shelter distance of a windbreak

A Casuarina windbreak is 8 m tall. Estimate the effective shelter distance on the leeward side.

Step 1: Effective shelter distance is roughly 10–20 times the tree height.

Step 2: 10 × 8 = 80 m to 20 × 8 = 160 m.

Answer: The windbreak protects roughly **80–160 m** leeward. Two belts 150–160 m apart would shelter a large field.

7.10 Agrostological (Pasture) Measures

Agrostological measures are the use of grasses and pasture species to stabilise the soil and reduce erosion, especially on land not suitable for cropping.

- **Grass cover** is the most effective protection against both splash and sheet erosion because it forms a dense mat of leaves and roots.
- **Grasses used:** Cenchrus ciliaris, Panicum, Chloris, Brachiaria, Vetiver, and for pasture, hybrid napier, Guinea grass and Cenchrus.
- **Pasture development** on degraded and steep land converts it from a source of erosion to a productive use (grazing and fodder).
- **Silvipasture:** trees plus grass on the same land.

7.11 Mulching and its Types

Mulch is a protective layer of crop residues, straw, leaves or synthetic material spread on the soil surface to reduce raindrop impact, runoff and evaporation.

- **Organic mulch:** crop residues, straw, husk, compost — adds organic matter as it decomposes.
- **Inorganic mulch:** plastic film (used in horticulture and drip systems; raises soil temperature and saves water).

- **Vertical mulch:** crop residues placed in a vertical pit to store water and recharge the profile.
- **Live mulch:** a low-growing cover crop (e.g., cowpea) grown as a ground cover.

7.12 Selection of Biological Measures

Table 7.2 — Choosing a biological measure by slope and land use

Slope	Land use	Suitable biological measure
0–2%	Arable	Good cropping systems, cover crops, mulching
2–5%	Arable	Contour cultivation, strip cropping
5–10%	Arable	Contour strip cropping, vegetative barriers
10–15%	Marginal arable	Vegetative barriers, contour strip cropping, agroforestry
>15%	Non-arable	Grass/pasture, agroforestry, afforestation
Dryland / coastal	Any	Windbreaks, shelterbelts, mulching

7.13 Watershed Development Approach

Biological measures work best when combined with mechanical measures in an integrated **watershed development** plan, in which the whole catchment (ridge to valley) is treated: ridge → afforestation and contour trenching; middle → contour bunding and strip cropping; valley → check dams and farm ponds. This is the basis of programmes such as PMKSY and the Tamil Nadu watershed schemes.

7.14 Cropping Systems for Conservation — Tamil Nadu Examples

Table 7.3 — Conservation-friendly cropping patterns in Tamil Nadu

System	Example	Benefit
Cover cropping	Cowpea / horsegram between main crops	Ground cover in the off-season
Green manuring	Sunhemp / daincha ploughed in	Organic matter, better structure
Intercropping	Groundnut + redgram; coconut + banana	Continuous cover, higher income
Strip cropping	Maize strips alternating with cowpea/grass	Runoff filtered, soil trapped
Agroforestry	Coconut/mango + fodder grass	Tree cover over the slope
Fodder/grasses	Hybrid napier on bunds and terrace risers	Bund protection + fodder

7.15 Role of Organic Matter

Organic matter improves soil aggregation, increases infiltration and water-holding capacity, and binds soil particles against both water and wind erosion. Its maintenance through green manuring, crop-residue recycling and farmyard manure is a biological conservation measure in itself.

Exam Focus — Lecture 7

Key points

- Erosion control measures: biological (vegetative) and mechanical (engineering).
- Contour cultivation: all operations across the slope; suits gentle-moderate slopes.
- Strip cropping: alternate close-growing and row-crop strips; four types.

- Cropping systems: cover crops, green manuring, mulching, intercropping, agroforestry.
- Vegetative barriers: vetiver, Cenchrus etc. form living barriers that trap sediment.
- Windbreaks/shelterbelts reduce wind speed for ~10–20 times their height.
- Shifting cultivation (jhum/podu/kumari) is highly erosive; replaced by terracing and agroforestry.

Short-answer questions

Q1. What is contour cultivation? State its benefits and limitations.
Model answer: Carrying out tillage and planting along the contour (across the slope). Benefits: slows runoff, reduces erosion, increases infiltration. Limitation: on very irregular or steep land, rows may break and concentrate water; must be combined with grassed waterways or bunds.

Q2. Define strip cropping and name its types.
Model answer: Growing crops in alternate strips of close-growing and row crops so the close strips trap soil and water. Types: contour, field, buffer and wind strip cropping.

Q3. What is a shelterbelt and how far does it protect?
Model answer: A shelterbelt is a belt of several rows of trees/shrubs planted across the prevailing wind to reduce wind velocity. It protects roughly 10–20 times its height on the leeward side.

Long-answer questions

L1. Describe the biological measures for controlling soil erosion on agricultural lands.
Outline: Classification → contour cultivation → strip cropping (types) → cropping systems (cover crops, green manure, mulch, intercropping, agroforestry) → vegetative barriers (vetiver system) → windbreaks/shelterbelts → shifting cultivation and its control → conclusion with TN examples.

L2. Explain strip cropping and cropping systems as erosion-control practices, with examples suitable for Tamil Nadu.
Outline: Definition → strip cropping types and widths → cropping systems → TN examples (groundnut/redgram strips, sunhemp green manure, coconut-based agroforestry, vetiver barriers in the uplands) → combined effect on erosion → conclusion.

8 Mechanical Measures — Bunds, Terraces, Trenches, Stone Walls

8.1 Purpose of Mechanical Measures

Biological measures slow water on the surface, but where slopes are steeper or runoff is concentrated, engineering structures are needed. **Mechanical (engineering) measures** intercept, store or safely dispose of runoff so that it cannot detach and carry away soil.

On **gentle to moderate slopes (up to about 8%)**, bunds are used. On **steep hill slopes**, terraces, trenches and stone walls are used.

8.2 Contour Bund

> **Contour bund** is an embankment of soil constructed along the contour (at the same elevation throughout) to impound runoff water behind it, so that it infiltrates and is stored for crops, and soil is not carried away.

- **Objective:** to conserve both soil and water — water is stored in the field, not led away.
- **Suited to:** low-rainfall areas (< 800 mm) and permeable soils on gentle slopes (up to about 6–8%).
- **Shape:** trapezoidal cross-section with a top width, side slopes, and a height that depends on the expected runoff depth.
- **Bund height** is normally kept low (0.5–1.0 m) because a contour bund stores water rather than conveying it.
- **Spacing (horizontal interval):** closer on steeper land.

Vertical interval (VI) between bunds (standard empirical forms):

```
VI (m) = 0.3 (S + 2)     [where S = land slope in per cent]  ... (commonly used)
or  VI = (S / 2) + 0.3   ... (another common textbook form)
Horizontal interval (HI) = VI / (S/100) = 100 × VI / S
Number of bunds = Slope length along the land / HI  (+1 for the topmost)
```

8.3 Graded Bund

> **Graded bund** is an embankment constructed along the contour but with a gentle longitudinal grade, so that excess runoff is led slowly along the bund to a safe outlet (a natural drain or grassed waterway) instead of being stored behind it.

- **Objective:** to dispose of surplus runoff safely while checking erosion.
- **Suited to:** high-rainfall areas (> 800 mm) and soils of low permeability where water cannot be stored in the field (water-logging risk).
- **Grade:** typically 0.2–0.5% (1 in 200 to 1 in 500) toward the outlet.
- Also called a **graded bund** or, when large, a **channel terrace**; the graded bund in heavy soils of the black-soil tracts is often called a **broad-based bund**.

8.4 Contour Bund vs Graded Bund

Table 8.1 — Contour bund vs graded bund

Point	Contour bund	Graded bund

Contour bund — cross-section

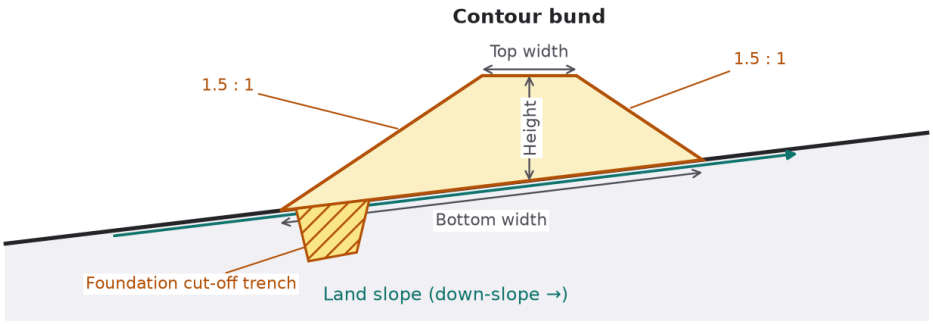


Figure 8.1 Contour bund — cross-section

Trapezoidal bund seated on the sloping ground, with foundation cut-off trench.

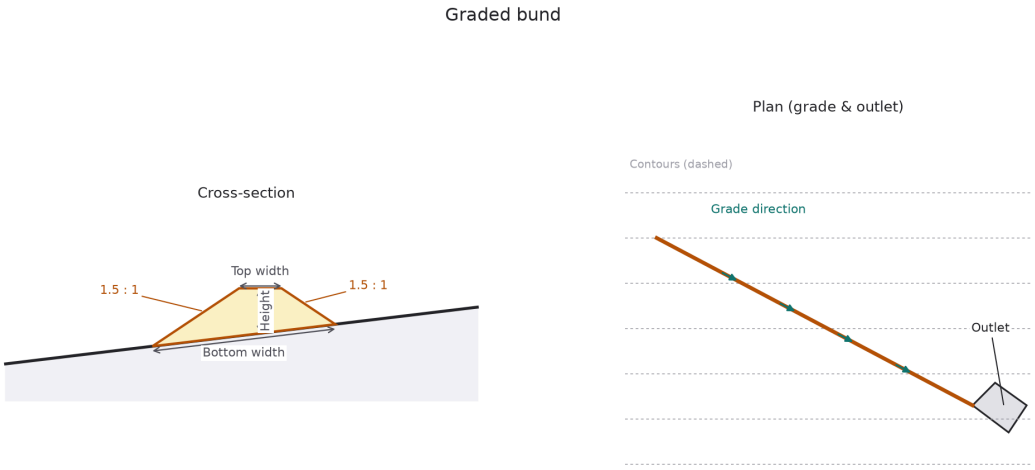


Figure 8.2 Graded bund

Cross-section with the grade direction and outlet shown in plan.

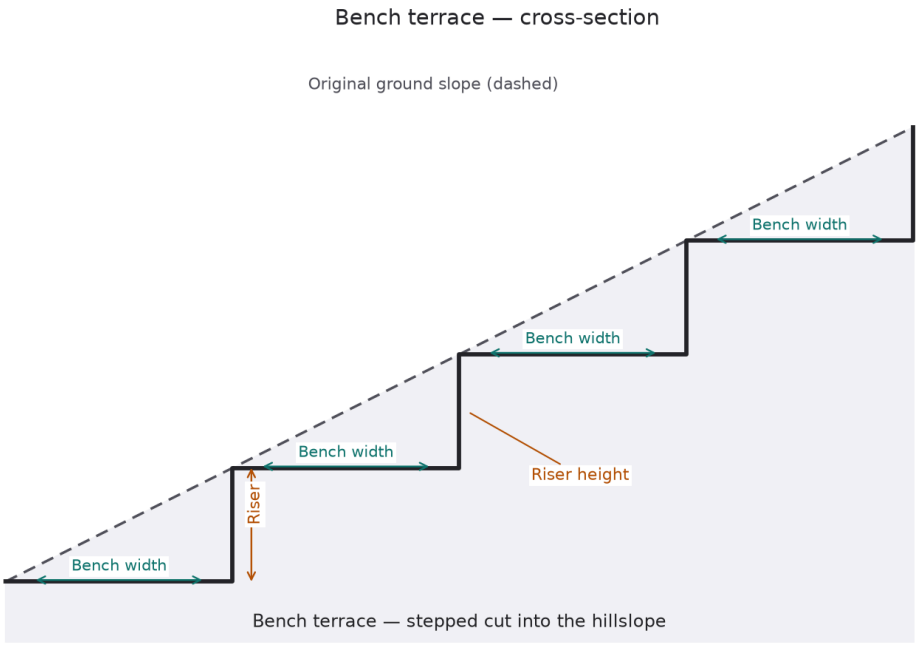


Figure 8.3 Bench terrace — cross-section

Stepped terraces cut into a hillslope with riser and bench width.

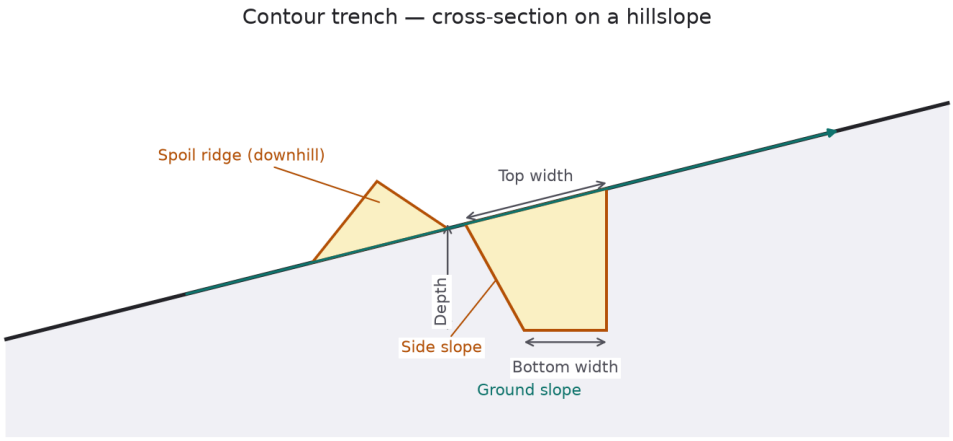


Figure 8.4 Contour trench — cross-section

Trench on a hillslope with downhill spoil ridge.

Alignment	Along the contour (level)	Along the contour but with a longitudinal grade
Runoff handling	Stored behind the bund (impounded)	Led along the bund to an outlet (disposed)
Best for	Low rainfall (<800 mm), permeable soils	High rainfall (>800 mm), low-permeability soils
Water-logging risk	Possible in heavy soils	Low (water is removed)
Length of bund	Can be long (level)	Kept to limited length to reach an outlet
Grade	Nil	0.2–0.5%
Main aim	Soil & water conservation	Safe disposal of surplus runoff

8.5 Bund Dimensions

Table 8.2 — Typical bund dimensions (standard ranges)

Parameter	Contour bund	Graded bund
Height (m)	0.5–1.0	0.6–1.2
Top width (m)	0.4–0.6	0.5–0.8
Side slope (upstream : downstream)	1:1 : 1.5:1 (H:V)	1:1 : 1.5:1
Bottom width (m)	≈ top width + (1.5–2.5 × height)	≈ top width + (1.5–2.5 × height)
Longitudinal grade	Nil	0.2–0.5%
Bund spacing	By VI formula, 10–60 m	By VI formula, closer on steep land
Side slopes of borrow pit	0.5:1 (H:V)	0.5:1 (H:V)

8.6 Broad Beds and Furrows

Broad beds and furrows (BBF) is a land configuration in which the field is formed into broad beds (for crops) separated by furrows (for drainage), usually laid along the contour on gentle slopes.

- Beds carry the crop; furrows collect and safely drain excess rain water.
- Suited to deep, medium-textured soils of gentle slopes in high-rainfall areas; used in the black-soil and coastal tracts.
- Reduces both erosion and temporary water-logging; improves aeration.
- Bed width and furrow depth depend on rainfall and soil; typical bed width 1.5–3 m with furrows about 0.3 m deep.

8.7 Basin Listing

Basin listing is a tillage practice in which a special lister forms a series of small, level basins (or pockets) in the furrow, each of which stores water in place for the crop.

- Creates micro-depressions that hold rain water where it falls.
- Useful in low-rainfall, dryland areas for moisture conservation and for establishing crops/grasses.
- Also used in reclamation of degraded land and for tree/grass establishment on slopes.

8.8 Random Tie Ridging

Random tie ridging is the practice of placing small earthen barriers (ties) at random intervals across furrows or rills to pond and store water, checking its flow down the furrow.

- Simple and cheap; done with a hoe or by hand.
- Useful on moderate slopes to prevent furrow water from gaining velocity.
- Also called **tie ridging**; "random" because ties are not at fixed regular intervals.

8.9 Mechanical Measures for Hill Slopes

(a) Contour Trench

Contour trench is a long, narrow, continuous ditch dug along the contour on steep hill slopes (usually with the excavated soil forming a ridge on the downhill side) to intercept and store runoff, allowing it to infiltrate and support tree/grass growth.

- **Suited to:** steep, non-arable hill slopes (roughly >10–15%) for afforestation and pasture.
- **Dimensions:** typically 0.3–0.6 m deep and 0.3–0.5 m wide; the spoil is placed on the lower edge.
- **Spacing:** by the VI formula; often 2–5 m on steep slopes.
- Trenches are staggered or continuous, and are planted with trees/grass on the ridge.

(b) Bench Terrace

Bench terrace is a step-like flat platform (bench) cut into a hill slope and protected by a riser or wall on the downhill side, so that each bench is nearly level and can be cultivated without erosion.

Table 8.3 — Types of terraces

Type	Description	Suited to
Bench terrace (level)	Nearly level platform, vertical/high riser	Steep slopes (15–33%), orchards, irrigated crops
Bench terrace (sloping)	Platform with a slight inward/outward slope	Steeper slopes where level benches are difficult
Contour/ridge terrace	Ridge with channel, along contour	Gentle to moderate slopes
Graded terrace	Channel terrace with a grade to an outlet	High rainfall, low-permeability soils
Broad-base terrace	Wide, gentle ridge-channel, machine-friendly	Deep soils, mechanised farms
Zing (or zinj) terrace	Ridge terraces on which crops are grown, common in black-soil areas	Medium-deep black soils

Bench terrace dimensions: bench width depends on slope, soil depth and the crop (wider benches on gentler slopes and for orchards); risers are protected by stone pitching or grass. In the hills of Tamil Nadu (Nilgiris, Shevaroys), bench terracing is used for tea, coffee and vegetables.

(c) Contour Stone Wall

> **Contour stone wall** is a wall of stones laid along the contour on stony hill slopes to intercept runoff and trap soil; over time, soil accumulates behind the wall, forming a natural bench.

- **Suited to:** steep, stony slopes with plenty of loose stone; common in the rocky uplands.
- **Construction:** stones placed without mortar, with a slight batter (lean into the slope) and a foundation trench.
- **Advantage:** uses local material, cheap, durable, forms a bench terrace gradually.
- **Combined with:** trenching and vegetative barriers for best results.

8.10 Text Diagrams — Bunds, Terrace and Trench

```
(a) CONTOUR BUND (section)              (b) GRADED BUND (section)
     top width 0.5 m                        top width 0.6 m

      ___________                             ___________
     /           \  1:1                      /           \  1:1
    /             \_____                    /             \_____
   /  stored water  \ 1.5:1                 /   runoff flows  \ 1.5:1
__/_________________\______             ___/____grade 0.3%_____\_____
^     original ground                           to outlet --->
|<-- spacing by VI formula -->|         |<-- spacing by VI formula -->|

(c) BENCH TERRACE (section on hill slope)
    original hill slope / (dashed)
                       /
       ____/___________/________   bench 1 (level)
       |riser|           |
    ___|_____|___________|_____   bench 2
       |riser|           |
    ___|_____|___________|_____   bench 3
       |riser|           |
       step-like platforms cut into the slope; risers pitched with stone/grass

(d) CONTOUR TRENCH (section)
      ridge (spoil)                 ridge
         /\                            /\
    ____/  \___trench___/\___  ____/  \___trench___/\
       (0.3-0.6 m deep)  intercepts runoff, water infiltrates
```

Fig. 8.1 — Cross-sections of contour bund, graded bund, bench terrace and contour trench

8.11 Worked Numerical — Bund Spacing and Earthwork

Worked Example 8.1 — Spacing of contour bunds

Contour bunds are to be laid on a uniform 4% slope. Find the vertical and horizontal intervals, and the number of bunds on a slope length of 240 m.

Step 1 (VI): VI = 0.3 (S + 2) = 0.3 (4 + 2) = 0.3 × 6 = **1.8 m**.

Step 2 (HI): HI = 100 × VI / S = 100 × 1.8 / 4 = **45 m**.

Step 3 (number): Number = 240 / 45 = 5.33 → take 5 intervals, so **6 bunds** (including the topmost) or 5 interior bunds.

Using the alternative form VI = S/2 + 0.3 = 2.3 m gives HI = 57.5 m and about 4–5 bunds. Always state which VI formula is used.

Worked Example 8.2 — Earthwork of a bund (volume)

A graded bund has a top width of 0.6 m, height 0.9 m and side slopes of 1:1 (H:V) on both sides. Find the cross-sectional area and the volume of earthwork for a 200 m length.

Step 1 (base width): Base = top + 2 × (side slope × height) = 0.6 + 2 × (1 × 0.9) = 0.6 + 1.8 = 2.4 m.

Step 2 (cross-sectional area, trapezium): A = ½(top + base) × height = ½(0.6 + 2.4) × 0.9 = ½ × 3.0 × 0.9 = **1.35 m²**.

Step 3 (volume): V = A × length = 1.35 × 200 = **270 m³**.

Earthwork for bund construction is normally quoted in m³; with a bulking factor (about 1.1–1.25) the loose volume to be handled is somewhat larger.

Worked Example 8.3 — Spacing of bench terraces

On a 20% slope, benches are to be formed with a riser height of 1.5 m. Find the horizontal spacing of the terrace risers.

Step 1: HI = 100 × VI / S, with VI = riser height = 1.5 m.

Step 2: HI = 100 × 1.5 / 20 = **7.5 m**.

The bench width would be about 7.5 m (minus riser allowance); narrower on steeper slopes, wider on gentler ones.

8.12 Gully-Control Structures and Check Dams

- **Check dam:** a small barrier across a gully or nala to reduce the velocity of flow, trap sediment and store water for recharge.
- **Types:** brushwood, loose-rock, gabion, masonry and concrete check dams.
- **Drop structure:** used where there is a sudden fall, to drop the water safely and dissipate energy.
- **Chute spillway:** a paved ramp to carry concentrated flow down a slope without erosion.

8.13 Grassed Waterway

> **Grassed waterway** is a natural or constructed channel, established with suitable grass, that safely carries concentrated runoff from a field to an outlet without eroding the soil.

- Designed for the peak runoff by the rational method (Q = CIA/360).
- The grass lining resists scouring; a permissible velocity of about 1.5–2 m/s is usually allowed for good grass cover.
- Widely used as the outlet for contour/graded bunds and terraces.

8.14 Text Diagram — Check Dam and Grassed Waterway

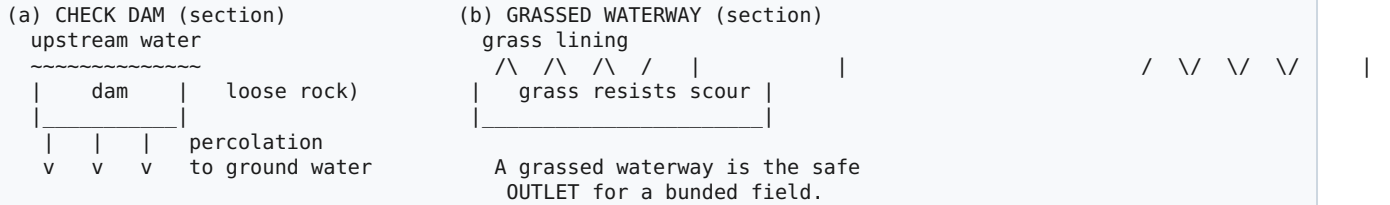


Fig. 8.2 — Check dam and grassed waterway

8.15 Comparison of Hill-Slope Measures

Table 8.4 — Hill-slope mechanical measures

Measure	Purpose	Slope suitability	Cost
Contour trench	Intercept & store runoff, support plantations	Steep (>15%)	Low
Bench terrace	Create cultivable level platforms	Steep (15–33%)	High
Contour stone wall	Trap soil, form natural benches	Steep, stony	Moderate
Half-moon / crescent terrace	Establish trees/shrubs on dry slopes	Steep, dry	Low
Check dam	Check gully advance, recharge	Drainage lines	Moderate

8.16 Worked Numerical — Grassed Waterway Design

Worked Example 8.4 — Peak runoff for a waterway

A bunded field of 8 ha has a runoff coefficient of 0.45 and experiences a design rainfall intensity of 90 mm/h. Find the peak runoff the waterway must carry.

Step 1: Q = C I A / 360.

Step 2: Q = (0.45 × 90 × 8) / 360 = 324 / 360 = **0.90 m³/s**.

The waterway is then sized so that the flow velocity stays below the permissible velocity for the chosen grass cover (about 1.5–2 m/s).

8.17 Worked Numerical — Number of bunds in a field

Worked Example 8.5 — Bunds on a 3% slope

A field has a uniform 3% slope and a slope length of 180 m. Find the bund spacing and the number of bunds.

Step 1 (VI): VI = 0.3(3 + 2) = 1.5 m.

Step 2 (HI): HI = 100 × 1.5 / 3 = 50 m.

Step 3 (number): 180 / 50 = 3.6 → 3 intervals, so **4 bunds** including the topmost (or 3 interior bunds).

Exam Focus — Lecture 8

Key points

- Mechanical measures intercept, store or dispose of runoff; used with biological measures.
- Contour bund stores water (low rainfall, permeable soil); graded bund disposes of water (high rainfall, heavy soil).
- Graded bunds have a longitudinal grade of 0.2–0.5% to an outlet.
- Broad beds and furrows, basin listing and random tie ridging are simple water-spreading measures.
- Hill slopes: contour trench, bench terrace and contour stone wall.
- VI = 0.3(S+2); HI = 100 VI / S.

- Bund cross-section area = ½(top + base) × height.

Short-answer questions

Q1. Differentiate contour bund and graded bund.
Model answer: A contour bund runs along the contour (level) and stores runoff behind it; it suits low-rainfall permeable soils. A graded bund has a gentle longitudinal grade (0.2–0.5%) and leads runoff to an outlet; it suits high-rainfall, low-permeability soils.

Q2. What is a bench terrace and where is it used?
Model answer: A bench terrace is a step-like nearly level platform cut into a hill slope with a riser on the downhill side. It is used on steep slopes (about 15–33%) for orchards and irrigated crops, e.g., tea/coffee in the Nilgiris.

Q3. Write the formulas for the vertical and horizontal interval between bunds.
Model answer: VI = 0.3(S + 2) m, where S is slope in per cent; HI = 100 × VI / S.

Long-answer questions

L1. Describe the mechanical measures used for erosion control on gentle slopes and explain when each is chosen.
Outline: Purpose of mechanical measures → contour bund → graded bund → comparison → broad beds and furrows → basin listing → random tie ridging → choice based on rainfall and soil → worked spacing/earthwork examples.

L2. Explain the mechanical measures used for controlling erosion on hill slopes, with diagrams.
Outline: Need for hill-slope measures → contour trench (dimensions, spacing) → bench terrace (types, dimensions, diagram) → contour stone wall (construction, advantages) → combination with vegetation → TN hill examples → conclusion.

9 Mid Semester Examination — Revision and Model Paper

This lecture is the mid-semester examination of the course. It covers Lectures 1 to 8 (surveying, levelling, area and volume, soil erosion, water and wind erosion, and biological and mechanical erosion control). This chapter consolidates the first half of the syllabus into a revision sheet and a model question paper.

9.1 Half-Syllabus Revision Map

Table 9.1 — Lecture-wise revision map for the mid-semester

Lecture	Core idea to remember	Must-know formula / value
1. Surveying, chain & cross staff	Chain survey = only linear measurement; principle of the triangle	Metric chain 20 m = 100 links × 0.2 m; True length = Measured × (L'/L)
2. Dumpy level & levelling	RL from BS/FS; two reduction methods	HC = RL + BS; ΣBS−ΣFS = ΣRise−ΣFall = Last−First RL
3. Area and volume	Trapezoidal (any no.), Simpson's (odd no.)	Trap: (d/2)[O_0+O_n+2Σi]; Simpson: (d/3) [(O_0+O_n)+4Σe+2Σo]
4. Soil erosion — causes & effects	Detachment → transport → deposition; geologic vs accelerated	Tolerable loss ≈ 5 t/ha/yr (medium-deep soils)
5. Water erosion & USLE	Splash → sheet → rill → gully; erosivity vs erodibility	A = R K LS C P
6. Wind erosion	Suspension, saltation, surface creep	Threshold ≈ 5–6 m/s; power ∝ v^3
7. Biological measures	Contour cultivation, strip cropping, barriers, shelterbelts	Shelter distance ≈ 10–20 × tree height
8. Mechanical measures	Contour vs graded bund; terraces and trenches	VI = 0.3(S+2); HI = 100 VI/S

9.2 Ten High-Yield Points for the Mid-Semester

1. Surveying fixes positions; levelling fixes elevations. Know both.
2. The chain correction formula: True = Measured × (L'/L).
3. In the level book, always apply the arithmetic check — it earns marks and catches errors.
4. Simpson's rule needs an odd number of ordinates; trapezoidal needs no such condition.
5. Erosivity is a property of rain, erodibility is a property of soil — a favourite one-mark difference.
6. The four stages of water erosion, in order, with one line each.
7. USLE: A = R K LS C P; the C and P factors are the farmer's levers.
8. Wind erosion: three modes and the threshold velocity; saltation is the largest share.
9. Biological vs mechanical measures — know the difference and examples of each.
10. Contour bund stores water; graded bund disposes of it.

9.3 Model Question Paper

Model Mid-Semester Paper — SWE 211 (Theory)

Time: 1½ hours Maximum marks: 40

Part A — Answer all (1 mark each, 10 marks)

1. Define surveying.
2. Give the length and number of links in a 20 m metric chain.
3. What is a change point?
4. State the condition for applying Simpson's rule.
5. Define erodibility.
6. Name the three modes of wind transport.
7. What is strip cropping?
8. Write the formula for horizontal interval between bunds.
9. Give one example of a vegetative barrier grass.
10. State the tolerable soil loss for medium-deep soils.

Part B — Answer any three (5 marks each, 15 marks)

1. Explain the uses of surveying in agriculture with examples.
2. Describe the stages of water erosion.
3. Differentiate contour bund and graded bund.
4. Explain the factors influencing wind erosion.

Part C — Answer any one (15 marks)

1. Describe chain surveying with principle, instruments and field procedure; draw the field book.
2. Explain the two methods of reducing levels with a worked example; apply the arithmetic check.

Suggested answers: Part A — (1) art and science of fixing relative positions of points; (2) 20 m, 100 links; (3) common point of two instrument settings with FS and next BS; (4) odd number of equally spaced ordinates; (5) susceptibility of soil to erosion; (6) suspension, saltation, surface creep; (7) alternate strips of close-growing and row crops; (8) HI = 100 VI/S; (9) vetiver; (10) about 5 t/ha/yr.

9.4 Worked Revision Numericals

Revision Numerical 1 — Level book (both methods)

Readings (BS, IS, FS): BS 1.500 on BM (RL 50.000); IS 1.100; IS 1.700; FS 2.000 with next BS 1.200; IS 1.400; FS 1.900. Reduce the levels by the Height of Collimation method and apply the check.

HC1 = 50.000 + 1.500 = 51.500.

RL of IS points = 51.500 − 1.100 = 50.400; 51.500 − 1.700 = 49.800.

RL at first FS = 51.500 − 2.000 = 49.500.

HC2 = 49.500 + 1.200 = 50.700.

RL of IS = 50.700 − 1.400 = 49.300; last RL = 50.700 − 1.900 = 48.800.

Check: ΣBS − ΣFS = (1.500 + 1.200) − (2.000 + 1.900) = 2.700 − 3.900 = −1.200. Last − First = 48.800 − 50.000 = −1.200. ✓

Revision Numerical 2 — Area by both rules

Offsets at 20 m intervals: 0, 4, 7, 9, 6, 3, 0 m. Find the area.

Trapezoidal: A = (20/2)[0 + 0 + 2(4+7+9+6+3)] = 10 × 2 × 29 = **580 m²**.

Simpson's: A = (20/3)[(0+0) + 4(4+9+3) + 2(7+6)] = (20/3)[0 + 64 + 26] = (20/3)(90) = **600 m²**.

Revision Numerical 3 — Soil loss by USLE

R = 60, K = 0.03, LS = 2.0, C = 0.25, P = 0.6. Find A.

A = 60 × 0.03 × 2.0 × 0.25 × 0.6 = **0.54 t/ha/yr**.

9.5 Common Mistakes to Avoid in the Mid-Semester

- Forgetting the level-book **arithmetic check** — always show it.
- Using **Simpson's rule with an even number of ordinates** without splitting the last strip.
- Mixing up **erosivity (rain)** and **erodibility (soil)**.
- Writing the stages of water erosion **out of order** — it must be splash, sheet, rill, gully.
- Confusing **contour bund (stores water)** with **graded bund (disposes water)**.
- Omitting **units** in the final answer (m, ha, cumec, t/ha/yr).

9.6 Quick-Fire Definitions (One Line Each)

Table 9.2 — One-line definitions for rapid revision

Term	One-line definition
Surveying	Fixing relative positions of points on/above/below the earth's surface.
Levelling	Finding the difference of elevation between points.
Reduced level	Height of a point above or below the datum.
Offset	Lateral distance of a detail from the chain line.
Soil erosion	Detachment and transport of soil by water, wind, ice or gravity.

Erosivity	Potential of rain/runoff to cause erosion.
Erodibility	Susceptibility of soil to erosion.
Gully	A channel too deep to be removed by tillage.
Contour bund	Embankment along the contour that impounds runoff.
Graded bund	Bund with a longitudinal grade leading runoff to an outlet.

9.7 Three Full Model Answers for the Mid-Semester

Model Answer — "Describe chain surveying and draw the field book."

Chain surveying is the method of measuring land by dividing the area into a network of triangles and measuring only the linear dimensions of the sides with a chain or tape; no angles are measured. Its principle is that a triangle is the only figure that can be plotted from the lengths of its sides alone, so a framework of triangles (built from a base line and subsidiary lines) fixes the whole area.

Instruments: chain/tape, arrows, ranging rods, cross staff, pegs, plumb bob, field book.

Procedure: (1) reconnaissance; (2) index sketch; (3) fix and measure the base line and main lines; (4) take perpendicular and oblique offsets to details; (5) measure check lines; (6) plot to scale and compute the area.

Field book: a single-line or double-line book; distances are entered from bottom to top and offsets to the left or right of the chain line. [See Fig. 1.1 for the layout.]

Model Answer — "Reduce the level book by the Rise and Fall method and apply the check."

Set out a table with columns for station, BS, IS, FS, rise, fall and RL. Compute the rise or fall between successive readings (rise if the reading decreases, fall if it increases), apply it to the previous RL to get the next RL, and finally check that $\Sigma BS - \Sigma FS = \Sigma Rise - \Sigma Fall =$ Last RL − First RL. [See Worked Example 2.1 for a full reduction.]

Model Answer — "Explain the biological and mechanical measures of erosion control."

Biological: contour cultivation, strip cropping, cropping systems (cover crops, green manuring, mulching, intercropping, agroforestry), vegetative barriers (vetiver, Cenchrus), windbreaks and shelterbelts, and control of shifting cultivation. They protect the surface and improve infiltration.

Mechanical: contour bunds (store water), graded bunds (dispose of water), broad beds and furrows, basin listing, random tie ridging on gentle slopes; and contour trenches, bench terraces and contour stone walls on hill slopes. They intercept, store or safely dispose of runoff.

The two are combined in a watershed development plan.

Exam Focus — Lecture 9

Key points

- The mid-semester covers Lectures 1–8.
- Definitions, comparisons and causes/effects dominate the theory marks.
- Numericals are direct one- or two-formula problems; memorise the formula sheet.
- Always apply the level-book arithmetic check.
- Know the four stages of water erosion and the three modes of wind transport in order.
- Contour bund vs graded bund and biological vs mechanical measures are frequent comparisons.

Short-answer questions

Q1. State any four uses of surveying in agriculture.
Model answer: Farm layout and land levelling; contour mapping for conservation; canal and drain alignment; computing area and volume of earthwork.

Q2. Write the arithmetic check for the level book.
Model answer: $\Sigma BS - \Sigma FS = \Sigma Rise - \Sigma Fall =$ Last RL − First RL.

Q3. Differentiate erosivity and erodibility.
Model answer: Erosivity is the potential of rain/runoff to cause erosion (agent property, R); erodibility is the susceptibility of soil to erosion (soil property, K).

Long-answer questions

L1. Reduce the given level book by both methods and apply the checks (a standard 15-mark numerical).
Outline: Set out the level-book table → HC method → Rise-and-Fall method → arithmetic checks → final RLs and slope interpretation.

L2. Describe the biological and mechanical measures of erosion control with examples (a standard 15-mark essay).
Outline: Classification → biological (contour cultivation, strip cropping, barriers, shelterbelts) → mechanical (contour/graded bund, terraces, trenches) → TN examples → conclusion.

10 Rainwater Harvesting — Runoff Water Harvesting and Runoff Computation

10.1 Rainwater Harvesting — Concept

Rainwater harvesting (RWH) is the collection and storage of rain water, from a catchment surface (rooftop, field, road or natural slope), for productive use or for recharging ground water, instead of allowing it to run off and be lost.

Rainwater harvesting is especially important in Tamil Nadu, where a large part of the state is rain-shadow or semi-arid and depends on the monsoon. It serves two purposes: **storage** for later use and **recharge** to replenish ground water.

Types of rainwater harvesting

- **Rooftop rainwater harvesting:** collects rain from house/building roofs into a tank or recharge pit.
- **Runoff (surface) water harvesting:** collects runoff from fields, watersheds, roads and catchments into farm ponds, tanks or check dams.
- **In-situ rainwater conservation:** retains rain where it falls — contour bunding, trenching, basin listing, mulching.
- **Ground water recharge structures:** pits, trenches, shafts and recharge wells (Lecture 11).

10.2 Runoff Water Harvesting

Runoff water harvesting is the collection of surface runoff from a catchment and its storage in structures such as farm ponds, tanks, check dams and percolation ponds, for irrigation, livestock, domestic use or recharge.

Components

- **Catchment (watershed):** the area that contributes runoff.
- **Collection / conveyance system:** diversion channels, waterways, pipes.
- **Storage structure:** farm pond, tank, check dam, percolation pond.
- **Distribution system:** pumps, pipelines, channels for use.
- **Spillway / outlet:** to pass surplus water safely.

Benefits

- Provides life-saving irrigation (protective irrigation) in dry spells and for a second crop.
- Recharges ground water and revives wells and bore wells.
- Reduces soil erosion and downstream flooding.
- Supports livestock, fish culture and domestic needs.

10.3 Catchment and Runoff

Catchment (watershed) is the area of land from which all the rain water drains to a common point (the outlet).

> **Runoff** is that part of rainfall which, after meeting interception, evaporation and infiltration, flows over the land surface to streams and storage structures.

The **runoff coefficient (C)** is the fraction of rainfall that becomes runoff. It depends on soil type, land use, slope, rainfall intensity and antecedent moisture.

10.4 Runoff Computation

(a) Rational Method

Rational method (peak runoff rate):

$Q = (C \times I \times A) / 360$ (Q in cumec, I in mm/h, A in ha)

or $Q = C \times I \times A$ (Q in m^3/s, I in m/s, A in m^2)

where C = runoff coefficient, I = rainfall intensity, A = catchment area.

(b) Volume of runoff

Runoff volume:

Runoff volume = C × Rainfall × Catchment area

$V\ (m^3) = C \times P(m) \times A(m^2)$ or $V = C \times P(mm) \times A(ha) \times 10$

(1 mm of rain over 1 ha = 10 m^3.)

(c) SCS Curve Number method (volume)

SCS-CN method:

$Q = (P - 0.2\ S)^2 / (P + 0.8\ S)$ for $P > 0.2\ S$; else $Q = 0$

$S = (25400 / CN) - 254$ (S in mm)

where P = rainfall (mm), Q = runoff (mm), CN = curve number (0–100).

Typical runoff coefficients

Table 10.1 — Typical runoff coefficients (C)

Surface / land use	Runoff coefficient
Rooftops / paved surfaces	0.75–0.95
Hard, bare, compacted soil	0.50–0.70
Clay / heavy soil, cultivated	0.40–0.60
Loam soil, cultivated	0.30–0.50
Sandy / permeable soil, cultivated	0.15–0.30
Pasture / grassland	0.10–0.40
Forest / dense vegetation	0.05–0.25

10.5 Text Diagram — Rainwater Harvesting Systems

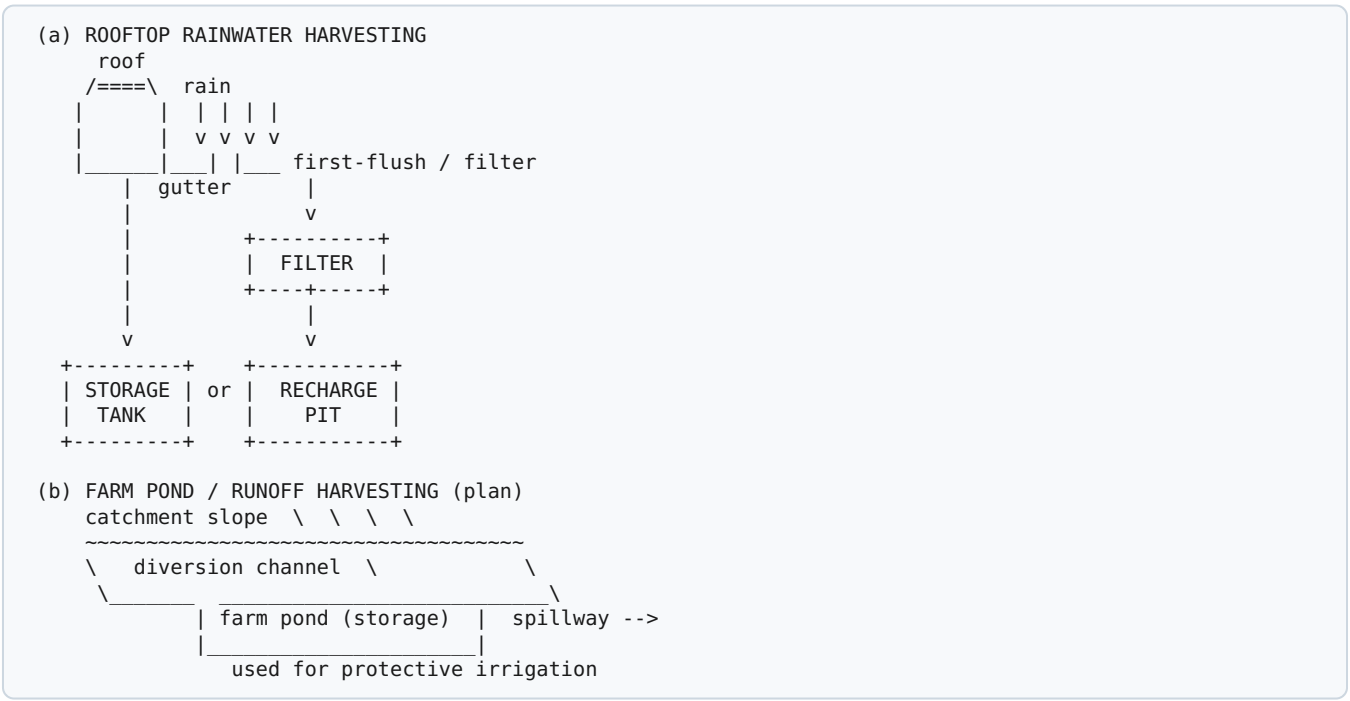


Fig. 10.1 — Rooftop harvesting and runoff water harvesting layout

10.6 Worked Numericals — Runoff Computation

Worked Example 10.1 — Runoff volume from a catchment

A 20 ha catchment with a runoff coefficient of 0.35 receives 600 mm of rainfall in a season. Find the volume of runoff available for harvesting.

Step 1: Runoff volume = C × P × A = 0.35 × 0.600 m × 200,000 m².

Step 2: = 0.35 × 0.600 × 200,000 = **42,000 m³**.

Alternative: runoff depth = 0.35 × 600 = 210 mm; volume = 210 mm × 20 ha × 10 = 42,000 m³. This is enough to irrigate about 14 ha at a 150 mm depth per irrigation, or to give several protective waterings.

Worked Example 10.2 — Peak runoff by the rational method

For a 50 ha catchment with C = 0.45, the rainfall intensity for the time of concentration is 80 mm/h. Find the peak runoff rate.

Step 1: Q = (C × I × A) / 360.

Step 2: Q = (0.45 × 80 × 50) / 360 = 1800 / 360 = **5.0 cumec** (m³/s).

This peak flow is used to size the spillway and the diversion channel of the harvesting structure.

Worked Example 10.3 — Rooftop harvesting for a farmhouse

A farmhouse has a roof area of 150 m². The annual rainfall is 900 mm and the runoff coefficient for the roof is 0.85. How much water can be harvested in a year, and how many days of supply is that at 100 litres per day?

Step 1 (volume): V = 0.85 × 0.900 m × 150 m² = **114.75 m³** = 114,750 litres.

Step 2 (days of supply): 114,750 / 100 = **1147 days** (about 3 years of domestic need).

A first-flush diversion (the first 1–2 mm of rain discarded) and a filter are essential to keep the stored water clean.

Worked Example 10.4 — SCS Curve Number runoff

A watershed has a curve number of 75. For a storm of 100 mm, find the runoff.

Step 1 (potential retention): S = (25400/75) − 254 = 338.67 − 254 = 84.67 mm.

Step 2 (initial abstraction): 0.2S = 16.93 mm; P (100) > 0.2S, so runoff occurs.

Step 3: Q = (P − 0.2S)²/(P + 0.8S) = (100 − 16.93)² / (100 + 67.74) = (83.07)² / 167.74 = 6901 / 167.74 = **41.1 mm**.

So of the 100 mm rain, about 41 mm becomes runoff and 59 mm is stored/infiltrated.

10.7 Catchment Yield and Design Rainfall

The amount of water a harvesting structure can collect depends on the **catchment yield** — the runoff produced by the design rainfall. For reliable design, a rainfall value exceeded in most years (e.g., a 75% dependable rainfall) is chosen rather than the highest recorded value.

Dependable rainfall & yield:

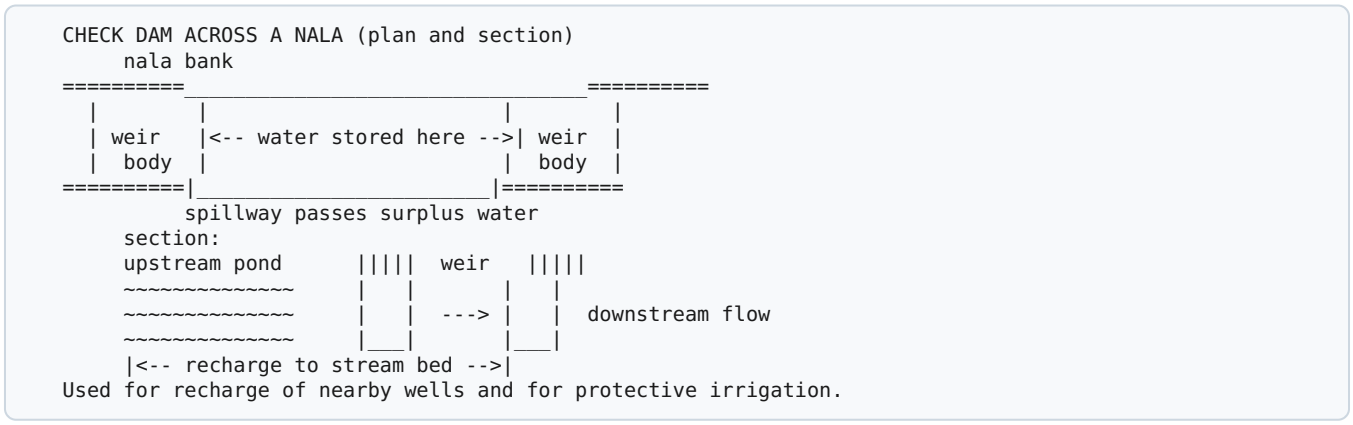
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75% dependable rainfall = the rainfall exceeded in 75 out of 100 years (arranged in descending order, take the value at the 75th position).
Catchment yield = C × design rainfall × catchment area
```

10.8 Storage vs Recharge Harvesting

Table 10.2 — Storage vs recharge harvesting

Point	Storage harvesting	Recharge harvesting
Water is	Stored in tanks/ponds	Led into the ground
Purpose	Direct use (irrigation, domestic)	Replenish ground water
Loss	Evaporation, seepage	None to evaporation after recharge
Structures	Farm ponds, tanks, check dams	Recharge pits, trenches, shafts, wells
Best where	Water needed on the surface	Over-exploited aquifers, wells to revive

10.9 Text Diagram — Check Dam for Runoff Harvesting



Used for recharge of nearby wells and for protective irrigation.

Fig. 10.2 — Check dam for runoff water harvesting and recharge

10.10 Rainwater Harvesting in Tamil Nadu

- **Rooftop harvesting** is mandatory for new buildings in many TN urban local bodies; farmhouses and schools use it for domestic supply and recharge.
- **Farm ponds and tanks** harvest runoff for protective irrigation in the dryland blocks of the state.
- **In-situ conservation** (contour bunding, trenching, basin listing) retains rain where it falls.
- **Check dams and percolation ponds** recharge the over-exploited aquifers.
- The **Kudimaramathu** programme revives traditional tank systems by de-silting and repairing them.

Worked Example 10.5 — Water harvesting potential of a school roof

A school has a roof of 500 m². The average annual rainfall is 800 mm and the roof runoff coefficient is 0.85. Find the annual harvest and the daily supply it can support at 500 litres/day.

Step 1: V = 0.85 × 0.800 × 500 = 340 m³ = 340,000 litres.

Step 2: 340,000 / 500 = **680 days** of supply.

Exam Focus — Lecture 10

Key points

- Rainwater harvesting = collecting and storing rain from a catchment for use or recharge.
- Types: rooftop, runoff (surface), in-situ conservation and recharge structures.
- Runoff coefficient = fraction of rainfall that becomes runoff.
- Rational method: Q = C I A / 360 (cumec, mm/h, ha).
- Runoff volume = C × P × A; 1 mm over 1 ha = 10 m³.
- SCS-CN: Q = (P − 0.2S)²/(P + 0.8S), S = (25400/CN) − 254.

Short-answer questions

Q1. Define rainwater harvesting and name its types.

Model answer: Collection and storage of rain water from a catchment for use or recharge.

Types: rooftop, runoff (surface) water harvesting, in-situ rainwater conservation and ground water recharge structures.

Q2. What is the runoff coefficient? Give typical values for a roof and for forest land.
Model answer: The runoff coefficient is the fraction of rainfall that becomes runoff. Typical: roof 0.75–0.95; forest 0.05–0.25.

Q3. Write the rational formula and explain each term.
Model answer: Q = C I A / 360, where Q = peak runoff (cumec), C = runoff coefficient, I = rainfall intensity (mm/h), A = catchment area (ha).

Long-answer questions

L1. Explain runoff water harvesting with its components and benefits, and describe how runoff is computed.
Outline: Concept → components (catchment, conveyance, storage, distribution, spillway) → benefits → runoff computation (rational method, volume method, SCS-CN) → worked examples → TN context (farm ponds under watershed schemes).

L2. Describe the methods of rainwater harvesting suitable for Tamil Nadu, with examples.
Outline: RWH concept → rooftop harvesting for farmhouses and schools → runoff harvesting in farm ponds and tanks → in-situ conservation (contour bunding, trenching) → recharge structures → benefits → conclusion.

11 Ground Water Recharge Techniques

11.1 Ground Water and the Need for Recharge

> **Ground water** is the water held in the saturated zone of soil and rock below the water table. **Recharge** is the process by which water moves downward from the surface into the saturated zone, replenishing ground water.

Ground water is over-exploited in many parts of Tamil Nadu (notably Coimbatore, Erode, Salem, Dharmapuri and the Cauvery delta margins), causing falling water tables, drying of wells and intrusion of saline water in coastal areas. **Artificial recharge** is the deliberate addition of surface water to the ground water reservoir to arrest this decline.

Objectives of artificial recharge

- Raise the water table and sustain well/bore-well yields.
- Store surplus monsoon runoff underground for the dry season.
- Prevent land subsidence and salt-water intrusion in coastal areas.
- Improve ground water quality by dilution.
- Reduce flood peaks and soil erosion.

11.2 Factors Governing Recharge

- **Availability of source water:** surplus runoff or treated water.
- **Permeability of the soil and aquifer:** sandy soils recharge fast; clay soils poorly.
- **Depth to water table:** a deeper water table gives more space to recharge.
- **Quality of source water:** should be free of excessive silt and pollutants.
- **Slope and topography:** flatter land favours infiltration; slopes favour trenching.
- **Geological structure:** presence of fractures and porous strata.

11.3 Methods / Techniques of Ground Water Recharge

(a) Surface spreading methods

- **Recharge pit:** a pit (typically 2×2×2 m to 3×3×3 m) filled with boulders, gravel and coarse sand, into which roof or surface runoff is led and from which it percolates.
- **Recharge trench:** a long, shallow, back-filled trench used where permeable strata lie close to the surface.
- **Percolation pond / tank:** a shallow pond for infiltrating stored water (Lecture 12).
- **Flooding / spreading basins:** allowing water to spread over a large area to infiltrate.
- **Stream-channel modification / check dams:** small dams across streams to hold water and increase infiltration (also called **recharge dams**).

(b) Sub-surface methods

- **Recharge shaft:** a vertical shaft (with or without a bore) used where an impermeable layer overlies the aquifer.
- **Recharge well / injection well:** a well into which water is injected directly into the aquifer.
- **Dug well recharge:** leading filtered surface water into an existing open well.
- **Defunct / dry bore-well recharge:** using old bore wells to inject water to deeper aquifers.

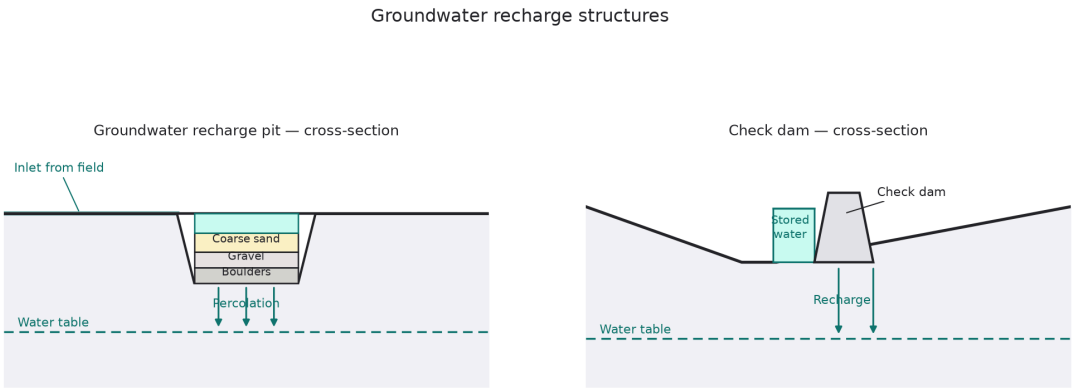


Figure 11.1 Groundwater recharge structures

Recharge pit and check dam cross-sections.

(c) Combination methods

- Recharge pit + shaft; trench + well; check dam + recharge well.

11.4 Recharge Structures — Comparison Table

Table 11.1 — Ground water recharge structures

Structure	Where used	Typical size	Method
Recharge pit	Rooftop / small catchments, permeable strata near surface	2×2×2 m to 3×3×3 m	Surface spreading
Recharge trench	Long strips where permeable layer is shallow	0.5–1 m wide, 1–2 m deep, 10–30 m long	Surface spreading
Percolation pond	Permeable soil, available land	Depends on catchment	Surface spreading
Recharge shaft	Impermeable layer over aquifer	1–3 m dia., 5–15 m deep	Sub-surface
Recharge/injection well	Direct aquifer recharge	As per aquifer depth	Sub-surface
Check dam / recharge dam	Streams and nalas	Small weir across channel	Surface + sub-surface
Dug-well recharge	Existing open wells	Existing well	Sub-surface

11.5 Text Diagram — Recharge Structures

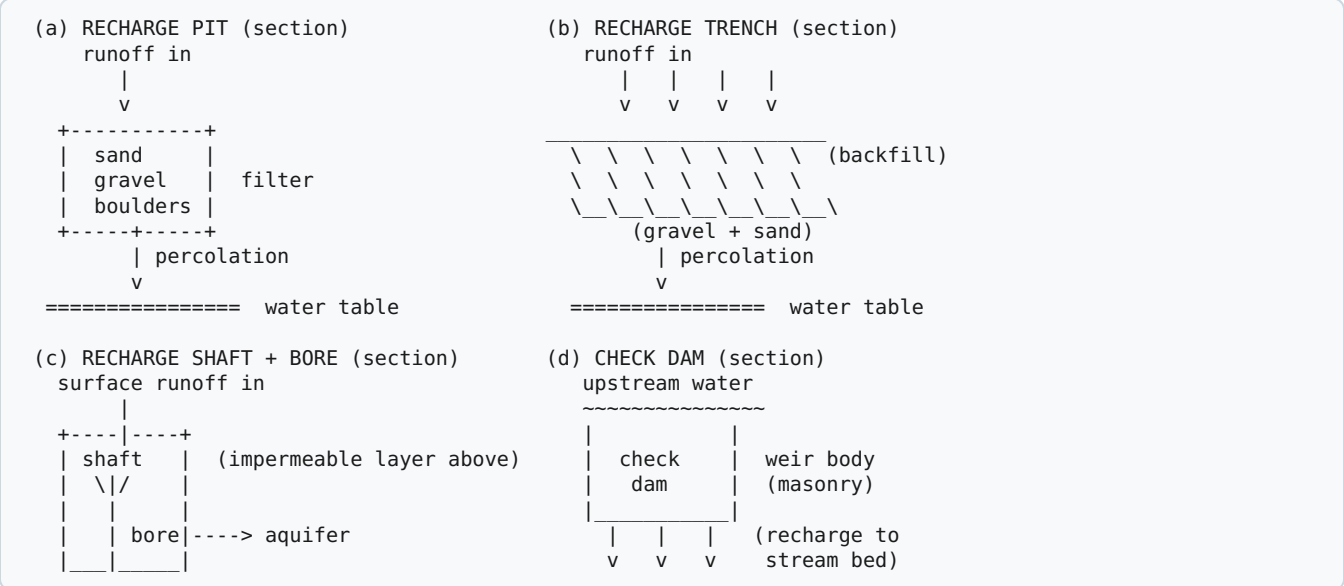


Fig. 11.1 — Recharge pit, trench, shaft and check dam

11.6 Design Considerations and Precautions

- **Filter media:** layers of boulders, gravel and coarse sand to remove silt and prevent clogging.
- **Desilting chamber:** to settle silt before water enters the pit/well.
- **First-flush diversion:** to discard the first dirty runoff.
- **Maintenance:** periodic removal of silt and replacement of filter media; clogging is the main failure mode.
- **Water quality:** do not recharge with sewage or highly polluted water.

- **Recharge capacity:** governed by the infiltration rate of the strata and the available head.

Indicative recharge rate:

```
Volume recharged per unit time ≈ Infiltration rate × Surface (or percolating) area
For a pit: Q = k × A (k = infiltration rate, A = effective percolating area of the pit walls and base)
```

Worked Example 11.1 — Capacity of a recharge pit

A recharge pit is 3 m × 3 m in plan and 3 m deep, filled with filter media to 2.5 m depth. If the void space in the filter media is 30%, find the water the pit can hold before percolation.

Step 1 (volume of media): 3 × 3 × 2.5 = 22.5 m³.

Step 2 (void volume): 22.5 × 0.30 = **6.75 m³** = 6750 litres.

This buffer water percolates into the ground over a few hours, recharging the aquifer below.

Worked Example 11.2 — Recharge from a check dam

A check dam holds a water spread of 500 m² over a permeable bed with an average infiltration rate of 0.5 m/day. Find the daily recharge.

Step 1: Recharge = infiltration rate × area = 0.5 × 500 = **250 m³/day**.

Over a 30-day storage period this is about 7500 m³ recharged — enough to significantly raise nearby well levels.

11.7 Site Selection for Recharge Structures

- Choose a site where the aquifer is over-exploited and wells have dried up.
- The soil/strata below should be permeable (sandy or fractured rock).
- There should be a dependable source of surplus runoff nearby.
- The depth to the water table should be adequate to accommodate recharge.
- Avoid sites where the source water is polluted or highly silt-laden.
- Prefer locations upstream of existing wells so that recharged water benefits them.

11.8 Recharge vs Injection

Table 11.2 — Surface spreading vs injection (sub-surface)

Point	Surface spreading	Injection (sub-surface)
Water enters	Vertically through the soil	Directly into the aquifer
Filtration	Natural, through soil (good quality)	Minimal; water must be clean
Clogging	Of the surface layer	Of the well screen (serious)
Structures	Pits, trenches, ponds, basins	Shafts, injection wells, bore wells
Best where	Permeable soil, shallow water table	Impermeable layer over aquifer, deep aquifer

11.9 Role in Watershed Development

Recharge structures are the "underground storage" component of watershed development. By capturing monsoon runoff that would otherwise flow away, they raise water levels in nearby open wells and bore wells, extend the period of well irrigation, and reduce the cost of pumping

— a major benefit in the hard-rock areas of Tamil Nadu (Coimbatore, Salem, Dharmapuri, Krishnagiri).

11.10 Worked Numerical — Recharge trench capacity

Worked Example 11.3 — Recharge from a trench

A recharge trench is 20 m long, 0.8 m wide and 1.5 m deep, back-filled with media of 35% void space. Find the water it can hold.

Step 1 (volume of trench): 20 × 0.8 × 1.5 = 24 m³.

Step 2 (void volume): 24 × 0.35 = **8.4 m³** = 8400 litres.

This buffer percolates into the ground over a few hours, recharging the aquifer along the trench.

11.11 Checklist for a Recharge Scheme

1. Assess the aquifer and the water-table trend.
2. Identify and measure the source of surplus runoff.
3. Select the structure type (pit/trench/shaft/well) by the strata.
4. Size the structure for the available runoff.
5. Provide desilting and first-flush arrangements.
6. Plan maintenance (de-silting, media replacement).

Exam Focus — Lecture 11

Key points

- Ground water recharge = downward movement of water into the saturated zone.
- Artificial recharge arrests falling water tables and stores monsoon surplus underground.
- Surface methods: pit, trench, percolation pond, spreading basin, check dam.
- Sub-surface methods: recharge shaft, injection/recharge well, dug-well and defunct bore-well recharge.
- Filter media and desilting prevent clogging, the main failure mode.
- Recharge depends on permeability, depth to water table and source-water quality.

Short-answer questions

Q1. What is artificial recharge and why is it needed?
Model answer: Artificial recharge is the deliberate addition of surface water to the ground water reservoir. It is needed to raise falling water tables, sustain well yields, store monsoon surplus underground and prevent saline intrusion in coastal areas.

Q2. Name any four ground water recharge structures.
Model answer: Recharge pit, recharge trench, recharge shaft, recharge/injection well (also percolation pond and check dam).

Q3. What is the function of the filter media in a recharge pit?
Model answer: The layers of boulders, gravel and coarse sand filter out silt and debris, preventing clogging of the pit and keeping the recharged water clean.

Long-answer questions

L1. Describe the techniques of ground water recharge with diagrams and state where each is used.

Outline: Concept and objectives → surface spreading methods (pit, trench, pond, check dam) → sub-surface methods (shaft, injection well, dug-well recharge) → diagrams → comparison table → design and maintenance → TN context.

L2. Explain the need for ground water recharge in Tamil Nadu and the structures suitable for hard-rock and coastal areas.

Outline: Over-exploitation and falling water tables → hard-rock areas (recharge shafts, pits, check dams) → coastal areas (to check saline intrusion) → role of watershed programmes → maintenance → conclusion.

12 Farm Ponds and Percolation Ponds

12.1 Farm Pond — Definition and Purpose

> **Farm pond** is a small water-storage structure constructed on a farm (usually by excavating or by embanking a natural depression) to store surface runoff for irrigation, livestock and other farm uses.

Farm ponds are a core component of watershed development and rainwater harvesting in Tamil Nadu, and are promoted under state and central schemes (for example, the **Pradhan Mantri Krishi Sinchayee Yojana (PMKSY)** and Tamil Nadu watershed programmes). They provide **life-saving (protective) irrigation** during dry spells.

Purposes / benefits

- Store runoff for protective irrigation and a possible second crop.
- Provide water for livestock, fish culture and domestic use.
- Recharge ground water (if permeable) and raise nearby well levels.
- Reduce soil erosion and downstream flood peaks.
- Provide a micro-climate and support horticulture/plantation around it.

12.2 Types of Farm Ponds

- **Embankment (dug-out-cum-embankment) pond:** formed by damming a natural depression or drainage line with an earthen embankment; water is stored by the embankment.
- **Excavated (dug-out) pond:** formed by excavating a pit in the ground where there is no natural site for an embankment; stores runoff collected from the surrounding area.
- **Spring-fed or stream-fed pond:** fed by a spring or a diversion from a stream.
- **Off-stream pond:** filled by diverting runoff from a channel or field, not lying on the main stream.

12.3 Percolation Pond

> **Percolation pond** is a shallow pond constructed specifically to impound runoff and allow it to percolate into the ground, thereby recharging ground water, rather than for storing water for direct use.

- **Purpose:** ground water recharge; reviving wells and bore wells in the vicinity.
- **Location:** on permeable soil or over fractured rock, often in a natural drainage line.
- **Design:** shallow (depth usually 1–3 m), with a large surface area to maximise infiltration; often provided with a silt trap.
- **Difference from a farm pond:** a farm pond stores water for use; a percolation pond infiltrates water for recharge.

12.4 Farm Pond vs Percolation Pond

Table 12.1 — Farm pond vs percolation pond

Point	Farm pond	Percolation pond

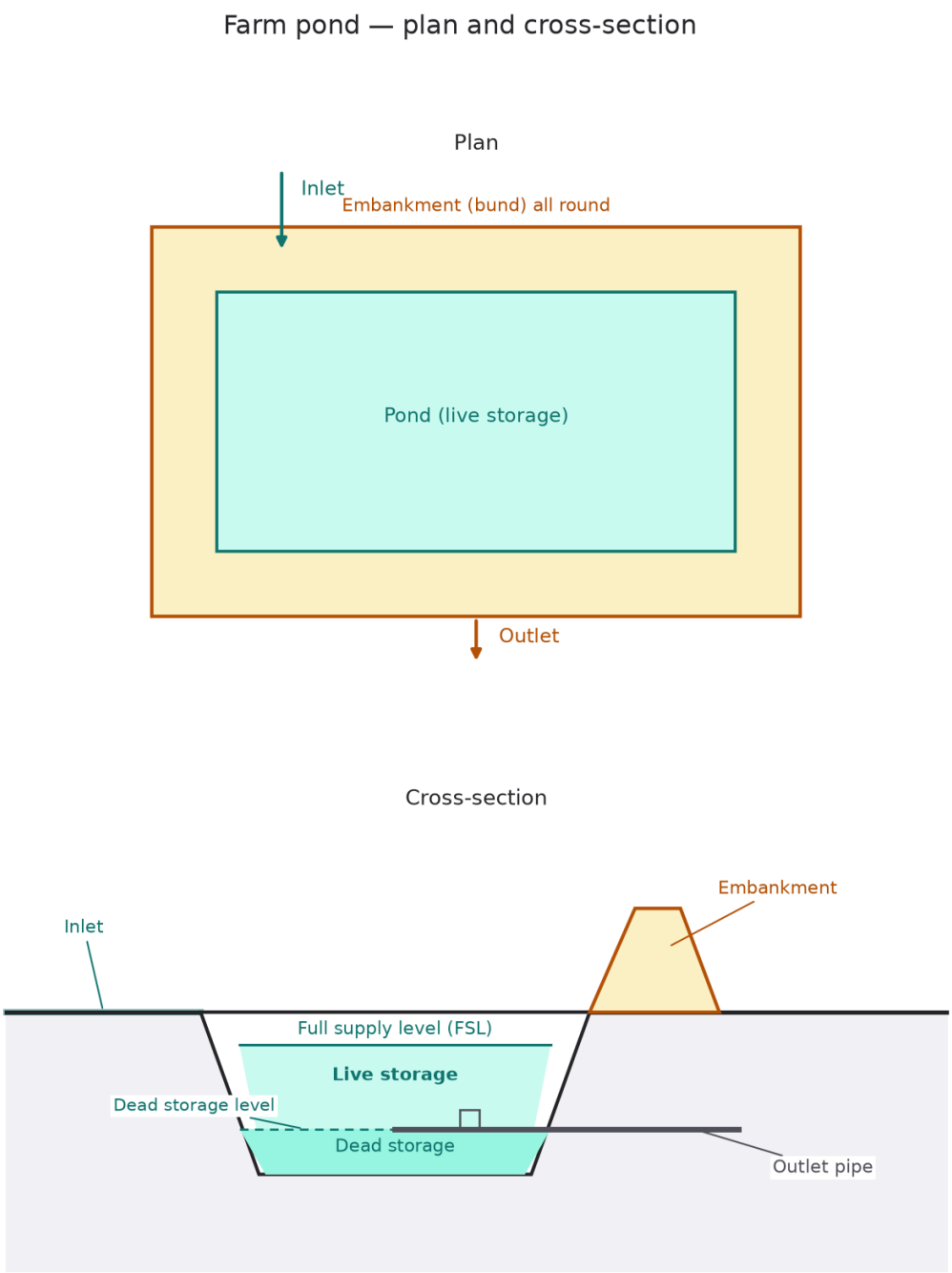


Figure 12.1 Farm pond — plan and cross-section

Embankment, inlet, outlet, live storage and dead storage.

Main purpose	Store water for irrigation/farm use	Recharge ground water
Water retention	Held for a long time	Allowed to percolate quickly
Depth	Deeper (2–5 m)	Shallow (1–3 m)
Base/soil	Fairly impermeable base to reduce seepage	Permeable base to encourage seepage
Seepage	Undesirable (loss)	Desirable (the objective)
Typical use	Protective irrigation, fish, livestock	Well recharge, base-flow improvement

12.5 Components of a Farm Pond

- **Catchment:** the area contributing runoff.
- **Embankment (bund):** earthen dam with top width, side slopes and freeboard.
- **Cut-off trench:** below the embankment to prevent seepage under it.
- **Spillway:** to pass surplus water safely (usually a grassed or masonry chute).
- **Outlet / sluice:** to draw water for use.
- **Silt trap:** to keep silt out of the storage.
- **Fencing and pitching:** for protection of the embankment.

12.6 Text Diagram — Farm Pond and Percolation Pond

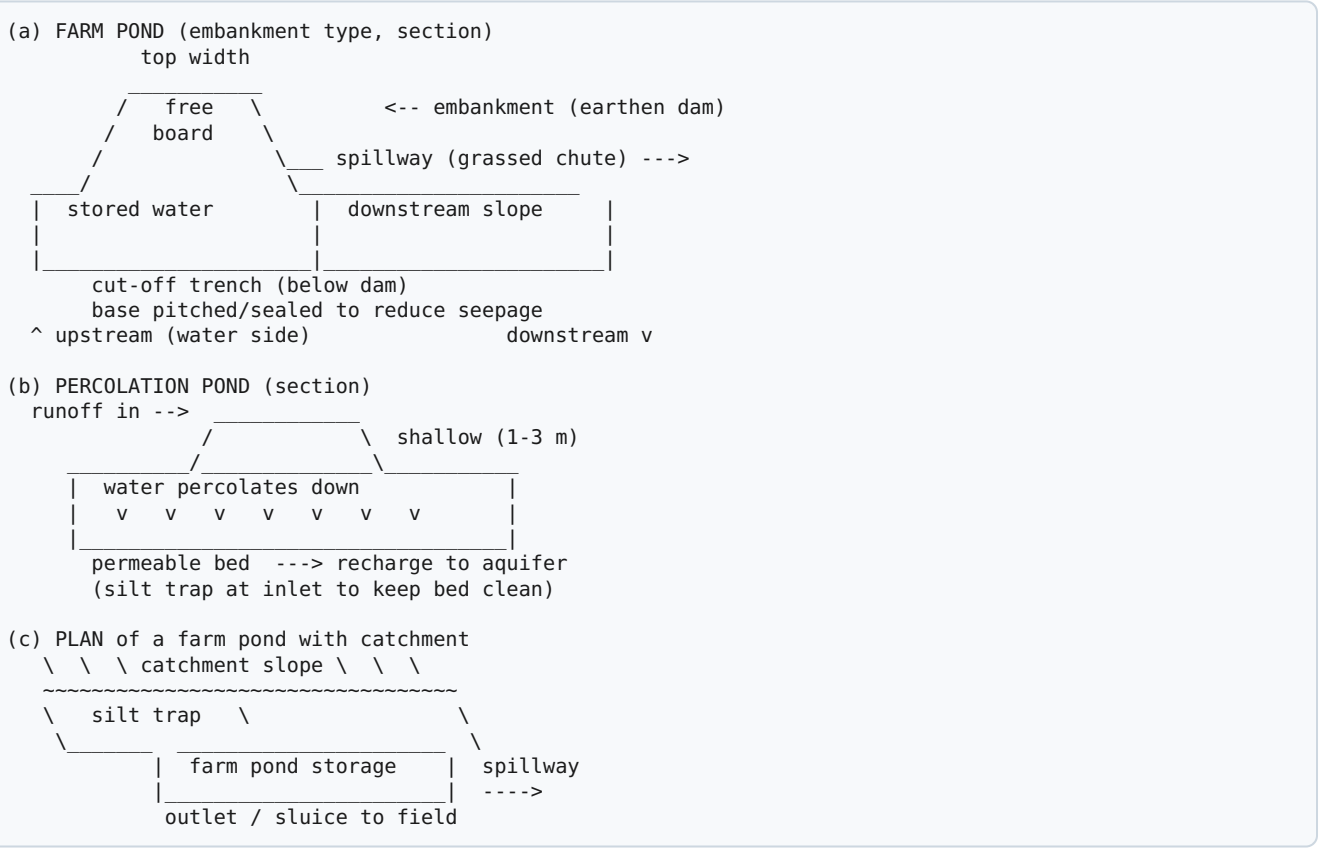


Fig. 12.1 — Farm pond and percolation pond (section and plan)

12.7 Design of a Farm Pond

Storage capacity of a farm pond (from contour areas):

$V = (h/3)[\ A_0 + A_n + 4(A_1+A_3+...) + 2(A_2+A_4+...)\ ]$ (prismoidal / Simpson's)

For a simple rectangular/trapezoidal pond: $V \approx \frac{1}{2}(A_0 + A_5) \times \text{depth}$

Catchment requirement:

```
Catchment area needed = Storage volume / (C × design rainfall × 10)  ... (ha)
```
Embankment volume: V = ½(top + base) × height × length

Worked Example 12.1 — Storage capacity of a farm pond

A farm pond has water-spread areas of 200, 600, 1100 and 1500 m² at depths of 0, 0.5, 1.0 and 1.5 m. Find the storage volume.

Given: h = 0.5 m; areas 200, 600, 1100, 1500 m² (4 areas, 3 intervals — odd number of intervals, so apply Simpson's to all and use the trapezoidal form for the last if needed). Since there are 4 ordinates (even), apply Simpson's to the first 3 and trapezoidal for the last strip.

Simpson's part (200, 600, 1100): V_1 = (0.5/3)[(200 + 1100) + 4(600)] = (0.5/3)[1300 + 2400] = 0.1667 × 3700 = 616.7 m³.

Trapezoidal last strip (1100 to 1500): V_2 = (0.5/2)(1100 + 1500) = 0.25 × 2600 = 650 m³.

Total: V = 616.7 + 650 = **1266.7 m³** (about 1.27 million litres).

Worked Example 12.2 — Catchment area needed for a farm pond

A farm pond of 2000 m³ capacity is to be filled from a catchment with a runoff coefficient of 0.30 in an area receiving 500 mm of rainfall. Find the catchment area required.

Step 1: Runoff per hectare = C × rainfall × 10 = 0.30 × 500 × 10 = 1500 m³/ha.

Step 2: Catchment needed = 2000 / 1500 = **1.33 ha**.

So a catchment of about 1.3 ha is needed to fill a 2000 m³ pond in a 500 mm rainfall year. In a lower-rainfall year, the pond fills partially.

Worked Example 12.3 — Embankment earthwork

A farm-pond embankment is 30 m long, 0.6 m top width, 1.5 m high with side slopes 1:1 both sides. Find the earthwork volume.

Step 1 (base width): 0.6 + 2(1 × 1.5) = 3.6 m.

Step 2 (area): ½(0.6 + 3.6) × 1.5 = ½ × 4.2 × 1.5 = 3.15 m².

Step 3 (volume): 3.15 × 30 = **94.5 m³**.

12.8 Site Selection and Precautions

- Choose a site with a good catchment, impermeable or suitable soil, and a natural depression.
- Avoid sites on very permeable soils for storage ponds (water is lost).
- Provide a safe spillway — the most common cause of pond failure is inadequate spillway capacity.
- Provide a cut-off trench and pitching to control seepage and erosion.
- Keep a silt trap and de-silt periodically.
- Protect the embankment with grass/turf and fencing.

12.9 Economics and Management of Farm Ponds

- **Cost:** mainly the earthwork (excavation/embankment) and the spillway; farm ponds are subsidised under PMKSY and state schemes.
- **Benefit:** protective irrigation raises and stabilises yields; recharge raises well levels; fish culture and horticulture add income.
- **Desilting:** remove accumulated silt periodically to restore capacity; spread it on the fields as fertile soil.
- **Multiple use:** irrigation, livestock, fish, and recharge from the same pond.
- **Seepage control:** where the base is permeable and storage is the aim, seal it; where recharge is the aim, keep it permeable (percolation pond).

12.10 Design Steps for a Farm Pond

1. Select the site (catchment, soil, natural depression, spillway possibility).
2. Estimate the catchment yield (C × rainfall × area).
3. Fix the storage capacity required (for the intended irrigation/livestock use).
4. Design the pond dimensions and compute storage by the contour-area method.
5. Design the embankment (height, top width, side slopes, freeboard).
6. Design the spillway for the peak inflow.
7. Provide a silt trap, outlet and pitching.

Protective irrigation from a farm pond:

```
Area irrigated = Storage volume / (depth of each watering)
Number of waterings = Storage / (area × depth per watering)
1 hectare-cm of water = 100 m³.
```

Worked Example 12.4 — Protective irrigation from a farm pond

A farm pond stores 1500 m³. If a protective watering of 5 cm depth is given to a 1 ha plot, how many waterings can be given?

Step 1 (water per watering): 1 ha × 5 cm = 10,000 m² × 0.05 m = 500 m³.

Step 2 (number of waterings): 1500 / 500 = **3 waterings**.

Three life-saving waterings at critical growth stages can make the difference between crop failure and a reasonable yield in a dry spell.

12.11 Text Diagram — Types of Farm Ponds

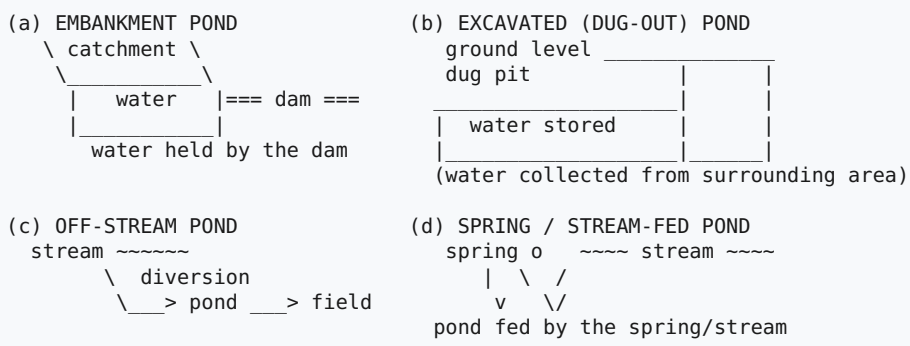


Fig. 12.2 — Types of farm ponds

Worked Example 12.5 — Catchment for a 3000 m³ pond

A pond of 3000 m³ capacity is to be filled from a catchment with C = 0.25 in a 600 mm rainfall area. Find the catchment area required.

Step 1: Runoff per ha = 0.25 × 600 × 10 = 1500 m³/ha.

Step 2: Catchment = 3000 / 1500 = **2.0 ha**.

Exam Focus — Lecture 12

Key points

- Farm pond stores runoff for farm use; percolation pond infiltrates it for recharge.
- Farm pond types: embankment, excavated, spring/stream-fed, off-stream.
- Components: catchment, embankment, cut-off trench, spillway, outlet, silt trap.
- Percolation ponds are shallow and on permeable beds; farm ponds are deeper with sealed bases.
- Storage volume by contour-area (prismoidal) method.
- Catchment needed = storage / (C × rainfall × 10).
- Spillway capacity is critical for safety.

Short-answer questions

Q1. Define a farm pond and state its purposes.

Model answer: A farm pond is a small storage structure on a farm to hold surface runoff. Purposes: protective irrigation, livestock and domestic water, fish culture, ground water recharge and erosion/flood reduction.

Q2. Differentiate a farm pond and a percolation pond.

Model answer: A farm pond stores water for use, is deeper (2–5 m) and has a fairly sealed base. A percolation pond is shallow (1–3 m) on a permeable bed and its purpose is to percolate water for ground water recharge.

Q3. Name the components of a farm pond.

Model answer: Catchment, earthen embankment, cut-off trench, spillway, outlet/sluice, silt trap, and pitching/fencing.

Long-answer questions

L1. Describe farm ponds — their types, components, design and benefits — with a diagram.

Outline: Definition and purpose → types → components (with section diagram) → design (storage capacity, catchment, embankment) → site selection → benefits → TN scheme context.

L2. Explain the role of farm ponds and percolation ponds in watershed development in Tamil Nadu.

Outline: Watershed concept → farm ponds for protective irrigation → percolation ponds for recharge → comparison → worked capacity/catchment examples → PMKSY/TN schemes → conclusion.

13 Irrigation — Measurement of Flow in Open Channels

13.1 Need for Measuring Flow

Efficient irrigation requires knowing how much water is actually flowing. Measurement of flow (discharge) is needed to:

- Allocate water fairly among farms and distributaries.
- Assess irrigation efficiency and water charges (abiana / water rate).
- Design channels, weirs, flumes and outlets.
- Monitor the performance of an irrigation or drainage system.

> **Discharge (Q)** is the volume of water passing a given cross-section per unit time, expressed in m^3/s (cumec) or litres/s.

13.2 Methods of Measuring Flow in Open Channels

- **Velocity-area method:** measure the cross-sectional area and the mean velocity; Q = A × V.
- **Weirs and notches:** flow over a raised obstruction (weir) or through an opening (notch).
- **Orifices:** flow through a submerged opening.
- **Flumes (Parshall, cut-throat):** flow through a calibrated converging-diverging section.
- **Current meters, floats, tracer methods** for velocity.

13.3 Velocity-Area Method

Velocity-area method:

```
Q = A × V
A = cross-sectional area of flow (m²)
V = mean velocity of flow (m/s)
For a rectangular channel: A = width × depth of flow.
For a trapezoidal channel: A = (b + z·y) y  [b = bed width, z = side slope (H:V), y = depth]
Mean velocity from a current meter: V = a + b N  (N = revolutions per second)
For subdivision: Q = Σ (A_i × V_i) over the cross-section.
```

13.4 Weirs and Notches

> **Weir** is a raised structure (usually masonry/concrete) built across an open channel over which water flows. **Notch** is the opening in a weir or tank wall through which water flows; a weir is a large notch, a notch is a small weir.

Common types

- **Rectangular weir/notch** — sharp-edged horizontal crest.
- **Cippoletti weir** — a trapezoidal weir with side slopes of ¼:1 (H:V) to compensate for end contractions, so no contraction correction is needed.
- **V-notch (triangular) weir** — a V-shaped notch, very accurate at low flows (commonly 90°).

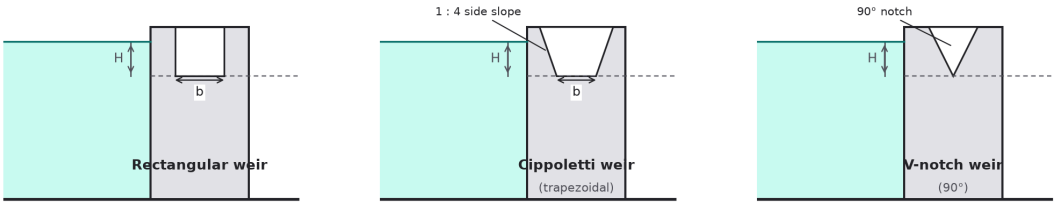


Figure 13.1 Flow measuring structures (weirs)

Rectangular, Cippoletti (trapezoidal) and V-notch weirs in elevation.

- **Broad-crested weir** — a wide crest; flow is critical at the crest.

13.5 Weir / Notch Formulas

Rectangular weir (general):

$Q = (2/3)\ C_d\ \sqrt{(2g)}\ L\ H^{(3/2)}$

With $C_d \approx 0.62$: $Q = 1.84\ L\ H^{(3/2)}$ (L, H in m; Q in m^3/s)

Rectangular weir with end contractions (Francis):

$Q = 1.84\ (L - 0.1\ n\ H)\ H^{(3/2)}$ (n = number of end contractions; 2 for a notch in mid-channel)

Cippoletti (trapezoidal) weir:

$Q = 1.86\ L\ H^{(3/2)}$ (no contraction correction needed)

V-notch (triangular) weir, general:

$Q = (8/15)\ C_d\ \sqrt{(2g)}\ \tan(\theta/2)\ H^{(5/2)}$

For $\theta = 90°$ and $C_d \approx 0.6$: $Q = 1.417\ H^{(5/2)}$

In all weir formulas H is the head (depth of water above the crest/apex), measured upstream, clear of the drawdown.

13.6 Orifice

Orifice is a small opening (usually in a plate or wall) through which water flows and which is completely submerged by the water on the upstream side.

Orifice discharge:

$Q = C_d \cdot a \cdot \sqrt{(2\ g\ h)}$

a = area of the orifice = $\pi d^2/4$

h = head of water above the centre of the orifice

C_d = coefficient of discharge (≈ 0.60–0.65 for a sharp-edged orifice)

For a mouthpiece or short tube, $C_d \approx 0.80–0.85$.

13.7 Parshall Flume

Parshall flume is a calibrated open-channel flow-measuring structure with a converging section, a narrow throat and a diverging section; discharge is obtained from a single upstream depth reading (free flow).

Parshall flume (free flow):

$Q = C \cdot H_a$ (also written $Q = K\ H_a$)

H_a = upstream (primary) depth reading

C (or K) and n depend on throat width (see table).

Free-flow condition: submergence ratio H_b/H_a below the transition value (0.5 for 1–3 in; 0.6 for 6–9 in; 0.7 for 1–8 ft).

Table 13.1 — Parshall flume free-flow constants (Q in m^3/s, H in m) [USBR Water Measurement Manual]

Throat width	Constant K (m^3/s)	Exponent n
2.54 cm (1 in)	0.0479	1.55

7.62 cm (3 in)	0.141	1.55
15.24 cm (6 in)	0.264	1.58
22.86 cm (9 in)	0.393	1.53
30.48 cm (1 ft)	0.624	1.522
60.96 cm (2 ft)	1.135	1.55
91.44 cm (3 ft)	1.612	1.566
182.88 cm (6 ft)	2.919	1.595

13.8 Measuring Structures — Comparison Table

Table 13.2 — Flow-measuring structures with formulas

Structure	Formula	Best suited to
Velocity-area method	Q = A × V	Any channel; large flows
Rectangular weir	Q = 1.84 (L − 0.1 n H) H^1.5	Moderate to large flows
Cippoletti weir	Q = 1.86 L H^1.5	Channels with side walls (no contraction correction)
V-notch (90°)	Q = 1.417 H^2.5	Small / low flows (accurate)
Orifice	Q = C_d a √(2gh)	Outlets, sluices, small openings
Parshall flume	Q = K H^n	Field channels; silt-laden water; free flow

13.9 Text Diagram — Weirs, Notch, Orifice and Flume

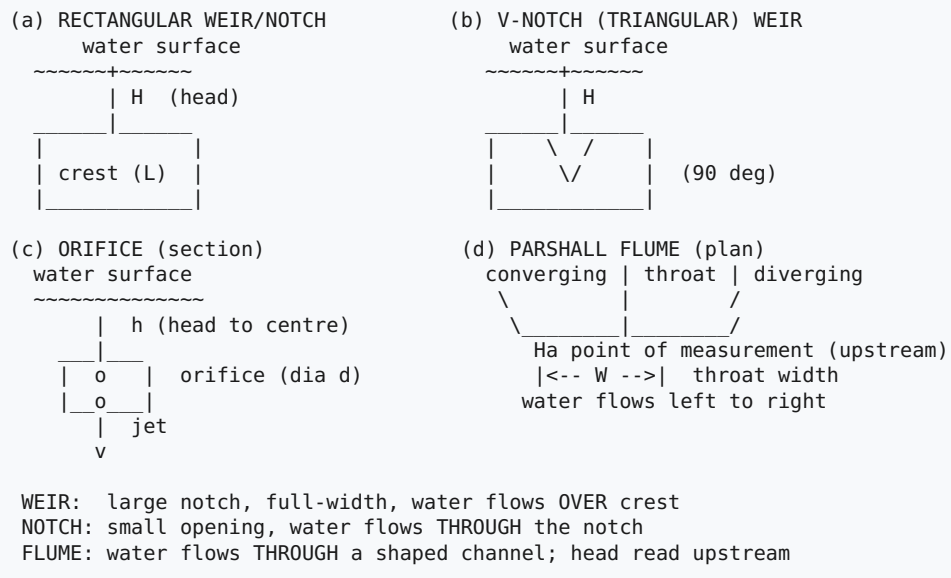


Fig. 13.1 — Rectangular weir, V-notch, orifice and Parshall flume

13.10 Worked Numericals

Worked Example 13.1 — Velocity-area method

A rectangular field channel is 1.5 m wide and the depth of flow is 0.60 m. The mean velocity measured by a current meter is 0.50 m/s. Find the discharge.

Step 1 (area): A = 1.5 × 0.60 = 0.90 m².

Step 2 (discharge): Q = A × V = 0.90 × 0.50 = **0.45 m³/s** (450 litres/s).

Worked Example 13.2 — Rectangular weir with end contractions

A rectangular weir 2.0 m long has a head of 0.30 m and two end contractions. Find the discharge (take C_d = 0.62).

Step 1: Q = 1.84 (L − 0.1 n H) H^1.5.

Step 2: H^1.5 = 0.30^1.5 = 0.1643.

Step 3: Q = 1.84 (2.0 − 0.1 × 2 × 0.30) × 0.1643 = 1.84 (2.0 − 0.06) × 0.1643 = 1.84 × 1.94 × 0.1643.

Step 4: Q = 1.84 × 0.3187 = **0.586 m³/s**.

Without end contractions (suppressed weir), Q = 1.84 × 2.0 × 0.1643 = 0.605 m³/s. Contractions reduce the effective length.

Worked Example 13.3 — Cippoletti weir

A Cippoletti weir has a crest length of 1.5 m and a head of 0.25 m. Find the discharge.

Step 1: Q = 1.86 L H^1.5.

Step 2: H^1.5 = 0.25^1.5 = 0.125.

Step 3: Q = 1.86 × 1.5 × 0.125 = **0.349 m³/s**.

Worked Example 13.4 — V-notch (90°) weir

A 90° V-notch has a head of 0.30 m. Find the discharge (C_d = 0.60).

Step 1: Q = 1.417 H^2.5.

Step 2: H^2.5 = 0.30^2.5 = 0.04930.

Step 3: Q = 1.417 × 0.04930 = **0.0699 m³/s** (69.9 litres/s).

Check by the general formula: Q = (8/15)×0.60×4.429×tan45°×0.04930 = 1.417×0.04930 = 0.0699 m³/s. ✓

Worked Example 13.5 — Discharge through an orifice

Water flows through a sharp-edged circular orifice of diameter 5 cm under a head of 1.5 m. Take C_d = 0.62. Find the discharge.

Step 1 (area): a = πd²/4 = 3.1416 × (0.05)²/4 = 0.001963 m².

Step 2 (velocity head): √(2gh) = √(2 × 9.81 × 1.5) = √29.43 = 5.425 m/s.

Step 3: Q = 0.62 × 0.001963 × 5.425 = **0.00660 m³/s** (6.60 litres/s).

Worked Example 13.6 — Parshall flume discharge

A 15.24 cm (6-inch) Parshall flume is in free flow with an upstream depth of 0.30 m. Find the discharge.

Step 1: For a 6-inch flume, Q = 0.264 H^1.58 (Q in m³/s, H in m).

Step 2: H^1.58 = 0.30^1.58 = 0.1493.

Step 3: Q = 0.264 × 0.1493 = **0.0394 m³/s** (39.4 litres/s).

Parshall flumes are popular in field channels because they pass silt and trash and need only one head reading. The free-flow constants above are from the USBR Water Measurement Manual.

13.11 Measurement of Head and Precautions

- The head (H) is measured upstream of the weir/notch, clear of the drawdown curve (at least 4–5 times H upstream).
- The crest must be level and sharp-edged for the standard coefficients to apply.
- The approach channel should be straight and free from turbulence.
- The weir must be **fully ventilated** (air under the nappe) for free flow.
- For a Parshall flume, read the head at the specified upstream point and ensure **free flow** (submergence below the transition value).

13.12 Notch / Weir vs Orifice vs Flume

Table 13.3 — Choosing a flow-measuring device

Device	Head-discharge relation	Best for	Main drawback
Rectangular weir	Q ∝ H^1.5	Moderate to large flows	Needs end-contraction correction
Cippoletti weir	Q ∝ H^1.5	Channels with side walls	Fixed side slope
V-notch weir	Q ∝ H^2.5	Small / low flows	Not for large flows
Orifice	Q ∝ H^0.5	Outlets, sluices	Head measured to centre
Parshall flume	Q ∝ H^n	Silt-laden field channels	Submergence correction if drowned

13.13 Text Diagram — Parshall Flume Dimensions

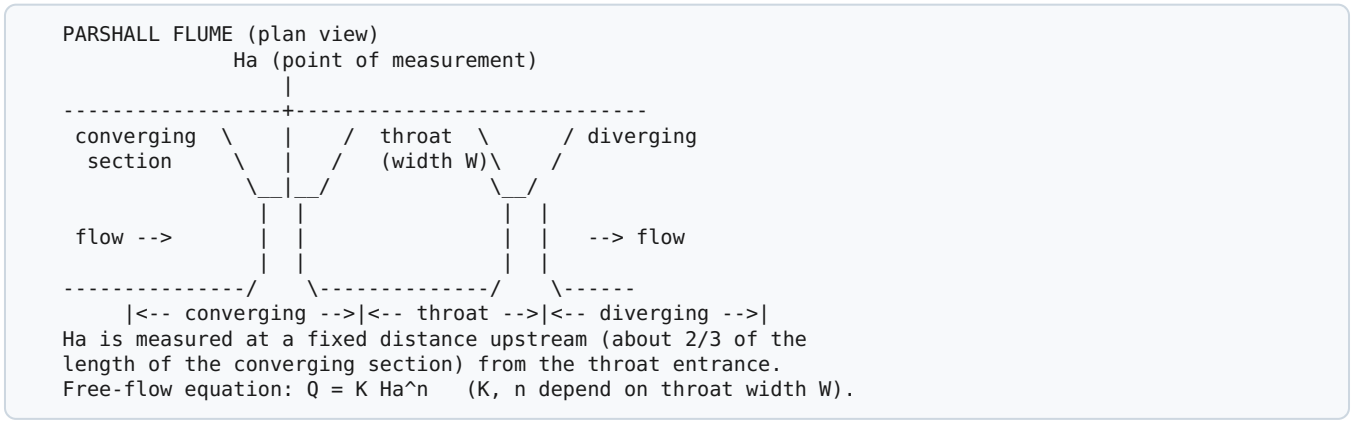


Fig. 13.2 — Parshall flume layout and point of head measurement

13.14 More Worked Numericals

Worked Example 13.7 — Rectangular weir without end contractions

A suppressed rectangular weir is 3.0 m long with a head of 0.40 m. Find the discharge.

Step 1: Q = 1.84 L H^1.5 (no contraction term).

Step 2: H^1.5 = 0.40^1.5 = 0.2530.

Step 3: Q = 1.84 × 3.0 × 0.2530 = **1.397 m³/s**.

Worked Example 13.8 — Discharge from a known area and velocity

A trapezoidal channel has a bed width of 1.0 m, a depth of 0.5 m and side slopes of 1:1 (H:V). The mean velocity is 0.4 m/s. Find the discharge.

Step 1 (area): A = (b + z y) y = (1.0 + 1 × 0.5) × 0.5 = 1.5 × 0.5 = 0.75 m².

Step 2: Q = A V = 0.75 × 0.4 = **0.30 m³/s**.

Worked Example 13.9 — Parshall flume with a 1-ft throat

A 30.48 cm (1-ft) Parshall flume reads a head of 0.30 m in free flow. Find the discharge.

Step 1: For a 1-ft flume, Q = 0.624 H^1.522.

Step 2: H^1.522 = 0.30^1.522 = 0.1600.

Step 3: Q = 0.624 × 0.1600 = **0.0998 m³/s** (about 100 litres/s).

Exam Focus — Lecture 13

Key points

- Discharge = volume per unit time; Q = A V for the velocity-area method.
- Weir = large notch (flow over crest); notch = small weir (flow through opening).
- Rectangular weir: Q = 1.84(L − 0.1 n H)H^1.5.
- Cippoletti weir: Q = 1.86 L H^1.5 (no contraction correction).
- V-notch 90°: Q = 1.417 H^2.5; best for low flows.
- Orifice: Q = C_d a √(2gh).
- Parshall flume: Q = K H^n; K and n depend on throat width.

Short-answer questions

Q1. Differentiate a weir and a notch.

Model answer: A weir is a raised masonry/concrete structure across a channel over which water flows; a notch is an opening through which water flows. A weir is a large notch and a notch is a small weir.

Q2. Write the formula for discharge over a rectangular weir with end contractions and explain the terms.

Model answer: Q = 1.84(L − 0.1 n H)H^1.5, where L = crest length, n = number of end contractions, H = head over the crest, Q in m³/s.

Q3. Why is a Cippoletti weir preferred in some channels?
Model answer: Its sloping sides (¼:1) compensate for end contractions, so no contraction correction is needed and discharge is obtained directly from Q = 1.86 L H^1.5.

Long-answer questions

L1. Describe the different methods of measuring flow in open channels, with formulas and a diagram.
Outline: Need for measurement → velocity-area method → weirs (rectangular, Cippoletti) and notches (V-notch) → orifice → Parshall flume → comparison table → worked examples.

L2. Explain the measurement of irrigation water by weirs and notches, and work out a numerical example for each.
Outline: Definition of weir/notch → rectangular weir formula with contractions → Cippoletti weir → V-notch → head measurement precautions → worked examples → conclusion.

14 Water Duty and Irrigation Efficiencies

14.1 Base Period, Crop Period and Duty

Crop period is the time from sowing to harvest of a crop. **Base period (B)** is the time from the first watering before sowing to the last watering before harvest, expressed in days.

Duty (D) of water is the area of land (in hectares) that can be irrigated by a continuous supply of one cubic metre per second (1 cumec) of water. It is expressed in **hectares per cumec** (ha/cumec).

Delta (Δ) is the total depth of water required by a crop during its base period, expressed in metres or centimetres; also called the consumptive use or the depth of watering.

14.2 Relation between Duty, Delta and Base Period

Duty – Delta – Base period:

$\Delta = 8.64\ B / D$ (Δ in m, B in days, D in ha/cumec)

$D = 8.64\ B / \Delta$

Also: $Q = A / D$ (A = area in ha, Q in cumec)

where $8.64 = (86{,}400\ \text{s/day} \times 1\ \text{day}) / (10{,}000\ \text{m}^2/\text{ha})$, i.e. 8.64.

Derivation: 1 cumec flowing for B days gives 86,400 × B m³. Spread over D hectares (= 10,000 D m²), the depth = 86,400 B / (10,000 D) = 8.64 B / D metres.

14.3 Factors Affecting Duty

- **Soil:** sandy soils need more water (low duty); clay soils hold water (higher duty).
- **Climate:** hot, dry, windy climate lowers duty; humid climate raises it.
- **Crop:** water-loving crops (sugarcane, rice) have low duty; dry crops (millets) higher duty.
- **Method of irrigation:** drip/sprinkler improve duty; flood irrigation lowers it.
- **Base period:** longer base period generally gives higher duty.
- **Efficiency of the system:** well-maintained channels raise duty.
- **Ground water table:** a high water table reduces the water requirement (raises duty).

14.4 Irrigation Efficiencies

Irrigation efficiency is the ratio (usually as a percentage) of the water beneficially used by the crop to the water diverted or delivered.

Table 14.1 — Types of irrigation efficiency

Efficiency	Formula	Meaning
Conveyance efficiency (E_c)	$E_c = (W_2/W_1) \times 100$	W_1 = water diverted at head; W_2 = water delivered to the field
Application efficiency (E_a)	$E_a = (W_s/W_2) \times 100$	W_s = water stored in the root zone; W_2 = water delivered to field
Storage efficiency (E_s)	$E_s = (W_s/W_n) \times 100$	W_n = water needed to fill the root zone to field capacity

Distribution efficiency (E_d)	$E_d = (1 - d/D) \times 100$	d = average deviation from mean depth; D = average depth stored
Water-use efficiency (E_u)	$E_u = (W_b/W_2) \times 100$	W_b = water beneficially used (incl. leaching, etc.)
Overall / project efficiency (E_o)	$E_o = E_c \times E_a$ (as fractions) $\times 100$	Combined conveyance and application efficiency

Typical values: conveyance efficiency of an unlined earthen channel 60–70%; well-lined channel 85–95%; application efficiency for surface (flood) irrigation 40–60%, sprinkler 70–80%, drip 85–95%. Overall project efficiency in many Indian systems is 30–45%.

14.5 Text Diagram — Water Losses and Efficiency

```
Head works (dam / weir)
     |  W1 = water diverted
     v
[CANAL]  ---- conveyance losses (seepage, evaporation, leakage) ----
     |  W2 = water delivered to the field   E_conveyance = W2/W1
     v
[FIELD]  ---- application losses (deep percolation, runoff, evaporation) ----
     |  Ws = water stored in root zone      E_application = Ws/W2
     v
[CROP]   beneficially used water

OVERALL EFFICIENCY  Eo = Ec x Ea = (W2/W1) x (Ws/W2) = Ws/W1
(the W2 cancels; overall efficiency = stored / diverted)

Drip/fertigation raises both Ea and crop water-use efficiency,
and hence the duty of water.
```

Fig. 14.1 — Conveyance and application losses and the overall efficiency

14.6 Worked Numericals

Worked Example 14.1 — Delta from duty and base period

The duty of water for a crop is 1000 ha/cumec and the base period is 120 days. Find the delta.

Step 1: $\Delta = 8.64\ B / D$.

Step 2: $\Delta = (8.64 \times 120) / 1000 = 1036.8 / 1000 =$ **1.037 m = 103.7 cm**.

Worked Example 14.2 — Duty from delta and base period

A crop requires a total depth of 90 cm of water over a base period of 100 days. Find the duty.

Step 1: $\Delta = 0.90$ m; $D = 8.64\ B / \Delta = (8.64 \times 100) / 0.90$.

Step 2: $D = 864 / 0.90 =$ **960 ha/cumec**.

Worked Example 14.3 — Discharge required for a given area

An area of 480 ha is to be irrigated with a duty of 960 ha/cumec. Find the discharge required at the canal head.

Step 1: $Q = A / D = 480 / 960 =$ **0.50 cumec**.

If the conveyance efficiency is only 70%, the discharge to be released at the head works would be 0.50 / 0.70 = 0.71 cumec.

Worked Example 14.4 — Overall irrigation efficiency

Water diverted at the head of a canal is 10 cumec; the water delivered to the fields is 7 cumec; the water stored in the root zone is 4.9 cumec. Find the conveyance, application and overall efficiencies.

Step 1 (conveyance): $E_c = (7/10) \times 100 =$ **70 %**.

Step 2 (application): $E_a = (4.9/7) \times 100 =$ **70 %**.

Step 3 (overall): $E_o = E_c \times E_a = 0.70 \times 0.70 = 0.49 =$ **49 %**.

Check: overall = stored/diverted = 4.9/10 = 49%. ✓

Worked Example 14.5 — Distribution efficiency

In a field the depths of water stored at several points are 60, 55, 65, 50 and 70 mm. Find the distribution efficiency.

Step 1 (mean depth): D = (60+55+65+50+70)/5 = 300/5 = 60 mm.

Step 2 (deviations): |60−60|=0, |55−60|=5, |65−60|=5, |50−60|=10, |70−60|=10; sum = 30.

Step 3 (average deviation): d = 30/5 = 6 mm.

Step 4: $E_d = (1 - d/D) \times 100 = (1 - 6/60) \times 100 = (1 - 0.10) \times 100 =$ **90 %**.

14.7 Improving Duty and Efficiency in Practice

- **Lining of channels** to cut seepage (raises conveyance efficiency).
- **Land levelling** and proper grading to spread water evenly (raises application efficiency).
- **Drip and sprinkler irrigation** (raises application efficiency and duty; in TN, drip with fertigation is widely promoted in sugarcane, banana and vegetables).
- **Rotational (warabandi) water supply** and volumetric measurement.
- **Mulching** and improved agronomic practices to reduce evaporation.
- **Conjunctive use** of canal and ground water.

14.8 Consumptive Use and Irrigation Requirement

> **Consumptive use (CU)** is the total water used by a crop for transpiration and tissue building, plus water evaporated from the adjacent soil.

Field irrigation requirement:

```
Net irrigation requirement (NIR) = Consumptive use − effective rainfall
Gross irrigation requirement (GIR) = NIR / application efficiency
Number of irrigations = NIR / (depth per irrigation)
```

If the gross requirement exceeds the net requirement, the difference is the water lost in conveyance and application — the reason efficiencies matter.

14.9 Relation between Duty and Efficiency

Duty and efficiency:

```
Improving efficiency raises the effective duty:
```

```
Effective duty = Actual duty × overall efficiency (as a fraction)
or, discharge required = (A / D) / E_o
```

Worked Example 14.6 — Discharge at the head works

An area of 600 ha is to be irrigated with a duty of 1200 ha/cumec at the field. If the overall efficiency of the system is 45%, find the discharge to be released at the head works.

Step 1 (field discharge): $Q_f = A / D = 600 / 1200 = 0.50$ cumec.

Step 2 (head discharge): $Q_h = Q_f / E_o = 0.50 / 0.45 =$ **1.11 cumec**.

More than twice the field requirement must be released because of system losses — the case for lining and efficient methods.

14.10 Text Diagram — Delta and Duty

```
DELTA (total depth of water applied over the base period)
  |<---------------- base period B (days) ---------------->|
  surface ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
  depth    |<-- delta (total depth, m) -->|
  profile  vvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvv

DUTY (area irrigated by 1 cumec flowing continuously)
  1 cumec  =====>  [================ D hectares ================]
  A higher duty means MORE area served by the same water (better).
```

Fig. 14.2 — Concept of delta and duty

14.11 Worked Numerical — Net and gross irrigation requirement

Worked Example 14.7 — Irrigation requirement of a crop

A crop has a consumptive use of 60 cm over its season and receives 20 cm of effective rainfall. If the application efficiency is 65%, find the net and gross irrigation requirements.

Step 1 (NIR): NIR = 60 − 20 = **40 cm**.

Step 2 (GIR): GIR = NIR / efficiency = 40 / 0.65 = **61.5 cm**.

So about 61.5 cm of water must be delivered to the field to meet the 40 cm net need — the extra 21.5 cm is lost in application.

14.12 Worked Numerical — Duty of a tank command

Worked Example 14.8 — Area irrigated by a tank

A tank releases 0.30 cumec for a base period of 90 days. If the delta for the crop is 60 cm, find the area irrigated.

Step 1: $D = 8.64\ B / \Delta = (8.64 \times 90) / 0.60 = 777.6 / 0.60 = 1296$ ha/cumec.

Step 2: Area = $Q \times D = 0.30 \times 1296 =$ **388.8 ha**.

This is the ayacut that the tank can support with the given release and delta.

Exam Focus — Lecture 14

Key points

- Duty = hectares irrigated by 1 cumec; delta = depth of water needed by the crop.

- Δ = 8.64 B / D (Δ in m, B in days, D in ha/cumec).
- Q = A / D for the discharge needed.
- Conveyance efficiency = delivered/diverted; application = stored/delivered.
- Overall efficiency = $E_c \times E_a$ = stored/diverted.
- Distribution efficiency = (1 − d/D) × 100.
- Drip/sprinkler and channel lining improve duty and efficiency.

Short-answer questions

Q1. Define duty and delta, and give the relation between them.
Model answer: Duty is the area (ha) irrigated by 1 cumec; delta is the total depth of water required by the crop. Relation: Δ = 8.64 B/D, where B is the base period in days.

Q2. Name any four factors affecting the duty of water.
Model answer: Soil type, climate, type of crop, method of irrigation (also base period, ground water table and system efficiency).

Q3. Define conveyance and application efficiency.
Model answer: Conveyance efficiency = (water delivered to field / water diverted) × 100. Application efficiency = (water stored in root zone / water delivered to field) × 100.

Long-answer questions

L1. Explain the terms duty, delta and base period, derive the relation between them, and work a numerical.
Outline: Definitions → derivation of Δ = 8.64B/D → factors affecting duty → worked example → practical significance.

L2. Describe the different irrigation efficiencies with formulas and explain how each can be improved.
Outline: Definitions (conveyance, application, storage, distribution, water-use, overall) with formulas → typical values → worked example → improvement measures (lining, levelling, drip/sprinkler, warabandi) → conclusion.

15 Agricultural Drainage — Need, Surface and Subsurface Systems, Drainage Coefficient

15.1 What is Agricultural Drainage?

Drainage is the removal of excess water and salts from the root zone of the soil, whether from the surface or from below the surface, to maintain favourable conditions for crop growth.

Excess water may come from heavy rain, over-irrigation, seepage from canals, high ground water table, or flooding. Drainage removes this surplus so that air can re-enter the soil pores.

15.2 Need for Drainage

- **Water-logging control:** excess water fills soil pores, excluding air and harming roots.
- **Aeration:** roots and soil organisms need oxygen; drainage restores it.
- **Salt removal:** drainage leaches and carries away harmful salts (essential in the Cauvery delta and coastal saline tracts).
- **Improved trafficability:** allows timely tillage and harvesting.
- **Higher yields:** prevents yield loss due to water-logging and salinity.
- **Health:** reduces mosquito breeding and water-borne disease.
- **Reclamation:** makes saline/alkaline and water-logged land cultivable.

Water-logging is the condition in which the soil pores are saturated with water and free air is absent. It raises the water table close to the surface, reduces oxygen and crop yield, and can bring salts up by capillary rise.

15.3 Classification of Drainage Systems

- **Surface drainage:** removes excess water flowing over the soil surface.
- **Subsurface (sub-soil) drainage:** removes excess water from within the soil, lowering the water table.
- **Vertical drainage:** pumping from wells/bore wells to lower the water table.
- **Bio-drainage:** using deep-rooted trees (e.g., Eucalyptus) to transpire excess soil water.

15.4 Surface Drainage Systems

Surface drainage is needed on flat, poorly drained land with heavy soils and high rainfall. Types:

- **Random ditch system:** ditches follow the natural depressions to remove water from scattered wet spots. Suited to undulating land.
- **Bedding system:** the field is formed into beds with dead furrows between them; water drains along the furrows. Suited to flat, heavy land.
- **Parallel ditch system:** parallel open ditches at regular intervals, usually with land graded toward them. Suited to flat land with uniform soil.
- **Field-ditch (or cross-slope) system:** ditches laid along the slope with outlets into a main drain. Suited to sloping land.

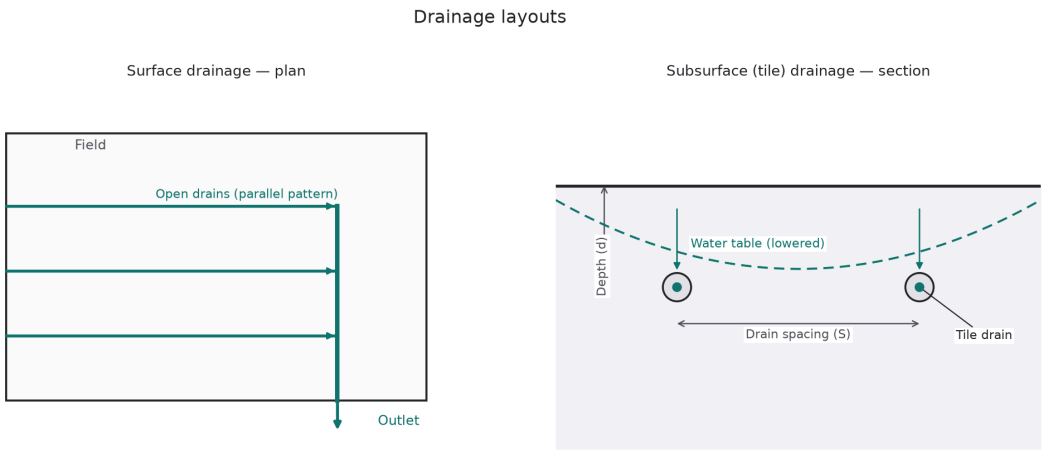


Figure 15.1 Drainage layouts

Surface (open drains) plan and subsurface (tile drain) section.

Surface drain design

Surface drainage capacity (rational method):

$Q = (C \times I \times A) / 360$ (Q in cumec, I in mm/h, A in ha)

Drain cross-section: $A = b \cdot y + z \cdot y^2$ (trapezoidal, z = side slope H:V)

$Q = A \times V$ (Manning's $V = (1/n)\ R^{(2/3)}\ S^{(1/2)}$)

15.5 Subsurface Drainage Systems

Subsurface drainage lowers the water table below the root zone. Types:

- **Tile drains:** buried pipes (clay, concrete or perforated plastic) laid at a designed depth and spacing to intercept and carry away ground water.
- **Mole drains:** unlined channels formed by pulling a mole plough through the soil; cheap but temporary; suited to heavy clay soils.
- **Open ditches (deep):** used where pipes are not economical.
- **Vertical drains (tube wells):** pumping to lower the water table.

Drain depth and spacing

- **Depth:** usually 0.9–1.5 m below the surface (deeper for deep-rooted crops).
- **Spacing:** depends on soil permeability, drain depth and the drainage coefficient; closer on less permeable soils. Spacing commonly ranges from 8–15 m (heavy clay) to 30–60 m (permeable soil).

Hooghoudt's equation (drain spacing):

$S^2 = (4\ k\ h^2 + 8\ k\ D\ h) / q$

S = drain spacing (m); k = hydraulic conductivity (m/day);
h = height of water table above the drain (m);
D = depth of the impermeable layer below the drain (m);
q = drainage coefficient (m/day).

15.6 Text Diagram — Drainage Layouts

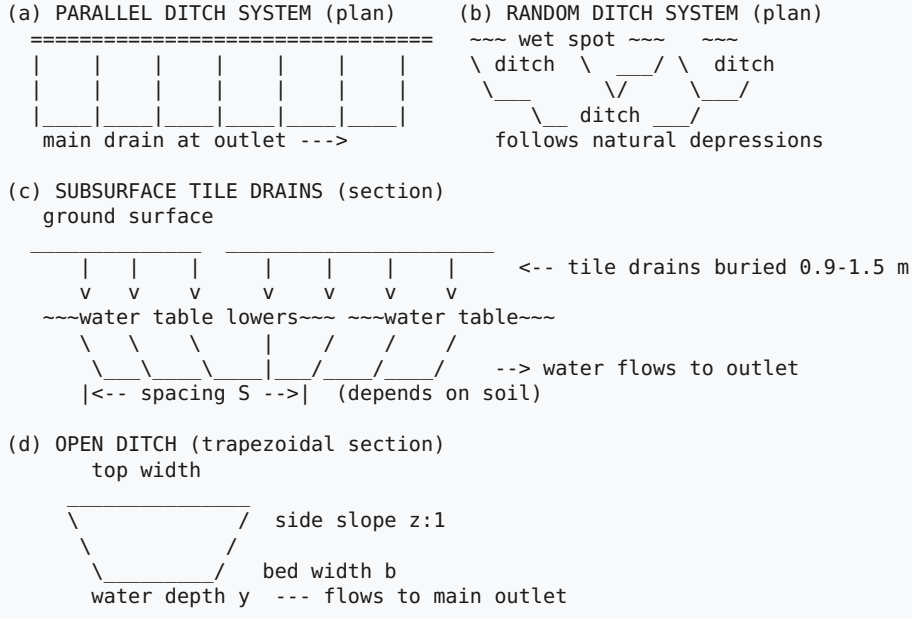


Fig. 15.1 — Surface drainage layouts and subsurface tile drains

15.7 Drainage Coefficient

Drainage coefficient (q) is the rate at which water must be removed from a drainage area by the drainage system to prevent water-logging; it is expressed as the depth of water removed in 24 hours (mm/day or cm/day).

Drainage coefficient & discharge:

```
q = (volume of water to be removed per day) / (drained area)     [mm/day]
Q (m³/s) = (q × A) / 8640
where q = drainage coefficient (mm/day), A = area (ha), 8640 = conversion (86,400 s/day ÷ 10).
For q in cm/day: Q = (q × A) / 864.
```

Table 15.1 — Typical drainage coefficients

Condition / region	Drainage coefficient (mm/day)
Arid / semi-arid areas	1–5
Sub-humid areas	5–10
Humid areas (moderate rainfall)	10–20
Humid tropical areas (heavy rainfall)	20–50
Rice (paddy) lands / tank command	Higher values used; often 20–50 or more
Irrigated lands (salinity control)	1–5 (leaching requirement)

The drainage coefficient is chosen from the peak rainfall/irrigation excess expected; a higher coefficient means a larger (costlier) drain. It is the single most important design parameter for a drainage system.

15.8 Worked Numericals

Worked Example 15.1 — Drainage discharge from a coefficient

A drainage system must remove a drainage coefficient of 10 mm/day from an area of 120 ha. Find the design discharge.

Step 1: Q = (q × A) / 8640.

Step 2: Q = (10 × 120) / 8640 = 1200 / 8640 = **0.139 m³/s** (139 litres/s).

Worked Example 15.2 — Drainage coefficient from a required discharge

A drain of capacity 0.25 m³/s serves a catchment of 200 ha. Find the drainage coefficient in mm/day.

Step 1: q = (Q × 8640) / A.

Step 2: q = (0.25 × 8640) / 200 = 2160 / 200 = **10.8 mm/day**.

Worked Example 15.3 — Drain spacing by Hooghoudt's equation

For a tile-drain system, the hydraulic conductivity is 0.6 m/day, the water table is to be lowered so that h = 0.5 m above the drains, the impermeable layer is D = 2.0 m below the drains, and the drainage coefficient is q = 0.01 m/day. Find the drain spacing.

Step 1: $S^2 = (4\ k\ h^2 + 8\ k\ D\ h) / q$.

Step 2: $4\ k\ h^2 = 4 \times 0.6 \times (0.5)^2 = 4 \times 0.6 \times 0.25 = 0.60$.

Step 3: $8\ k\ D\ h = 8 \times 0.6 \times 2.0 \times 0.5 = 4.80$.

Step 4: $S^2 = (0.60 + 4.80) / 0.01 = 5.40 / 0.01 = 540$.

Step 5: $S = \sqrt{540} =$ **23.2 m**.

So tile drains about 23 m apart, at the designed depth, will lower the water table as required.

15.9 Surface vs Subsurface Drainage

Table 15.2 — Surface vs subsurface drainage

Point	Surface drainage	Subsurface drainage
Removes	Water flowing over the soil surface	Water within the soil (lowers water table)
Best for	Flat, heavy, poorly drained land, high rainfall	High water table, irrigated and saline land
Structures	Open ditches, beds and furrows	Tile drains, mole drains, deep ditches
Land used	Takes land out of production (ditches)	Underground; no land lost
Cost	Lower	Higher
Maintenance	Frequent (silt, weeds)	Lower (but clogging possible)

15.10 Land Drainage for Salinity Control

- In irrigated areas, salts accumulate in the root zone as water evaporates.
- **Leaching requirement:** the extra water that must be applied to wash salts below the root zone.
- Leaching works only if there is **adequate drainage** to carry the salty water away; otherwise the water table rises and water-logging worsens.
- This is the basis of the "irrigation + drainage" combination needed in the Cauvery delta and coastal saline tracts.

Leaching requirement (simplified):

$LR = (EC_i) / (EC_d)$ (as a fraction)

EC_i = electrical conductivity of the irrigation water; EC_d = of the drainage water.

15.11 Text Diagram — Drainage and Salinity Control

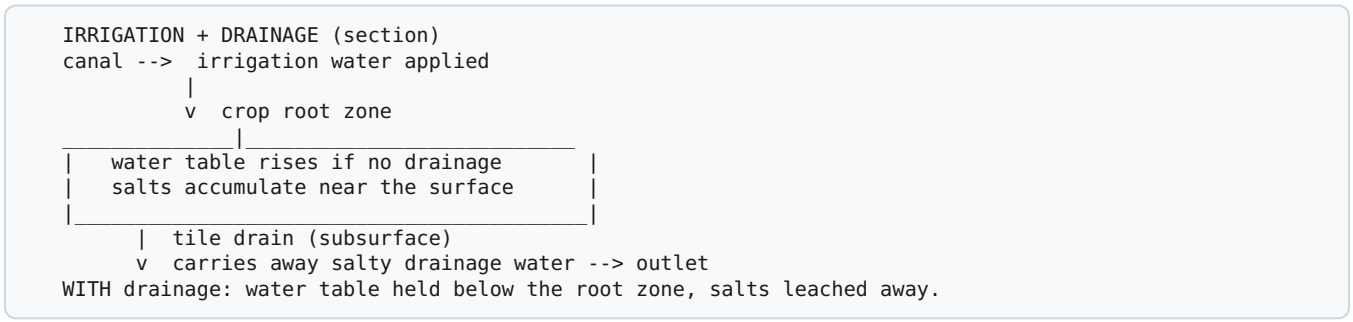


Fig. 15.2 — Drainage for salinity control in irrigated land

15.12 Drainage Maintenance

- De-silt open drains and remove weeds regularly.
- Clean tile drains and their outlets; check for root entry and clogging.
- Maintain the designed grade so that water does not stagnate.
- Protect outlets and junctions against erosion.

15.13 Worked Numerical — Drain discharge in cm/day

Worked Example 15.4 — Drainage discharge (cm/day)

A drainage system has a coefficient of 1.5 cm/day over an area of 80 ha. Find the design discharge.

Step 1: For q in cm/day, Q = (q × A) / 864.

Step 2: Q = (1.5 × 80) / 864 = 120 / 864 = **0.139 m³/s**.

Note that 1.5 cm/day = 15 mm/day, and indeed (15 × 80)/8640 = 0.139 m³/s — the two forms agree.

15.14 Mole Drainage

Mole drainage is a form of subsurface drainage in which unlined, continuous circular channels (moles) are formed by pulling a bullet-shaped mole plough through heavy clay soil at a depth of about 0.5–0.8 m.

- Cheap and quick, but the mole channels are temporary (may last a few years).
- Suited to heavy clay soils that hold a mole channel; not for sandy or loose soils.
- Usually combined with an open outlet drain at the lower end.

Exam Focus — Lecture 15

Key points

- Drainage removes excess water and salts from the root zone.
- Needs: control water-logging, restore aeration, remove salts, improve yields.
- Surface types: random, bedding, parallel, field-ditch.
- Subsurface types: tile drains, mole drains, deep ditches, vertical drainage.
- Drainage coefficient = depth of water removed per day (mm/day).
- Q = q A / 8640 (q in mm/day, A in ha).

- Hooghoudt's equation gives drain spacing.

Short-answer questions

Q1. Define drainage and state its need in agriculture.
Model answer: Drainage is the removal of excess water and salts from the root zone. Needs: control water-logging, restore soil aeration, leach harmful salts, improve trafficability and yields, and reclaim saline/water-logged land.

Q2. Define drainage coefficient.
Model answer: The drainage coefficient is the rate at which water must be removed from a drainage area to prevent water-logging, expressed as the depth of water removed in 24 hours (mm/day or cm/day).

Q3. Name the types of surface drainage systems.
Model answer: Random ditch system, bedding system, parallel ditch system and field-ditch (cross-slope) system.

Long-answer questions

L1. Explain the need for agricultural drainage and describe the surface and subsurface drainage systems with diagrams.
Outline: Need → water-logging and its effects → surface systems (random, bedding, parallel, field-ditch) → subsurface systems (tile, mole, deep ditch, vertical) → depth and spacing → comparison table → conclusion.

L2. Explain the drainage coefficient and how a drainage system is designed, with a worked example.
Outline: Definition and units → typical values → Q = qA/8640 → Hooghoudt's equation for spacing → worked examples → choice of coefficient → conclusion.

16 Types of Wells and Sizes — Suitability

16.1 What is a Well?

> **Well** is a vertical opening or structure dug or drilled into the ground to tap ground water from an aquifer for irrigation, domestic or industrial use.

An **aquifer** is a saturated geological formation that can store and transmit water in usable quantities. Wells are the main means of exploiting ground water, which is especially important in Tamil Nadu where surface supplies are limited.

16.2 Classification of Wells

(a) By method of construction

- **Dug (open) well:** excavated by hand or machine, large diameter (usually 2–10 m), shallow (5–30 m); taps the shallow water table.
- **Bored well:** made by a hand or power auger; small diameter, shallow; used for domestic supply.
- **Driven (sand-point) well:** a pipe with a screened point driven into sandy soil; shallow, low yield.
- **Tube well:** a long pipe of small diameter (usually 10–30 cm) drilled deep (30–300 m or more) into an aquifer; high yield.
- **Dug-cum-bore well:** a dug well with a bore hole at its bottom to reach a deeper aquifer; combines a large storage with a deeper source.

(b) By type of aquifer / pressure

- **Water-table (unconfined) well:** taps the free water table of an unconfined aquifer.
- **Artesian (confined) well:** taps a confined aquifer under pressure; water may flow out at the surface (flowing well).
- **Cavity well:** a shallow bore that ends in a cavity in hard rock; water enters through the cavity bottom.

16.3 Well Types — Comparison Table

Table 16.1 — Types of wells and their suitability

Type	Diameter	Depth	Yield	Best suited to
Dug (open) well	2–10 m	5–30 m	Low to moderate	Shallow water table, alluvial/hard-rock areas; domestic + small irrigation
Bored well	5–20 cm	up to ~10–20 m	Very low	Domestic supply in soft soil
Driven (sand-point) well	2.5–5 cm	up to ~10–15 m	Very low	Sandy, shallow aquifers; temporary supply
Tube well	10–30 cm	30–300 m	Moderate to high	Deep aquifers, large-scale irrigation; hard-rock and alluvial areas
Dug-cum-bore well	Dug 2–6 m + bore	Combined	Moderate	Where a dug well is insufficient; deep fracture zones

Artesian (flowing) well	As drilled	Deep	Varies	Confined aquifers under pressure
Cavity well	Small bore	Shallow	Low	Hard-rock cavities

16.4 Text Diagram — Open Well and Tube Well Sections

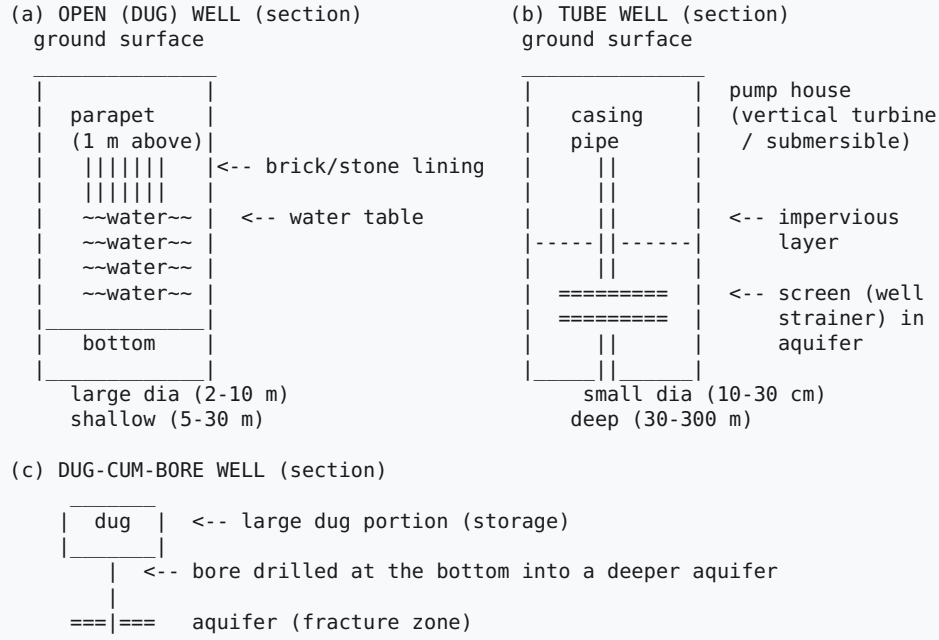


Fig. 16.1 — Open well, tube well and dug-cum-bore well sections

16.5 Well Hydraulics — Key Terms

- **Static water level:** the water level in a well when no pumping is done.
- **Pumping (dynamic) water level:** the water level while pumping is in progress.
- **Drawdown:** the difference between static and pumping water levels.
- **Cone of depression:** the funnel-shaped depression of the water table around a pumping well.
- **Radius of influence:** the horizontal distance from the well to the edge of the cone of depression.
- **Specific capacity:** the yield per unit drawdown.

Specific capacity:

Specific capacity = Yield / Drawdown (m³/h per m, or litres/s per m)

Yield (Q) ≈ Specific capacity × Drawdown

Well yield by Thiem's equation (unconfined, steady):

$Q = \pi k (H^2 - h^2) / \ln(R/r)$

k = permeability, H = original water-table height, h = water height in well, R = radius of influence, r = well radius.

16.6 Suitability of Wells

Table 16.2 — Choosing a well type

Situation	Suitable well
Shallow water table, small area, alluvial soil	Open (dug) well

Deep aquifer, large irrigation area	Tube well
Hard-rock area with fracture zones at depth	Tube well or dug-cum-bore well
Dug well inadequate, deeper source needed	Dug-cum-bore well
Domestic supply in soft soil, shallow	Bored or driven well
Confined aquifer under pressure	Artesian well

16.7 Worked Numericals

Worked Example 16.1 — Specific capacity and yield

An open well has a static water level 8 m below the ground. While pumping at a steady rate the water level falls to 11 m. If the specific capacity of the well is 0.6 m³/h per m of drawdown, find the yield.

Step 1 (drawdown): Drawdown = 11 − 8 = 3.0 m.

Step 2 (yield): Yield = specific capacity × drawdown = 0.6 × 3.0 = **1.8 m³/h** (about 1800 litres/h).

Worked Example 16.2 — Volume of water stored in an open well

A dug well has an internal diameter of 4 m and a depth of water of 5 m. Find the volume of water stored.

Step 1 (area): $A = \pi d^2/4 = 3.1416 \times 16/4 = 12.57$ m².

Step 2 (volume): V = A × depth = 12.57 × 5 = **62.85 m³** (62,850 litres).

This stored volume gives a dug well its buffer capacity, allowing it to meet peak demand even when the inflow is slow.

Worked Example 16.3 — Radius of influence (empirical)

For a well, the drawdown is 3 m and the coefficient of permeability is 10 m/day. Using the common empirical relation R = 3000 × s × √k (R, s in m, k in m/s), estimate the radius of influence. (Here use the simplified form R ≈ 3000 s √k with k in m/s.)

Step 1 (convert k): k = 10 m/day = 10/86,400 m/s = 1.157×10^{-4} m/s.

Step 2 (√k): $\sqrt{(1.157 \times 10^{-4})} = 0.01076$.

Step 3: R = 3000 × 3 × 0.01076 = **96.8 m**.

Empirical radius-of-influence formulas vary (Sichardt's and others); treat the value as an estimate, not a precise figure.

16.8 Tube Well Components

- **Casing pipe:** the outer pipe that keeps the bore hole open.
- **Well screen (strainer):** the perforated/slotted section opposite the aquifer that admits water while holding back sand.
- **Gravel pack:** a graded gravel envelope around the screen to improve flow and prevent sand entry.
- **Reducer and riser pipe:** connect the screen to the pump.

- **Bore-hole seal / cement plug:** prevents contamination from upper strata.

16.9 Well Development and Yield Factors

- **Well development:** the process of removing fines from around the screen (by surging, washing) to increase the yield and reduce sand pumping.
- **Yield depends on:** aquifer permeability and thickness, drawdown, well diameter and screen length.
- **Over-pumping** lowers the water table and can dry up the well; sustainable yield must be respected.

16.10 Text Diagram — Cone of Depression

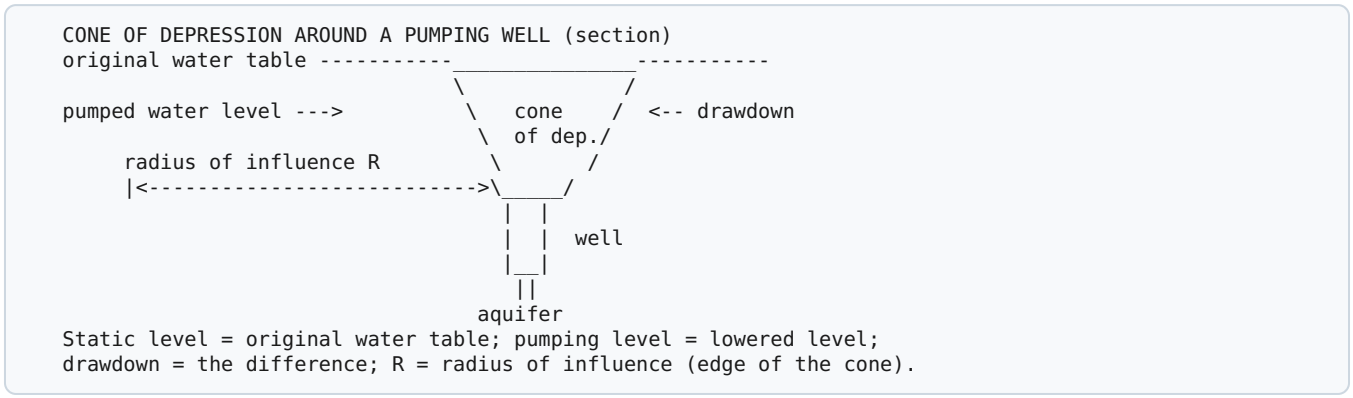


Fig. 16.2 — Cone of depression and radius of influence

16.11 Worked Numerical — Yield from specific capacity

Worked Example 16.4 — Permissible yield of a tube well

A tube well has a specific capacity of 40 litres/s per m of drawdown. The permissible drawdown is 5 m. Find the yield.

Step 1: Yield = specific capacity × drawdown.

Step 2: Yield = 40 × 5 = **200 litres/s** (0.20 cumec).

A yield of 0.20 cumec is large; more commonly tube wells in hard-rock areas yield 5–30 litres/s. The value depends on the aquifer.

16.12 Open Well vs Tube Well — Selection

Table 16.3 — Choosing between an open well and a tube well

Criterion	Open well	Tube well
Diameter	Large (2–10 m)	Small (10–30 cm)
Depth	Shallow (5–30 m)	Deep (30–300 m)
Yield	Low to moderate	Moderate to high
Storage in the well	Yes (large volume)	Negligible
Cost of construction	Lower per well but large volume of digging	Higher, needs a rig
Best for	Shallow water table, domestic + small irrigation	Deep aquifers, large irrigation

Exam Focus — Lecture 16

Key points

- A well taps ground water from an aquifer; aquifer = water-bearing formation.
- Types by construction: dug, bored, driven, tube, dug-cum-bore.
- Types by aquifer: water-table (unconfined), artesian (confined), cavity.
- Open well: large diameter, shallow, low-moderate yield; tube well: small diameter, deep, high yield.
- Drawdown = static level − pumping level; specific capacity = yield/drawdown.
- Cone of depression and radius of influence describe the effect of pumping.

Short-answer questions

Q1. Differentiate an open well and a tube well.
Model answer: An open (dug) well is large in diameter (2–10 m) and shallow (5–30 m) with low-to-moderate yield; a tube well is small in diameter (10–30 cm) and deep (30–300 m) with moderate-to-high yield.

Q2. Define drawdown and specific capacity.
Model answer: Drawdown is the difference between the static water level and the pumping water level. Specific capacity is the yield per unit drawdown (m^3/h per m).

Q3. Name any three types of wells with their suitability.
Model answer: Open well — shallow water table, small areas; tube well — deep aquifers, large irrigation; dug-cum-bore well — where a dug well is inadequate and a deeper source is needed.

Long-answer questions

L1. Describe the types of wells with their sizes and suitability, and draw an open well and a tube well section.
Outline: Definition and aquifer → classification (by construction and by aquifer) → comparison table (diameter, depth, yield, suitability) → diagrams → conclusion with TN context.

L2. Explain well hydraulics (static level, drawdown, cone of depression, specific capacity) and how well yield is assessed.
Outline: Definitions → specific capacity formula → Thiem's equation → radius of influence → worked examples → practical significance.

17 Types of Pumps and Working Principles

17.1 What is a Pump?

> **Pump** is a machine that lifts water from a lower level to a higher level, or moves water from one place to another, by adding energy to the water.

Pumps are essential for lifting ground water from wells, lifting water from canals, rivers, farm ponds and tanks, and for pressurising drip and sprinkler systems.

17.2 Basic Terms in Pumping

- **Suction lift:** the vertical distance from the water surface in the source to the centre of the pump impeller (the suction side).
- **Delivery head:** the vertical distance from the pump to the point of discharge, plus losses.
- **Total head (H):** the sum of suction lift, delivery head and friction losses in the pipes and fittings.
- **Static head:** the vertical difference between the source water level and the discharge level.
- **Friction head:** head lost due to friction in pipes, bends and valves.
- **Discharge (Q):** the volume of water pumped per unit time (litres/s or m^3/s).
- **Priming:** filling the pump casing and suction pipe with water before starting, to remove air (needed for centrifugal pumps).
- **NPSH (net positive suction head):** the minimum pressure head at the pump inlet needed to prevent cavitation.
- **Cavitation:** formation and collapse of vapour bubbles causing noise, vibration and impeller damage; avoided by limiting suction lift.

17.3 Classification of Pumps

- **Rotodynamic (centrifugal) pumps:** energy is added by a rotating impeller — centrifugal, mixed-flow, axial-flow (propeller) pumps.
- **Positive displacement pumps:** energy is added by trapping a fixed volume and pushing it — reciprocating (piston/plunger) and rotary pumps.
- **Special pumps:** jet pumps, air-lift pumps, submersible pumps (centrifugal of special construction).

17.4 Working Principle of the Centrifugal Pump

The centrifugal pump works on the principle of the centrifugal force. A rotating **impeller** with curved vanes spins inside a casing. Water entering at the centre (eye) of the impeller is thrown outward by centrifugal force to the casing, and its pressure rises. The casing guides the water to the delivery pipe, and the low pressure created at the eye draws in more water from the suction pipe.

- Must be **primed** before starting (the casing and suction pipe filled with water).
- Practical suction lift is limited to about 6–7 m (theoretical maximum ≈ 10.3 m at sea level, less in practice).
- Suited to large discharges at moderate heads.

17.5 Working Principle of the Submersible Pump

A submersible pump is a multistage centrifugal pump coupled directly to an electric motor, and the whole unit is lowered into the bore well below the water level. Water enters the pump through the motor housing, is pushed upward through the stages, and rises through the delivery pipe to the surface. Because the pump is submerged, priming and suction-lift problems are eliminated, and it can work at high heads.

- No suction lift problem; suited to deep bore wells.
- Multistage construction develops high heads.
- Widely used in Tamil Nadu for bore-well irrigation, drinking water and drip systems.

17.6 Pump Types — Head and Discharge Ranges

Table 17.1 — Types of pumps with typical head and discharge ranges

Pump type	Principle	Typical head (m)	Typical discharge	Best suited to
Centrifugal (single-stage)	Rotodynamic	10–50	Moderate to high (e.g., 5–100 L/s)	Shallow lifts, canal/tank/river lifting, general irrigation
Multistage centrifugal	Rotodynamic	30–150	Moderate	Higher heads at moderate flow
Submersible	Rotodynamic (multistage)	30–300	Low to moderate	Deep bore wells
Vertical turbine	Rotodynamic	20–200	Moderate to high	Deep wells, large supply
Propeller / axial-flow	Rotodynamic	2–10	Very high	Low-head, high-flow (drainage, lift irrigation)
Reciprocating (piston)	Positive displacement	High (>50, up to 100+)	Low	High heads, small flow
Jet pump	Rotodynamic + jet	Up to ~40–60	Low	Deep small-diameter wells, domestic
Air-lift pump	Compressed air	Limited	Low to moderate	Sandy, dirty water; bore wells

17.7 Text Diagram — Centrifugal and Submersible Pumps

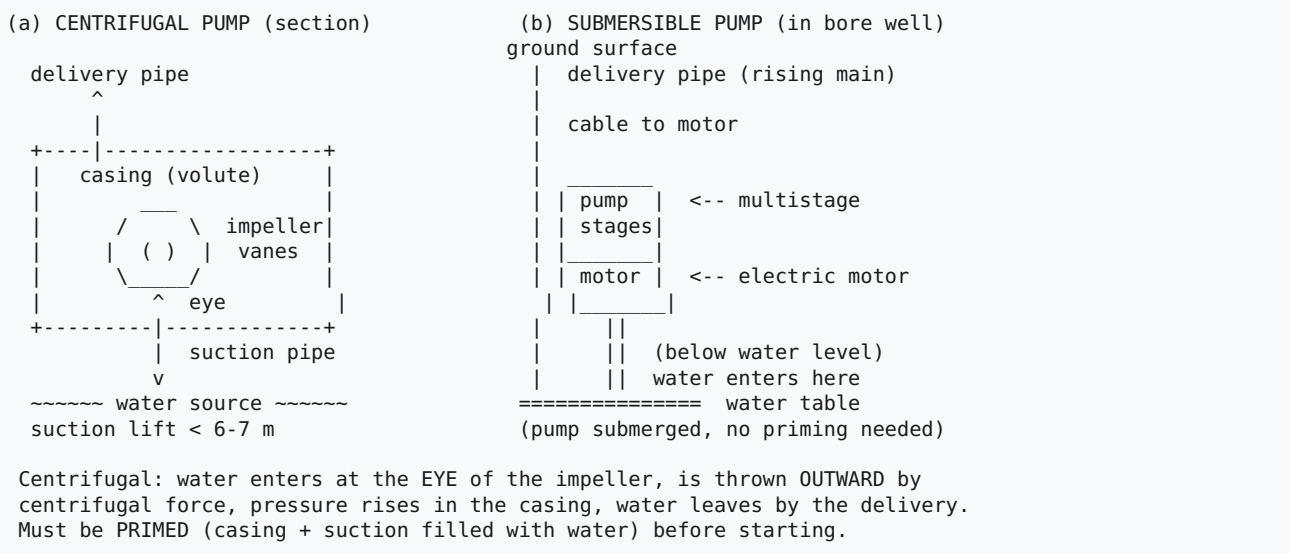


Fig. 17.1 — Centrifugal pump and submersible pump sections

17.8 Pump Power and Efficiency

Water horsepower (metric):

$WHP = \gamma\ Q\ H\ /\ 75 = (1000 \times Q \times H)\ /\ 75 = 13.33\ Q\ H$

Q = discharge (m^3/s), H = total head (m), γ = 1000 kgf/m^3.

Pump (brake) horsepower:

$BHP = WHP\ /\ \eta_p$ (η_p = pump efficiency, e.g., 0.6–0.8)

Power input to the motor (shaft input):

$\text{Input power} = BHP\ /\ \eta_m$ (η_m = motor efficiency)

Overall efficiency:

$\eta_o = WHP\ /\ \text{Input power} = \eta_p \times \eta_m$

In SI: $\text{Power (kW)} = 9.81\ Q\ H\ /\ \eta_o$

17.9 Worked Numericals

Worked Example 17.1 — Water horsepower

A pump lifts 50 litres/s against a total head of 20 m. Find the water horsepower.

Step 1 (convert Q): Q = 50 L/s = 0.050 m^3/s.

Step 2: WHP = 13.33 × Q × H = 13.33 × 0.050 × 20.

Step 3: WHP = 13.33 × 1.0 = **13.33 hp** (about 9.8 kW).

Worked Example 17.2 — Brake horsepower and input power

For the pump in Example 17.1, the pump efficiency is 70% and the motor efficiency is 85%. Find the brake horsepower and the input power.

Step 1 (BHP): BHP = WHP / η_p = 13.33 / 0.70 = **19.05 hp**.

Step 2 (input power): Input = BHP / η_m = 19.05 / 0.85 = **22.41 hp** (about 16.5 kW).

So a motor of about 22–25 hp would be selected (allowing a small margin). In kW, the input is about 16.5 kW.

Worked Example 17.3 — Overall efficiency and power in kW

A pump discharges 30 litres/s against a head of 30 m and draws 15 kW at the motor terminals. Find the overall efficiency and the water power.

Step 1 (Q): Q = 0.030 m^3/s.

Step 2 (water power in kW): P_n = 9.81 Q H = 9.81 × 0.030 × 30 = 8.83 kW.

Step 3 (overall efficiency): η_o = 8.83 / 15 = 0.589 = **58.9 %**.

Worked Example 17.4 — Selecting a pump for drip irrigation

A drip system needs 8 litres/s at a pressure of 2.5 kg/cm² (about 25 m head) plus a 5 m lift and 5 m friction loss. Find the total head and the water horsepower.

Step 1 (total head): H = operating pressure head + lift + friction = 25 + 5 + 5 = 35 m.

Step 2 (Q): Q = 8 L/s = 0.008 m³/s.

Step 3 (WHP): WHP = 13.33 × 0.008 × 35 = 13.33 × 0.28 = **3.73 hp**.

With a pump efficiency of 60%, BHP = 3.73/0.60 = 6.2 hp; a 7.5 hp pump set is typically chosen. Drip with fertigation improves water-use efficiency and is widely used in TN for sugarcane, banana and vegetables.

17.10 Selection and Maintenance of Pumps

- **Match the pump to the duty point** (head and discharge) — do not over- or under-size.
- Keep the suction lift low (below about 6–7 m for centrifugal pumps) to avoid cavitation.
- Use the correct pipe size to keep friction losses low.
- Prime centrifugal pumps before every start.
- Maintain the impeller, bearings and foot valve; check for wear and leakage.
- Use energy-efficient pumps and, where possible, solar pumps (popular in TN for irrigation).

17.11 Text Diagram — Pump Characteristic Curves

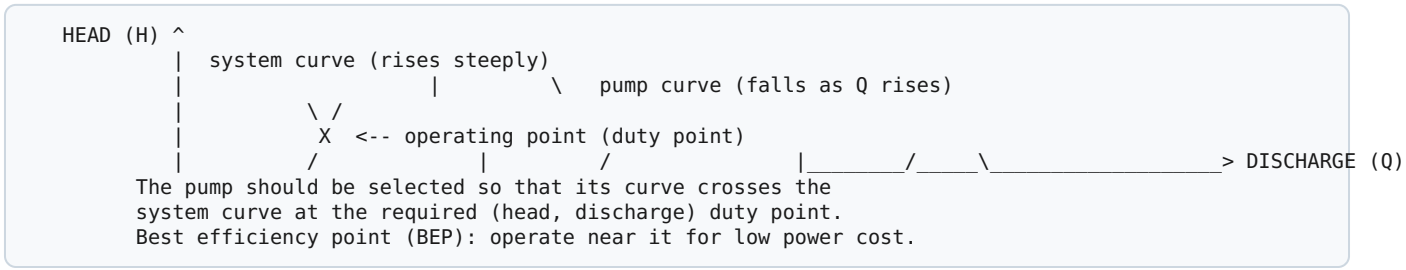


Fig. 17.2 — Pump and system curves and the duty point

17.12 Solar Pumps and Energy

- **Solar (photovoltaic) pumps** use solar panels to drive the motor; no fuel or electricity cost; widely promoted in Tamil Nadu for irrigation.
- Suited to sunny areas and small-to-medium irrigation; output varies with sunlight.
- **Energy cost** of pumping depends on head and efficiency — a 1% gain in efficiency saves a significant amount of power over a season.

17.13 Troubleshooting Common Pump Problems

Table 17.2 — Common pump problems and remedies

Problem	Likely cause	Remedy
No water delivered	Not primed / air leak / foot valve stuck	Prime the pump; check suction pipe and foot valve
Reduced discharge	Worn impeller / clogged screen / high lift	Repair impeller; clean screen; reduce lift
Noise and vibration (cavitation)	Excessive suction lift / high temperature	Lower the pump or reduce suction lift; check NPSH

Motor overload	Over-discharge / low head / voltage drop	Throttle to duty point; check supply voltage
Excessive power use	Poor efficiency / wrong duty point	Match pump to duty; maintain bearings

17.14 Worked Numerical — Pumping cost and energy

Worked Example 17.5 — Energy required to lift water

A pump lifts 40 litres/s against a total head of 25 m with an overall efficiency of 55%. Find the power input and the energy used in 100 hours of operation.

Step 1 (water power): $P_n = 9.81 \times Q \times H = 9.81 \times 0.040 \times 25 = 9.81$ kW.

Step 2 (input power): $P_i = 9.81 / 0.55 =$ **17.8 kW**.

Step 3 (energy in 100 h): $E = 17.8 \times 100 =$ **1780 kWh**.

At a typical agricultural tariff this energy cost can be worked out; raising efficiency or reducing head directly reduces the cost.

17.15 Reciprocating Pump — Working Principle

A reciprocating pump has a piston moving to and fro inside a cylinder. On the suction stroke the piston moves back, creating a vacuum that opens the suction valve and draws water in. On the delivery stroke the piston moves forward, closing the suction valve and forcing water through the delivery valve into the pipe. It gives a nearly constant discharge at high heads but is bulky and suited to small flows.

Discharge of a single-acting reciprocating pump:

```
Q = A · L · N / 60     (A = piston area, L = stroke, N = rpm; Q in m³/s)
For a double-acting pump: Q = 2 A L N / 60
```

Exam Focus — Lecture 17

Key points

- A pump adds energy to water to lift or move it.
- Two main classes: rotodynamic (centrifugal) and positive displacement.
- Centrifugal pump: works on centrifugal force; must be primed; suction lift < 6–7 m.
- Submersible pump: multistage centrifugal, submerged in bore well, no priming needed.
- WHP = 13.33 Q H (metric); BHP = WHP/η_p; overall $\eta = \eta_p \times \eta_m$.
- Total head = suction lift + delivery head + friction loss.
- NPSH and cavitation limit the suction lift.

Short-answer questions

Q1. Explain the working principle of a centrifugal pump.

Model answer: A rotating impeller throws water outward by centrifugal force, raising its pressure in the casing and delivering it to the pipe; the low pressure at the impeller eye draws in more water. The pump must be primed before starting.

Q2. What is priming and why is it needed?

Model answer: Priming is filling the pump casing and suction pipe with water before

starting, to remove air so that the pump can create suction and lift water; it is needed for centrifugal pumps.

Q3. Write the formula for water horsepower and brake horsepower.
Model answer: WHP = 13.33 Q H (Q in m³/s, H in m); BHP = WHP / pump efficiency.

Long-answer questions

L1. Describe the types of pumps with their working principles, head and discharge ranges, and draw a centrifugal pump section.
Outline: Classification → centrifugal working principle → submersible, turbine, propeller, reciprocating, jet → head/discharge table → diagram → selection → conclusion.

L2. Explain the terms suction lift, total head, cavitation and NPSH, and work out the horsepower requirement of a pump.
Outline: Definitions → total head composition → cavitation and NPSH → WHP, BHP, overall efficiency formulas → worked examples → pump selection → conclusion.

A Glossary of Key Terms

Term	Meaning
Accelerated erosion	Rapid erosion, mainly due to human activity, far exceeding the rate of soil formation.
Aquifer	A saturated geological formation that can store and transmit water in usable quantities.
Back sight (BS)	The first levelling reading, taken on a point of known reduced level.
Base period	Time from the first watering before sowing to the last watering before harvest (days).
Bench terrace	A step-like nearly level platform cut into a hill slope for cultivation.
Catchment	The area of land from which all rain water drains to a common outlet.
Change point (CP)	A point common to two instrument settings, on which a fore sight and the next back sight are taken.
Contour bund	An embankment along the contour that impounds runoff water in the field.
Cone of depression	The funnel-shaped lowering of the water table around a pumping well.
Delta (Δ)	Total depth of water required by a crop over its base period.
Drainage coefficient	Rate of water removal needed to prevent water-logging (mm/day or cm/day).
Drawdown	Difference between static and pumping water levels in a well.
Duty (D)	Area of land (ha) irrigated by a continuous supply of 1 cumec.
Erodibility (K)	Susceptibility of a soil to erosion.
Erosivity (R)	Potential of rain and runoff to cause erosion.
Graded bund	A bund along the contour with a longitudinal grade, leading runoff to an outlet.
Gully erosion	Erosion forming deep, wide channels too large to be removed by tillage.
Levee / bund	An earthen embankment.
NPSH	Net positive suction head; minimum inlet pressure head to prevent cavitation.
Offset	Lateral distance of a detail from the chain line.
Parshall flume	A calibrated converging-throat flume for measuring open-channel flow.
Percolation pond	A shallow pond for infiltrating runoff to recharge ground water.
Reduced level (RL)	Height of a point above or below the datum.
Rill erosion	Erosion forming small channels removable by ordinary tillage.
Runoff	The part of rainfall that flows over the land surface after losses.
Runoff coefficient (C)	Fraction of rainfall that becomes runoff.
Salinity / salt leaching	Removal of accumulated salts from the root zone by applying excess water with drainage.
Sheet erosion	Uniform removal of a thin layer of soil by unchanneled flow.
Specific capacity	Yield per unit drawdown of a well.
Splash erosion	Detachment of soil by the impact of raindrops.
USLE	Universal Soil Loss Equation: A = R K LS C P.
Water-logging	Saturation of soil pores with water, excluding air.
Watershed	All the land draining to a common point; a hydrological unit for development.
Weir / notch	A raised structure (weir) or opening (notch) for measuring flow.

Windbreak / shelterbelt	A strip (or belt of rows) of trees/shrubs planted to reduce wind velocity.

B Formula Sheet — One-Page Revision

Surveying and Levelling

RF = plan distance / ground distance
True length = Measured length × (L'/L) (chain correction)
Height of Collimation: HC = RL + BS; RL = HC − reading
ΣBS − ΣFS = ΣRise − ΣFall = Last RL − First RL (check)
Slope (%) = (Δh / HD) × 100; Slope(deg) = $\tan^{-1}(\Delta h/HD)$

Area and Volume

Trapezoidal (area): $A = (d/2)[O_0 + O_n + 2(O_1+O_2+...+O_{n11})]$
Simpson's (area): $A = (d/3)[(O_0+O_n) + 4\Sigma\text{even} + 2\Sigma\text{odd-interior}]$ (odd no. of ordinates)
Trapezoidal (volume): $V = L[(A_0+A_n)/2 + A_1 + ... + A_{n11}]$
Prismoidal (volume): $V = (L/3)[(A_0+A_n) + 4\Sigma\text{odd} + 2\Sigma\text{even}]$
1 ha = 10,000 m² = 2.471 acres; 1 acre = 4046.86 m²

Erosion

USLE: A = R × K × LS × C × P (t/ha/yr)
Wind erosion power ∝ (velocity)³
Soil depth lost = soil mass / (area × bulk density)
Tolerable soil loss T ≈ 5 t/ha/yr (medium-deep soils)

Bunds and Terraces

Vertical interval: VI = 0.3 (S + 2) m (S = slope in %)
Horizontal interval: HI = 100 × VI / S
Bund cross-section area = ½(top width + base width) × height
Base width = top width + 2 × (side slope × height)

Runoff and Water Harvesting

Rational method: Q = C I A / 360 (Q cumec, I mm/h, A ha)
Runoff volume: V = C × P × A; 1 mm over 1 ha = 10 m³
SCS-CN: $Q = (P - 0.2S)^2 / (P + 0.8S)$; $S = (25400/CN) - 254$
Catchment needed = Storage / (C × rainfall × 10)

Flow Measurement

Velocity-area: Q = A × V
Rectangular weir: Q = 1.84 (L − 0.1 n H) H^(3/2)
Cippoletti weir: Q = 1.86 L H^(3/2)
V-notch (90°): Q = 1.417 H^(5/2) = (8/15) $C_d\sqrt{(2g)}\tan(\theta/2)$ H^(5/2)

Orifice: $Q = C_d\ a\ \sqrt{(2gh)}$
Parshall flume: Q = K H^n

Duty and Efficiency

$\Delta = 8.64\ B / D$ (Δ m, B days, D ha/cumec); $D = 8.64\ B / \Delta$
$Q = A / D$
Conveyance eff. $E_c = (W_2/W_1)\times 100$; Application eff. $E_a = (W_s/W_2)\times 100$
Distribution eff. $E_d = (1 - d/D) \times 100$; Overall $E_o = E_c \times E_a$

Drainage

$Q\ (m^3/s) = q \times A / 8640$ (q in mm/day, A in ha)
Hooghoudt: $S^2 = (4\ k\ h^2 + 8\ k\ D\ h) / q$

Wells and Pumps

Specific capacity = Yield / Drawdown
Volume of open well = $\pi d^2/4 \times$ depth
Water horsepower (metric): WHP = 13.33 Q H (Q m^3/s, H m)
BHP = WHP / η_p; Input = BHP / η_m; Overall $\eta = \eta_p \times \eta_m$
Power (kW) = 9.81 Q H / η_o

Useful Unit Conversions

Quantity	Conversion
1 cumec	= 1 m^3/s = 1000 litres/s = 35.31 cusec
1 ha	= 10,000 m^2 = 2.471 acres
1 hectare-metre	= 10,000 m^3
1 mm rain over 1 ha	= 10 m^3
1 metric horsepower	= 75 kgf·m/s = 735.5 W
1 t/ha	= 1000 kg/ha = 100 g/m^2
1 acre	= 4046.86 m^2 = 0.4047 ha

C Reference List and Further Reading

Standard texts and course material for SWE 211 — Fundamentals of Soil and Water Conservation Engineering.

Core textbooks

1. **Michael, A.M. and Ojha, T.P.** *Principles of Agricultural Engineering, Vol. II* (Soil and Water Conservation Engineering, Irrigation and Drainage). Jain Brothers, New Delhi.
2. **Garg, S.K.** *Irrigation Engineering and Hydraulic Structures.* Khanna Publishers, New Delhi.
3. **Suresh, R.** *Soil and Water Conservation Engineering.* Standard Publishers Distributors, New Delhi.
4. **Schwab, G.O., Fangmeier, D.D., Elliot, W.J. and Frevert, R.K.** *Soil and Water Conservation Engineering.* John Wiley & Sons, New York.
5. **Murthy, V.V.N.** *Land and Water Management Engineering.* Kalyani Publishers, New Delhi.
6. **Murthy, V.V.N. and Jha, M.K.** *Land and Water Management.* Kalyani Publishers, New Delhi.
7. **Reddi, G.H. Sankara** and **Reddi, Y.V.** *Water Resources Engineering.* Standard Publishers.
8. **Punmia, B.C., Lal, P.B.B. and Jain, A.K.** *Surveying, Vols. I and II.* Laxmi Publications.
9. **Kanetkar, T.P. and Kulkarni, S.V.** *Surveying and Levelling.* Pune Vidyarthi Griha.

Course and reference material

1. **TNAU** — *SWE 211 Fundamentals of Soil and Water Conservation Engineering*, course manual and lecture notes, Tamil Nadu Agricultural University, Coimbatore.
2. **ICAR** — *e-Course / e-Krishi Shiksha material* on Soil and Water Conservation Engineering and Irrigation and Drainage Engineering (ecoursesonline.iasri.res.in).
3. **USDA-SCS / USBR** — *Water Measurement Manual* (Bureau of Reclamation, U.S. Department of the Interior), for Parshall flume and weir coefficients.
4. **Wischmeier, W.H. and Smith, D.D.** *Predicting Rainfall Erosion Losses — A Guide to Conservation Planning* (USDA Agriculture Handbook No. 537), for the USLE.
5. **FAO** — *Irrigation and Drainage Papers* (e.g., No. 24, Drainage Design Factors; No. 33, Water Quality for Agriculture).
6. **Ramdas, L.A.** and other Indian works on dryland agriculture and soil conservation.

Government schemes and reports (Tamil Nadu / India)

- Pradhan Mantri Krishi Sinchayee Yojana (PMKSY) — watershed development and "Har Khet Ko Pani".
- Tamil Nadu Watershed Development Agency (TAWDEVA) — state watershed programmes.
- Department of Agriculture, Government of Tamil Nadu — soil and water conservation and farm pond guidelines.
- Central Ground Water Board (CGWB) — Manual on Artificial Recharge of Ground Water.

Note on values: Where a parameter varies with soil, crop, climate or structure, a standard published range has been given rather than a single number. Students should quote the range and state the assumption used. Numerical examples use standard textbook constants (g = 9.81 m/s², ρ_w = 1000 kg/m³, 1 hp = 735.5 W).